

EMV Support Manual

BASE24-atm[®]



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Preface

The EMV product is an add-on to the BASE24-atm and BASE24-pos products. The ***BASE24-atm EMV Support Manual*** provides information about configuring EMV to operate in conjunction with the BASE24-atm product and describes the processing of BASE24-atm EMV transactions. For information about the BASE24-pos EMV program, see the ***BASE24-pos EMV Support Manual***.

Audience

This manual is intended for personnel who must understand the implementation of the BASE24-atm EMV product.

Prerequisites

This manual assumes the reader is very familiar with the BASE24-atm product. Only the additions and changes that have been made for the EMV module are described in the manual. Where appropriate, references for more information are made to other BASE24 manuals.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information.

This manual contains references to the following BASE24 publications:

- The ***BASE24 Base Files Maintenance Manual*** provides a comprehensive explanation of the BASE24 screens that access the base and switch files.
- The ***BASE24 Device Control Manual*** describes how to perform device control functions using the screens accessed from the BASE24 Device Control Terminal (DCT) Menu.
- The ***BASE24 Logical Network Configuration File Manual*** describes how institutions can define files, processes, and spooler locations at the network, product, process, and device level.
- The ***BASE24 Tokens Manual*** contains specifications for tokens and token processing.
- The ***BASE24 Transaction Security Services Processing Guide*** describes the Transaction Security Services process.
- The ***BASE24-atm DIAS Interchange Interface Manual*** provides EMV information regarding the BASE24-atm DIAS Interchange Interface.
- The ***BASE24-atm Files Maintenance Manual*** describes the screens that access the files specific to the BASE24-atm product.
- The ***BASE24-atm NCR 5XXX Device Support Manual*** provides a description of the CNFG5070 Configuration File.
- The ***BASE24-atm Netherlands BGC Interchange Interface User's Guide*** provides EMV information regarding the Netherlands BGC Interchange Interface.
- The ***BASE24-atm Transaction Processing Manual*** describes how BASE24 processes ATM transactions.
- The ***BASE24-pos EMV Support Manual*** describes how EMV operates with the BASE24-pos product.
- The ***BASE24 LINK (LIS5) Interchange Interface Manual*** provides EMV information regarding the BASE24 LINK (LIS5) Interchange Interface.

In addition, this manual references the following ISO standards:

- ISO 7816, ***Identification cards—Integrated circuit(s) cards with contacts***
- ISO 8583 (1987), ***Bank Card Originated Messages—Interchange Message Specifications—Content for Financial Transactions***
- ISO 3166, ***Codes for the Representation of Names of Countries***
- ISO 4217, ***Codes for the Representation of Currencies and Funds***

This manual also references the following vendor manuals:

- Diebold's *EMV User's Guide* for additional information regarding terminal configuration.
- The *NCR NDC+ Programmer's Reference Manual* and the *NCR NDC+ and Apra Advance NDC EMV Integrated Circuit Card (ICC) Reference Manual* for additional information about device states.
- The *Wincor Nixdorf NDC/Diebold D91x Message Format Extension for EMV* for specific information about the configuration file parameters.

Software

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (that is, CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

“Conventions Used in This Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, “Introduction,” provides an overview of the BASE24 EMV add-on product.

Section 2, “Files Maintenance Screens and Logical Network Configuration File Assigns and Params,” provides details of BASE24 screens and Logical Network Configuration File (LCONF) assigns and params that support BASE24-atm EMV extensions.

Section 3, “Transaction Processing,” describes the entities in the EMV transaction path and provides diagrams that illustrate EMV transaction flows.

Section 4, “Authorization Processing,” describes additional authorization processing that is performed on EMV transactions.

Section 5, “Device Handler Processing,” introduces the functionality provided by device-specific BASE24-atm Device Handler EMV extension modules.

Section 6, “Interchange Interface Processing,” describes the EMV processing that is performed by Interchange Interface processes.

Appendix A, “Diebold Device Support,” describes the device-specific configuration screens required to support BASE24-atm EMV transactions on Diebold ATMs.

Appendix B, “NCR Device Support,” describes the configuration of the NCR EMV-related state screens that you can access through the DCT to support EMV transactions on NCR ATMs.

Appendix C, “Shared D912 Device Support,” describes device-specific configuration screens required to support BASE24-atm EMV transactions on ATMs that support Wincor Nixdorf’s D912 message formats.

Appendix D, “Shared NDC+ Device Support,” introduces the EMV support for ATMs that support Wincor Nixdorf’s NDC+ message formats.

Readers can use the index by field name to locate information about a particular screen field and the index by data name to locate information about a particular field from a file or record structure.

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, AT-CC998-01 for the ***BASE24-atm EMV Support Manual***) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Dec-2009 for December, 2009) indicates the issue of the manual. The software release information (for example, R6.0v10 for release 6.0, version 10) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

Conventions Used in This Manual

This section explains how field descriptions are documented in this manual.

Field Descriptions

Each field appearing on a files maintenance screen is listed by name and then described. Field descriptions briefly summarize the contents, purpose, and permissible values, as shown in the following example taken from the EMV-specific screen of the Card Prefix File (CPF).

Example

EMV CHECK — Determines whether the Authorization process should perform EMV authorization for this card prefix. Valid values are as follows:

Y = Yes, perform EMV authorization.

N = No, do not perform EMV authorization. Process transactions with this card prefix as a magnetic stripe transaction.

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: Y
Data Name: CPF.EMVCPF.EMV-CHK

Explanation

Each field description is completed by one or more of the following items of information:

Item	Description
Example	Illustrates a possible entry for the field to further clarify the value or values that can be entered.

Item	Description
Field Length	Specifies the size of the field and the type of characters that can be entered. This length refers to the field and valid values for the file maintenance screen, not the field in the DDLs. Possible values are alphabetic, alphanumeric, hexadecimal, and numeric. The term alphanumeric includes all alphabetic, numeric, and special characters that can be entered from a keyboard without using a control sequence. The term hexadecimal includes all numbers and the letters A through F. When a field value cannot be modified by the operator, the field length is System Protected.
Occurs	Indicates the number of times the field can be displayed on the screen. This information is provided only when the field can be displayed multiple times.
Required Field	Specifies whether a value has to be entered in the field. Possible values are Yes and No. Some fields are required only under certain conditions. In this case, the entry is Yes, followed by the conditions that determine when the field is required.
Default Value	Specifies the value that is automatically placed in the field when the screen is first displayed or when the F8 key is pressed to clear the screen.
Data Name	Provides the DDL name associated with the field appearing on the screen. Data names are included in the documentation to assist in communicating screen and field issues to your technical staff. Note that screen data is not always stored in the BASE24 database as it appears on the screen or as it is described in the field description. If you need information on how screen data is actually stored, consult the DDLs.

1: Introduction

The BASE24-atm EMV product is an add-on to the BASE24-atm product. This section is an overview of the BASE24-atm EMV program and provides a brief introduction to the terminology used.

Europay, MasterCard, and Visa (EMV) have developed integrated circuit card (ICC) credit and debit specifications to facilitate interoperability across competing payment systems. That is, customers with EMV debit and credit cards could use any terminal that accepts EMV cards.

The BASE24-atm EMV product has been developed to support processing for both ICC and magnetic stripe transactions from a single terminal.

Integrated Circuit Card Technology

An integrated circuit card (ICC) is a plastic card, usually the size of a credit card, that contains an embedded microprocessor chip. This chip is capable of storing large amounts of cardholder information and can contain multiple applications. The terms chip card and smart card are sometimes used interchangeably with integrated circuit card. This manual uses both the terms integrated circuit card and ICC when describing the card that contains the embedded chip.

EMV cards have both a microprocessor chip and a magnetic stripe. Currently all debit and credit card schemes use the magnetic stripe with the microprocessor chip.

Card Standards

There are significant differences in the types of integrated circuit cards or EMV cards available. Almost all magnetic stripe cards are manufactured to a similar specification. However, each ICC manufacturer has offerings with a wide variety of chip memory sizes and processing capabilities.

Linked to the memory and processing potential is the operating system that resides on the chip itself. The operating system provides communications, file management, and security as well as application support. One of the goals of the EMV consortium is to establish a standard operating and application development system.

The following are technical standards that define the EMV card and its interface to the reading device.

- For bank payments, the card type conforms to ISO standards. The primary standard is the ISO 7816 standard, *Identification cards—Integrated circuit(s) cards with contacts*.
- The EMV 1996 (v3.1.1) specification defines the business requirements for the use of the technology. By agreeing to use the ISO 7816 standard, EMV has established the principle of allowing a single terminal to support all ICC and non-ICC schemes.
- The EMV 2000 (v4.0), EMV 2004 (v4.1), and EMV 2008 (v4.2) specifications update the business requirements initially defined in the EMV 1996 specification.

Common Core Definitions

Common core definitions (CCD) are specified in EMV version 4.1, and establish a set of common data definitions and processes between the EMV chip card and the issuer host. Both Visa and MasterCard mutually recognize chip card applications conforming to these definitions for use on their respective branded cards. These definitions address the external data flowing between the chip card and the issuer and certain minimum card behavior.

Common core definitions should simplify chip implementation. With these definitions, issuers receive and return chip data in one common format from both Visa and MasterCard branded cards. With previous implementations of VIS and M/Chip, there has been considerable equivalent functionality. However, both Visa and MasterCard, in some instances, have chosen different implementations to accomplish the same functionality. These differences have resulted in issuers having to deal with each implementation separately. Common core definitions go a long way to defining a common implementation to be used by Visa, MasterCard, and other payment organizations choosing to adopt the CCD as their chip implementation specifications. These changes are focused from the external aspects of the card through delivery of the transaction (authorization and clearing) to the issuer.

Common Payment Application

The common payment application (CPA) is a CCD-compliant application specification developed by Visa and MasterCard. CPA takes CCD a step further, and defines the internals of the entire chip card application. While CCD focuses on the external interface of the card to issuer data and some processes, it does not provide a truly common payment application from a card functionality perspective. CPA is a complete CCD-compliant card application specification. CPA provides common functionality for the entire payment application. It specifies common data element definitions, card risk management options, common tags, card behaviors, common personalization, etc., that meet the requirements of EMV and CCD. CPA enables a single application implementation to be personalized with the same data elements and tags, including common risk management controls, for cards from all payment schemes supporting the CCD and CPA specifications. The CPA specification is owned, managed, and maintained by EMVCo, with endorsement and acceptance by both Visa and MasterCard.

CPA cards, by definition, are CCD-compliant. Issuers can also choose to support cards, which are CCD-compliant, but not necessarily CPA-compliant. It is important to note that use of CPA is optional on the part of the issuer.

EMV Implementations Supported by BASE24-atm

The BASE24-atm EMV product currently supports the following EMV implementations. The version number and date identify the EMV standard on which the implementation was originally based. Because versions of the EMV Standard are backwards-compatible, all implementations are still EMV-compliant.

Implementation	Specification Identifier	
	Version Number	Date
Common Core Definitions (CCD)	v4.1	2004
MasterCard M/Chip 2.1	v3.1.1	1996
MasterCard M/Chip 4.0	v4.0	2000
United Kingdom ICC Specification (UKIS)	v3.1.1	1996
Visa (VIS) version 1.3.2	v3.1.1	1996
Visa (VIS) version 1.4.0	v4.0	2000

Why Use an EMV Card?

The following are key reasons for banks to introduce integrated circuit cards for credit and debit cards:

- Reduce fraud losses
- Provide end-to-end security
- Save on processing costs
- Offer new value-added services

Magnetic stripe cards are relatively easy to replicate, while the complexities of the ICC, and the embedded security provided by the chip, make replication of an ICC card very difficult.

Traditionally, transaction security has been performed in security zones along the transaction path for magnetic-stripe card-based transactions. For example, a different MAC would be used for each message passed between systems, where each system is considered a separate security zone, and the PIN block would be

translated every time a request message leaves a security zone. With EMV cards, the security is end-to-end (i.e., between the card and the issuer), rather than between various zones of the transaction path. Therefore, the function of an acquirer in an EMV transaction is basically to ensure that the data that comes in at one end is the same as the data that goes out at the other end.

The move to EMV technology enables banks to use the data storage and processing capabilities of the microchips to improve card and cardholder validation, as well as reduce communications costs by allowing increased offline authorization.

EMV Transaction Support

Integrated circuit cards have the usual attributes of a payment card: embossing, signature panel, hologram, and a magnetic stripe, allowing EMV-based payment methods to operate alongside existing traditional magnetic-stripe-based payment methods.

Four levels of transaction support provided by the BASE24-atm EMV product are based on the terminal capability, how the card details were read, and whether ICC data elements are included in the transaction message. Each level results in different processing requirements, as shown in the table below.

EMV-Capable Terminal?	Card Details Read from Chip or Magnetic Stripe?	Chip Data Elements Included?	Processing Requirements
No	Magnetic stripe	No	Standard BASE24 processing
Yes	Magnetic stripe	No	Add EMV Status token indicating EMV-capable terminal, but not a chip-read transaction. ¹
Yes	Microprocessor chip	No	Add EMV Status token indicating EMV-capable terminal and a chip-read transaction. ²

EMV-Capable Terminal?	Card Details Read from Chip or Magnetic Stripe?	Chip Data Elements Included?	Processing Requirements
Yes	Microprocessor chip	Yes, included in transaction message	Add EMV Status token indicating EMV-capable terminal and a chip-read transaction. Also add one or more tokens containing the chip data elements. EMV tokens are described in the <i>BASE24 Tokens Manual</i> .

- ¹ In this situation, the card may be fraudulent and can result in fallback transaction processing.
- ² In this situation, the card may be fraudulent. However, this situation could occur because of Visa's Easy Entry program or MasterCard's Partial Grade program. The Easy Entry and Partial Grade options allow for the replication of only Track 2 magnetic stripe information on the chip (i.e., the chip does not contain any other chip data elements).

The type of transaction described in the last row of the table above is defined as an EMV transaction. The other three types of transactions are not EMV transactions.

Files Maintenance

The following files contain information about the BASE24 EMV product. Section 2 describes the EMV-specific information in these files.

- The Institution Definition File (IDF) contains flags identifying the segments that can be included in the files belonging to the institution. For EMV authorization processing, a field on IDF screen 5 enable support of EMV transactions.
- The Card Prefix File (CPF) defines each card prefix that can be processed within a BASE24 logical network. For EMV authorization processing, screen 11 holds the CAM authentication control data for the card prefix and screen 12 holds EMV information specific to BASE24-atm.
- The Cardholder Authorization File (CAF) contains one record for each cardholder whose card-issuing institution uses the Positive or Positive Balance Authorization method. Screen 13 is specific to EMV authorization processing, and holds the CAM authentication control data for the ICC.
- The ICC KEY File (KEYI) supports the key storage requirements for EMV. The KEYI holds multiple authentication keys (up to a maximum of 24) for an individual KEYI group. In addition, this file contains the key used for message integrity, verifying script commands using a message authentication code (MAC).
- The Logical Network Configuration File (LCONF) is the central location of file information and parameters for a BASE24 system. There is one LCONF for each logical network. The BASE24 EMV product modules use the standard LCONF assigns and params plus some EMV-specific assigns and params.
- The ATM Terminal Data (ATD) File contains an EMV-specific screen. This screen enables you to specify the capabilities of each EMV terminal.

Maintaining the CAF

The CAF refresh record definition contains both mandatory and optional segments. Mandatory segments are those which must be present on tape for their entire length if the financial institution supports the segment. Optional segments must be present on the tape if the financial institution supports the segment, but the record can contain only the segment length field, set to the length of itself, instead of the entire segment. If only the length field of an optional segment is on the input file, that segment is not added to the disk record. If it is already present on disk, it will be deleted.

The Base and EMV segments of the CAF contain an application transaction counter (ATC) field. The ATC field you choose to update and check during transaction authorization affects your CAF maintenance procedures.

The EMV segment is processed as an optional segment, and therefore the segment on the CAF record is added or updated only if the entire segment is present in the CAF extract record. If the entire EMV segment is present in the refresh record, then the data fields in the CAF EMV segment (except the ATC field) are added or updated.

When a record is added (during either partial refresh or full refresh), the FILLER field in the EMV segment is initialized to ASCII zeros.

Note that a full CAF refresh resets the application transaction counter (ATC) value to zero in the Base segment and the EMV segment in records that contain that segment. Within the BASE24-atm Authorization process, if the ATC limit is set to the value zero, ATC limit checking is not performed on EMV transactions. If the ATC value from the card is less than or equal to zero, the transaction is declined. After a full file refresh the ATC in the CAF is reset to zero. The ATC limit checking is bypassed for the next EMV transaction and the CAF ATC (Base or EMV segment, depending on which is checked and updated) is set to the value on the card. This logic avoids the requirement for impacting ATC values during a full refresh, which would require considerable additional processing.

If contactless magnetic stripe transactions are also using the Base CAF ATC field to perform Dynamic Card Verification, it may be necessary to run a separate process to rebuild the Base CAF ATC after a full file CAF refresh.

From Host Maintenance Process

The From Host Maintenance process ISO message format allows user fields to be sent in the message from the host to BASE24 for information defined in the EMV segment of the CAF or PBF.

Authorization Processing

This section provides an overview of authorization processing for EMV transactions. Section 4 describes this processing in more detail.

BASE24-atm EMV authorization provides support for the following authorization levels and methods:

- Level 1 (online only) as an optional prescreen check
- Level 2 (offline only):
 - Negative with Usage Accumulation Authorization method
 - Positive Authorization method
 - Positive with Balance Authorization method
 - Negative Authorization method
- Level 3 (online/offline):
 - As an optional prescreen check
 - As stand-in processing for all authorization methods listed above

For more detailed information about these authorization levels and methods, see the *BASE24-atm Transaction Processing Manual*.

Card Authentication Method

EMV processing provides banks with the capability to capture and verify security-related data held on the microchip on the credit or debit card. This data is known as the card authentication method (CAM) data.

Part of the BASE24 processing of EMV cards includes calls to hardware security modules to verify card data. A separate set of security utilities, called CAM utilities, must be used in conjunction with the Base security utilities to support this verification.

On request messages, an authorization request cryptogram is generated by the card and the terminal and used by the issuer to authenticate the card. On response messages, an authorization response cryptogram is generated by the issuer and used by the card to authenticate the issuer. One call is made to the security module to verify the request cryptogram and generate the response cryptogram.

BASE24-atm allows issuers to validate the Card Verification Digits (CVD) during EMV authorization processing, where the CVD in the Track 2 equivalent data is different from the CVD on the magnetic stripe. Both CAM processing and card verification provide the business function of authenticating the card, but either or both authorization functions can be performed in a single transaction.

Issuer Script Commands

Issuer script commands allow the issuer to alter the data or functionality of cards already in circulation. The terminal forwards a script command to the card, if returned from the issuer in a response to an online authorization request. The card uses the script data to reconfigure itself.

The BASE24 system supports the following script commands:

- CARD BLOCK (an irreversible disabling of the microchip)
- PUT DATA (updates the card data)
- PIN CHANGE and PIN UNBLOCK (change or unblock an offline PIN)

Script data passed between the issuer and the card is secured with a message authentication code (MAC) generated by a security module. Issuer script data is supported either in pass-through mode or as response scripts generated by the BASE24 system.

Fallback Checking

If the microchip cannot be read on an ICC at an EMV-capable terminal, the system falls back or defaults to read the information from the magnetic stripe. The FALLBACK field in the CPF is used to configure whether to include fallback checking as a prescreen check.

Status Field Checking

Status field checking is performed using preconfigured edit files (CVRTBL and TVRTBL) that are compiled into the Authorization process. These files identify the field information to be checked and the required actions.

ATC Checking

The application transaction counter (ATC) is a value maintained by the microchip and updated for each transaction performed by the card application. A corresponding value is held in the CAF. The check is designed to detect fraudulent cards.

Dynamic card verification also uses an ATC. As a result, it is possible to maintain the ATC value on the Base segment or EMV segment of the CAF. However, this is typically done only if a card is likely to be used for both EMV and contactless magnetic stripe transactions. The dynamic card enhancement must be purchased to support this combination. If the card will be used only for EMV transactions, the ATC should be maintained only on the EMV segment.

Offline PIN Management

EMV cards have the capability to hold the cardholder's PIN within the ICC. This allows offline PIN verification (i.e., where the PIN entered by the cardholder is verified by the ICC) to be used as a Cardholder Verification Method (CVM) for electronic transactions. Many payment terminals are not enhanced to offer offline PIN verification, and so online PIN verification (i.e., where the PIN entered by the cardholder is verified by the issuer's system) continues to be widely used.

Where both online and offline PIN verification can be used as the CVM for an EMV card, cardholder confusion can result if the offline PIN value (i.e., the PIN held on the ICC) is different from the online PIN value (i.e., the PIN held on the issuer's system). Also, if a cardholder exceeds the allowable bad PIN tries when entering the offline PIN, a mechanism is required to allow the bad PIN try counter to be reset. Therefore, it is critical that card issuers provide a management function for the offline PIN.

The management of the offline PIN function on the card includes the following two administrative functions for card issuers:

- Support for offline PIN changes, where cardholders want to change their PINs to new values.
- Support for offline PIN unblocking, where the PIN Try Counter on the card has reached zero and must be reset to the normal value (the PIN Try Limit).

Both of these tasks are technically achieved by delivering an EMV script message to the card during an authorization response message, although it is impractical for this to take place during a routine point-of-sale transaction. Instead, ATMs are the ideal secure terminal at which to offer this service.

In order to ensure that card issuers can successfully manage the offline PIN function (e.g., ensuring that the offline PIN is synchronized with the online PIN known to the issuer), the BASE24-atm EMV product provides an offline PIN management service.

BASE24-atm support for PIN management comprises both PIN change and PIN unblock (offline PIN try counter reset) processing.

Note: You must use the Transaction Security Services process with the BASE24-atm EMV add-on product to support the generation of PIN CHANGE scripts because the actual PIN block sent in the script has to be encrypted, and this processing is supported only in the Transaction Security Services process. No encryption is required for PIN unblock scripts because the PIN block itself is not included in the script. For more information on the offline PIN management function support provided by the Transaction Security Services process, refer to the *BASE24 Transaction Security Services Processing Guide*.

Additional Routing and Authorization Functions

The BASE24 system also supports the following additional routing and authorization functions.

- Prefix routing of EMV transactions using a prefix route field in the EMV segment of the CPF
- Processing EMV card transactions as standard Track 2 card transactions when card issuers start distributing integrated circuit cards before being able to support transactions initiated by users with those cards

Hardware Security Module Support

To be implemented as an issuer, the BASE24-atm EMV add-on product must have access to a hardware security module (HSM). The BASE24-atm EMV add-on product supports the use of Atalla or Thales e-Security (Racal) security modules. Whichever vendor is chosen, the security module must have a version of firmware that is capable of supporting the EMV add-on product.

Atalla

EMV 2000 processing without common core definitions (CCD) or common session key derivation is supported on the following Atalla HSMs: Axx100-V or Axx100 (AKB) running version 2.62 or later firmware. Support for common core definitions or common session key derivation requires Atalla Axx100-VAR or Axx100-AKB HSMs running version 2.80 or later firmware.

Thales e-Security

For Thales e-Security (Racal) HSMs, the minimum firmware version depends on hardware model and EMV key derivation method, as shown in the following table.

EMV Key Derivation Method	Hardware Model	Firmware Version
1996	7x00	5.05
2000	7x00	7.0
CCD or common session key derivation	Not available	
1996	7x10	5.05
2000	7x10	3.0
CCD or common session key derivation	Not available	
1996	8xxx	5.05
2000	8xxx	2.0
CCD or common session key derivation	8xxx	2.1, 2.2, 2.3

The following table shows which HSM commands must be supported if the BASE24-atm EMV product is configured to perform specific cryptographic processing.

Cryptographic Process	Atalla Command	Thales e-Security Command	
		EMV 1996	EMV 2000
ARQC verification/ARPC generation	350	KQ	KW
EMV MAC generation	352	KU	KY
PIN change	351	KU	KY

Binary Data Support

EMV security processing requires the use of binary data. This requirement impacts certain HSMs and their interfaces with the BASE24-atm product.

The Thales e-Security HSM uses binary data for certain control commands. As a result, the interface between BASE24-atm and the Thales e-Security HSM must be configured with a protocol that supports the transmission of binary data (e.g., TCP/IP).

The Atalla NSP uses binary-coded hexadecimal data instead of binary data for control commands. As a result, the interface between BASE24-atm and the Atalla NSP can use protocols that do not support the transmission of binary data (e.g., async, bisync) in addition to protocols that support the transmission of binary data.

Transaction Security Services Process Requirements

The BASE24-atm Authorization process supports the following normal (EMV1996) cryptographic processing in an Atalla or Thales e-Security hardware module, with or without the use of the Transaction Security Services process.

- ARQC verification and ARPC generation
- EMV MAC generation

However, you must use the Transaction Security Services process with the BASE24-atm EMV product for the following functions:

- ARPC generation using Method 2 described in EMV version 4.1¹
- Card key derivation using the Option B algorithm defined in EMV version 4.1¹
- Issuer script processing with the Secure Messaging Format 1 defined in EMV version 4.1¹
- Offline PIN management; specifically PIN change script generation
- Session key derivation defined in EMV 2000
- Session key derivation using the Common method defined in EMV version 4.1¹

¹ Issuers requiring support for the CCD EMV scheme or common session key derivation must have at least version 06.4 of TSS.

For more information on the Transaction Security Services process and the functions it performs, refer to the ***BASE24 Transaction Security Services Processing Guide***.

Token Processing

The data required to authorize transactions is passed between the various processing modules in BASE24-atm Standard Internal Message (STM) format. These messages comprise a set of core (fixed) fields that are present in all internal messages, and a number of sets of function-specific fields known as tokens.

A token is a set of related fields added to an internal message as a method for passing data between modules in which new functionality has been implemented. EMV-related data is passed between the various processing modules within a BASE24 system by means of data tokens.

The following tokens contain information for the BASE24-atm EMV product.

- The EMV Status token contains control information, not necessarily specific to EMV transactions.
- The EMV Request Data token contains the 13 minimum data elements defined by EMV.
- The EMV Discretionary Data token contains other data elements defined in more than one application message standard where EMV support has been implemented.
- The EMV Response Data token contains data elements for response processing, and script command flags.
- The EMV Script Data token contains issuer script commands.
- The EMV Issuer Script Results token contains the results of the EMV Issuer Script processing received from the ATM.
- The EMV Supplementary Data token is not currently used by BASE24-atm, but is available to carry supplementary EMV data received from a terminal.

For more information about BASE24 tokens, including descriptions of tokens used by the BASE24-atm EMV product, see the ***BASE24 Tokens Manual***.

BASE24-atm Device Handler EMV Support

The following BASE24-atm device handlers support EMV processing:

- Diebold 10XX/478X Device Handler, which has two add-on EMV support modules that can be bound with it. Both add-on modules can be bound with the same Device Handler process.
 - The Diebold 10XX/478X EMV module provides EMV support for Diebold 10XX/478X ATMs and ATMs that emulate the Diebold 10XX/478X message.
 - The Shared Diebold 912 EMV module provides EMV support for ATMs that use Wincor Nixdorf's D912 message formats.
- NCR 5XXX Device Handler, which has two add-on EMV support modules that can be bound with it. Both add-on modules can be bound with the same Device Handler process.
 - The NCR 5XXX EMV module provides EMV support for NCR 5XXX ATMs and ATMs that emulate the NCR 5XXX message.
 - The Shared NDC+ EMV module provides EMV support for ATMs that use either of Wincor Nixdorf's NDC message formats, including the original Wincor Nixdorf NDC message format that is compatible with NCR's CAM II implementation and the later version as defined in Wincor Nixdorf's "NDC/Diebold D91x Message Format Extension for EMV."
- Tidel Device Handler¹
- Triton Device Handler¹

¹ Additional device-specific information is not provided in this manual for the Tidel and Triton Device Handlers.

Refer to section 5 and the appendixes for more information about device support.

BASE24 Interchange Interface EMV Support

The following Interchange Interfaces support BASE24-atm EMV processing:

- AMEX GNS Interface
- BIC ISO Interchange Interface
- DIAS Interchange Interface¹
- EPS-Net Interchange Interface¹
- JCB ISO Interchange Interface
- LINK Interchange Interface¹
- MasterCard Debit Switch Interchange Interface
- Netherlands BGC Interchange Interface¹
- SPAN2 Interchange Interface Interface¹
- VisaNet Interface

¹ Additional interchange interface-specific information is not provided in this manual for the DIAS, EPS-Net, JCB ISO, LINK, Netherlands BGC, and SPAN2 Interchange Interfaces. Refer to the *BASE24-atm DIAS Interchange Interface Manual*, *BASE24 EPS-Net Interchange Interface Manual*, *BASE24 LINK (LIS5) Interchange Interface Manual*, or the *BASE24-atm Netherlands BGC Interchange Interface User's Guide* for EMV information regarding these interchange interfaces. The JCB ISO Interchange Interface is not documented.

Refer to section 6 for more information about interchange interface processing.

EMV Support in the BASE24 External Message

The BIC ISO Interchange Interface and the BASE24 ISO Host Interface use existing tokens to carry EMV information internally and place these tokens in data element S-126 of the BASE24 external message.

Both interface processes use the standard technique for configuring the EMV-specific tokens to be included in external messages. Refer to the *BASE24 Tokens Manual* for information on configuring the tokens used by these interface processes.

2: Files Maintenance Screens and Logical Network Configuration File Assigns and Params

This section provides details of BASE24 screens and Logical Network Configuration File (LCONF) assigns and params that support the BASE24-atm EMV extensions. For information about other BASE24-atm screens, see the *BASE24 Base Files Maintenance Manual* and the *BASE24-atm Files Maintenance Manual*. For additional information about LCONF assigns and params, see the *BASE24 Logical Network Configuration File Manual*.

Institution Definition File (IDF)

The Institution Definition File (IDF) contains one record for each institution participating in the logical network and defines processing for each institution. The IDF contains routing tables for transaction routing within a BASE24 product and the parameters for cards, dates, processing control and sharing.

IDF Screen 5

IDF screen 5 contains flags identifying the segments that can be included in the files belonging to this institution. For EMV authorization processing, a field on the screen enables support of EMV transactions. This section describes the field on IDF screen 5 that has been modified to support EMV transaction processing.

```

BASE24-BASE  INSTITUTION FILE          LLLL          YY/MM/DD  HH:MM  05 OF 42

                FIID:                  FI-NAME:

                FIID AUTH FILE SEGMENT INDICATORS (01 OF 02)

Y BASE          N ATM          N POS          N EMV
N PBF CR LINE   N PBF SHORT NM  N SSB BASE   N SSB CHECK
N ADDR VERIF    N PREAUTH HOLD  N NON-CURR DSP N ENHANCED PREAUTH

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP

```

FIID AUTH FILE SEGMENT INDICATORS

A series of flags indicate which segments the institution's authorization files contain. These flags permit each institution to carry only the information needed for the BASE24 products supported by that institution. The following field is used to indicate EMV support.

EMV — Indicates whether EMV processing is supported. Valid values are as follows:

Y = Yes, EMV processing is supported.

N = No, EMV processing is not supported.

Field Length: 1 alphabetic character

Required Field: Yes

Default Value: N

Data Name: IDF.IDFBASE.FIID-SEG-MAP

Card Prefix File (CPF)

The Card Prefix File (CPF) defines each card prefix that can be processed within a BASE24 logical network. One record must exist in the CPF for each card prefix to be processed. If one prefix is used with multiple Primary Account Number (PAN) lengths, multiple records must exist in the CPF—one for each PAN length.

CPF records define the characteristics of each prefix and also contain prefix-specific parameters that allow institutions to define portions of their authorization processing that can be controlled at the prefix level. These parameters include expiration date checks, card and PIN verification controls, withdrawal limits, and credit account minimum standard increments.

CPF Screen 3

For EMV authorization processing, CPF screen 3 contains one field that can be used to control ATC checking. CPF screen 3 is shown below, followed by a description of the ATC CHECK field.

```

BASE24-BASE      CARD PREFIX      LLLL      YY/MM/DD  HH:MM  03 OF 22
PREFIX:          PAN LENGTH: 00      FIID:

      AUTHORIZATION INFORMATION

      EXPIRATION DATE PROCESSING FLAG:  0  (EXP DATE NOT REQUIRED)
      EXPIRATION DATE COMPARISON:  0  (COMPARISON NOT REQUIRED)
      CARD VALIDITY PERIOD (MONTHS):  000
      CHECK IF HOST ONLINE - SERVICE CODE:  N
      SERVICE CODE CHECKING ACTION INDEX:  1  (1-4)

      DYNAMIC CARD VERIFICATION INFORMATION
      DYN CARD VERIF KEY LOCATOR:          DCV CHECK TYPE:
      TRK1 DCVD OFST:      DCVD LEN:      ATC OFST:      ATC LEN:      UNPRED NUM OFST:
      TRK2 DCVD OFST:      DCVD LEN:      ATC OFST:      ATC LEN:      UNPRED NUM OFST:
      ATC CHECK:          CHECK IF HOST ONLINE DCV:
                                   ATC:

      BAD ATC ACTION:          BAD DCV ACTION:

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                   F12-HELP

```

ATC CHECK — A flag indicating whether the application transaction counter (ATC) check is to be performed. Setting this field causes the ATC fields in the Base segment of the CAF to be updated. Valid values are as follows:

- 0 = Do not perform the ATC check.
- 1 = Perform the ATC check on EMV transactions. The ATC CHECK TYPE field on CPF screen 11 must be set to a value of 8 or 9 for this check to be performed.
- 2 = Perform the ATC check on contactless magnetic stripe transactions.
- 3 = Perform the ATC check on EMV and contactless magnetic stripe transactions. The ATC CHECK TYPE field on CPF screen 11 must be set to a value of 8 or 9 for this check to be performed on EMV transactions.

Field Length: 1 numeric character
 Required Field: Yes
 Default Value: 0
 Data Name: ATC-CHK

CPF Screen 11

For EMV authorization processing, screen 11 holds the CAM authentication control data for the card prefix. This section describes the fields present on CPF screen 11.

```

BASE24-EMV      CARD PREFIX      LLLL      YY/MM/DD  HH:MM  11 OF 22
PREFIX:          PAN LENGTH: 00      FIID:

                        EMV INFORMATION
                        EMV CHECK: Y (Y/N)      KEYI GROUP:
CAM CHECK TYPE: 0 (CHECK DISABLED)      ATC CHECK TYPE: 0 (DISABLED)
      EMV IAD FORMAT: 0 (UNDEFINED)      STATUS CHECK ACTION INDEX: 1 (1-6)
ALTERNATE IAD FORMAT: 0 (UNDEFINED)      ALTERNATE ACTION INDEX: 1 (1-6)
SCRIPT TEMPLATE TAG: 0 (USE LCONF VALUE)      SCRIPT MAC LENGTH: 0
CHECK IF HOST ONLINE -      CAM: N (Y/N)      FALLBACK: N (Y/N)
                        TVR/CVR: N (Y/N)      REASON ONLINE CODE: N (Y/N)
BAD CAM ACTION - DATA RELIABLE: 1 (DENY AND RETURN)
BAD CAM ACTION - DATA UNRELIABLE: 1 (DENY AND RETURN)
      EMV PREFIX ROUTING:
CARD VERIFY - CHECK TYPE: 0 (NONE)      EMV CV DATE: 0501
                        CHECK METHOD: 0 (STANDARD/STANDARD)      CV DATA: 000
UPDATE COUNTERS -      DECLINED TRANSACTION: 0 (*****
                        APPROVED FINANCIAL TRANSACTION: 0 (*****
                        APPROVED NON-FINANCIAL TRANSACTION: 0 (*****

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                        F12-HELP
  
```

EMV INFORMATION

EMV CHECK — Determines whether the Authorization process should perform EMV authorization for this card prefix. Valid values are as follows:

Y = Yes, perform EMV authorization.

N = No, do not perform EMV authorization. Process transactions with this card prefix as a magnetic stripe transaction.

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: Y
Data Name: CPF.EMVCPF.EMV-CHK

KEYI GROUP — A value that matches the KEYI GROUP field in the KEYI. This field is not required if the BASE24 system is acquiring EMV transactions and passing them through to a host or interchange for authorization. However, this field is required if the BASE24 system is authorizing EMV transactions for this prefix or doing any prescreening (e.g., verification of request cryptograms).

Field Length: 16 alphanumeric characters
Required Field: Yes, if the BASE24 system is authorizing or prescreening EMV transactions for this prefix
Default Value: No default value
Data Name: CPF.EMVCPF.KEYI-GRP

CAM CHECK TYPE — The type of CAM verification to be performed. Valid values are as follows:

0 = Do not perform CAM check.

1 = Perform CAM check only if CAM data is reliable.

2 = Perform CAM check whether CAM data is reliable or unreliable.

9 = Process transaction as magnetic stripe instead of chip.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: CPF.EMVCPF.CAM-CHK-TYP

ATC CHECK TYPE — Determines whether the application transaction counter (ATC) check is to be performed and the location of the ATC. Valid values are as follows:

- 0 = Do not perform ATC checking for EMV transactions.
- 1 = Perform ATC checking for EMV transactions. Use this value when the card does not support contactless magnetic stripe transactions, or when the card maintains a separate ATC value for EMV transactions and contactless magnetic stripe transactions.
- 8 = Perform ATC checking for EMV transactions and maintain one ATC value for both EMV transactions and contactless magnetic stripe transactions. Use this value only if the card does not support contact EMV transactions, but does support contactless transactions that include an EMV cryptogram (for which a CPF EMV segment is required, but a CAF EMV segment is not). The ATC CHECK field on CPF screen 3 must be set to a value of 1 or 3 for this check to be performed.
- 9 = Perform ATC checking for EMV transactions and maintain one ATC value for both EMV transactions and contactless magnetic stripe transactions. Use this value only if there is one ATC on the card that is used for both contactless magnetic stripe transactions and EMV transactions. The ATC CHECK field on CPF screen 3 must be set to a value of 1 or 3 for this check to be performed.

Field Length: 1 numeric character
Required Field: Yes
Default Value: Y
Data Name: CPF.EMVCPF.ATC-CHK

EMV IAD FORMAT — The primary format of the issuer application data (IAD) used for this card type. Two IAD formats can be used for a card prefix, enabling cards with the same prefix to be migrated from one format to another as they are reissued.

A number of predefined formats for issuer application data exist, as specified by external entities (e.g., Visa, MasterCard, and American Express), but issuer-specific formats also can be defined. The data typically contains information that is used in the derivation of the authorization cryptogram, and the detailed layout is therefore required for issuer processing. Decoding procedures are included for recognized formats. Custom encoding and decoding procedures are required for proprietary formats.

The following data elements, which are included in the Visa, MasterCard, and American Express definitions, are used in EMV authorization processing and generation of the authorization cryptograms:

- Derivation key index
- Cryptogram version number
- Card verification results

Valid values are as follows:

- | | | |
|--------|---|---|
| 0 | = | Undefined. The format is unknown. The contents are transported as passthrough data, but no checks requiring the IAD are performed by the BASE24 system. |
| 1 | = | Visa/UKIS. This is the IAD format defined by Visa in the Visa Integrated Circuit Card (ICC) Specification. |
| 2 | = | M/Chip 2.1. This is the IAD format recommended for MasterCard MCPA M/Chip 2.1 cards. |
| 3 | = | M/Chip 4. This is the IAD format recommended for MasterCard MCPA M/Chip 4 cards. |
| 4 | = | CCD-A. This is the IAD format mandated for CCD (IAD Format A) cards. |
| 5 | = | AMEX. This is the IAD format defined for American Express cards. |
| 6 to 9 | = | Reserved for other national or international scheme formats. |
| A to Z | = | Proprietary format defined for the issuer. (Custom formatting procedures are required.) |

If this field is set to 1 (Visa/UKIS), the Visa format of issuer application data is used to extract the fields.

If this field is set to 2 (M/Chip 2.1), the MCPA M/Chip 2.1 format of issuer application data is used to extract the fields.

If this field is set to 3 (M/Chip 4), the MCPA M/Chip 4 format of issuer application data is used to extract the fields.

If this field is set to 4 (CCD-A), the CCD (IAD format A) format of issuer application data is used to extract the fields.

If this field is set to 5 (AMEX), the AMEX format of issuer application data (IAD) is used to extract the fields from the IAD to populate the EMV tokens. The AMEX format of the IAD matches the Visa format.

If this field is set to any other value, the data fields cannot be extracted because the format is currently unknown. If the fields are extracted, they are passed to the CAM utilities for request cryptogram verification and response cryptogram generation.

Field Length: 1 alphabetic or numeric character
Required Field: Yes
Default Value: 0
Data Name: CPF.EMVCPF.EMV-ISS-APPL-DATA-FRMT

STATUS CHECK ACTION INDEX — Identifies which of the preset rules for status field processing are to be used for this card type with the primary IAD format. These preset rules are defined in the CVR and TVR tables. For more information on these tables, see the topic “Status Field Verification” in section 4. The range of valid values is 1–6.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 1
Data Name: CPF.EMVCPF.ACTION-TABLE-INDEX

ALTERNATE IAD FORMAT — The alternate format of the issuer application data (IAD) used for this card type. Two IAD formats can be used for a card prefix, enabling cards with the same prefix to be migrated from one format to another as they are reissued.

A number of predefined formats for issuer application data exist, as specified by external entities (e.g., Visa, MasterCard, and American Express), but issuer-specific formats also can be defined. The data typically contains information that is used in the derivation of the authorization cryptogram, and the detailed layout is therefore required for issuer processing. Decoding procedures are included for recognized formats. Custom encoding and decoding procedures are required for proprietary formats.

The following data elements, which are included in the Visa, MasterCard, and American Express definitions, are used in EMV authorization processing and generation of the authorization cryptograms:

- Derivation key index
- Cryptogram version number
- Card verification results

Valid values are as follows:

- | | | |
|--------|---|---|
| 0 | = | Undefined. The format is unknown. The contents are transported as passthrough data, but no checks requiring the IAD are performed by the BASE24 system. |
| 1 | = | Visa/UKIS. This is the IAD format defined by Visa in the Visa Integrated Circuit Card (ICC) Specification. |
| 2 | = | M/Chip 2.1. This is the IAD format recommended for MasterCard MCPA M/Chip 2.1 cards. |
| 3 | = | M/Chip 4. This is the IAD format recommended for MasterCard MCPA M/Chip 4 cards. |
| 4 | = | CCD-A. This is the IAD format mandated for CCD (IAD Format A) cards. |
| 5 | = | AMEX. This is the IAD format defined for American Express cards. |
| 6 to 9 | = | Reserved for other national or international scheme formats. |
| A to Z | = | Proprietary format defined for the issuer. (Custom formatting procedures are required.) |

If this field is set to 1 (Visa/UKIS), the Visa format of issuer application data is used to extract the fields.

If this field is set to 2 (M/Chip 2.1), the MCPA M/Chip 2.1 format of issuer application data is used to extract the fields.

If this field is set to 3 (M/Chip 4), the MCPA M/Chip 4 format of issuer application data is used to extract the fields.

If this field is set to 4 (CCD-A), the CCD (IAD format A) format of issuer application data is used to extract the fields.

If this field is set to 5 (AMEX), the AMEX format of issuer application data (IAD) is used to extract the fields from the IAD to populate the EMV tokens. The AMEX format of the IAD matches the Visa format.

If this field is set to any other value, the data fields cannot be extracted because the format is currently unknown. If the fields are extracted, they are passed to the CAM utilities for request cryptogram verification and response cryptogram generation.

Field Length: 1 alphabetic or numeric character
Required Field: Yes
Default Value: 0
Data Name: CPF.EMVCPF.EMV-ISS-APPL-DATA-FRMT-ALT

ALTERNATE ACTION INDEX — Identifies which of the preset rules for status field processing are to be used for this card type with the alternate IAD format. These preset rules are defined in the CVR and TVR tables. For more information on these tables, see the topic “Status Field Verification” in section 4. The range of valid values is 1–6.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 1
Data Name: CPF.EMVCPF.ACTION-TABLE-INDEX

SCRIPT TEMPLATE TAG — Specifies the template tag to be used when an issuer script is sent to the card. Valid values are as follows:

- 0 = Use the value from the LCONF parameter. See the topic “Issuer Script Commands” in section 4 for additional information about the available parameters.
- 1 = Use a value of 71.
- 2 = Use a value of 72.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 1
Data Name: CPF.EMVCPF.SCRIPT-TPLT-TAG

SCRIPT MAC LENGTH — Specifies the MAC length to be used when an issuer script is sent to the card. Valid values are as follows:

- 0 = Use the value from the LCONF parameter. See the topic “Issuer Script Commands” in section 4 for additional information about the available parameters.
- 4 = Use a length of 4 bytes.
- 6 = Use a length of 6 bytes.
- 8 = Use a length of 8 bytes.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: CPF.EMVCPF.SCRIPT-MAC-LGTH

CHECK IF HOST ONLINE

The following four fields determine what prescreening checks the Authorization process should perform if the host is online:

CAM — Determines whether the Authorization process should perform an authorization request cryptogram prescreening check. Valid values are as follows:

- Y = Yes, perform a prescreen check.
- N = No, do not perform a prescreen check.

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: N
Data Name: CPF.EMVCPF.PRE-SCRN-CHK

FALLBACK — Determines whether the Authorization process should perform an EMV fallback prescreening check. Valid values are as follows:

- Y = Yes, perform a prescreen check.
- N = No, do not perform a prescreen check.

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: N
Data Name: CPF.EMVCPF.PRE-SCRN-CHK-FALLBACK

TVR/CVR — Determines whether the Authorization process should perform a TVR and CVR prescreening check. Valid values are as follows:

Y = Yes, perform a prescreen check.

N = No, do not perform a prescreen check.

Field Length: 1 alphabetic character

Required Field: Yes

Default Value: N

Data Name: CPF.EMVCPF.PRE-SCRN-CHK-TVR-CVR

REASON ONLINE CODE — Determines whether the Authorization process should perform a Reason Online Code prescreening check. Valid values are as follows:

Y = Yes, perform a prescreen check.

N = No, do not perform a prescreen check.

Field Length: 1 alphabetic character

Required Field: Yes

Default Value: N

Data Name: CPF.EMVCPF.PRE-SCRN-CHK-RSN-ONL-CDE

Note: This field is used for BASE24-pos EMV product transactions only.

BAD CAM ACTION – DATA RELIABLE — Indicates what actions the Authorization process should take if the request cryptogram fails verification when the CAM data is reliable. Valid values are as follows:

0 = Continue processing the transaction

1 = Decline the transaction and return the card

2 = Decline the transaction and retain the card

3 = Refer the transaction (This value applies to BASE24-pos only.)

If the request cryptogram verification fails, an authorization response cryptogram is not returned (i.e., the EMV Response token is not included in the response message) regardless of the value of this flag.

Field Length: 1 numeric character

Required Field: Yes

Default Value: 1

Data Name: CPF.EMVCPF.BAD-CAM-ACT-RELIABLE

BAD CAM ACTION—DATA UNRELIABLE — Indicates what actions the Authorization process should take if the request cryptogram fails verification when the CAM data is unreliable. Valid values are as follows:

- 0 = Continue processing the transaction
- 1 = Decline the transaction and return the card
- 2 = Decline the transaction and retain the card
- 3 = Refer the transaction (This value applies to BASE24-pos only.)

If the request cryptogram verification fails, an authorization response cryptogram is not returned (i.e., the EMV Response token is not included in the response message) regardless of the value of this flag.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 1
Data Name: CPF.EMVCPF.BAD-CAM-ACT-UNRELIABLE

EMV PREFIX ROUTING — A code indicating the routing group for this card prefix. Prefix routing groups are used to group prefixes for different types of authorization processing. The EMV prefix routing group number can be used to define a routing group to be used for EMV transactions, which can be different from the default routing group for this prefix, as specified in the base segment of the CPF. Valid values are as follows:

- 0–9 = A prefix routing group number. Any number specified should have a corresponding group number entry in the IDF.
- A = Any prefix not to be included in a special routing group.
- Blank = Use the default value for this prefix, as set in the base segment of the CPF (i.e., EMV cards will use the same routing group as non-EMV cards).

Field Length: 1 alphabetic or numeric character
Required Field: No
Default Value: No default value
Data Name: CPF.EMVCPF.EMV-PREFIX-RTE

CARD VERIFY – CHECK TYPE — Specifies the circumstances under which card verification is to be performed on EMV transactions. Valid values are as follows:

- 0 = Do not perform card verification.
- 1 = Perform card verification only when an ARQC is not present.
- 2 = Perform card verification when an ARQC is not present or when the ARQC has not been verified successfully.
- 3 = Always perform card verification.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: CPF.EMVCPF.EMV-CV-CHK-TYP

EMV CV DATE — Specifies the earliest card expiration date for which card verification is to be performed on EMV transactions. Valid values are dates in the format YYMM.

Field Length: 4 numeric characters
Required Field: Yes
Default Value: 0501
Data Name: CPF.EMVCPF.EMV-CV-EFF-DAT

CHECK METHOD — Specifies the algorithm and data to be used when card verification is performed on EMV transactions. Valid values are as follows:

- 0 = Use non-EMV card verification.
- 1 = Replace the service code with the value in the CV DATA field on this screen for cards expiring on or after the date in the EMV CV DATE field and use non-EMV card verification for cards expiring before the date in the EMV CV DATE field.
- 2 = Use non-EMV card verification for cards expiring on or after the date in the EMV CV DATE field and do not perform card verification for cards expiring before the date in the EMV CV DATE field.
- 3 = Replace the service code with the value in the CV DATA field on this screen for cards expiring on or after the date in the EMV CV DATE field and do not perform card verification on EMV transactions for cards expiring before the date in the EMV CV DATE field.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: CPF.EMVCPF.EMV-CV-CHK-MTHD

CV DATA — Contains any additional data required when card verification is performed on EMV transactions. Valid values are 000 through 999. For Visa issuers, the value is always 999.

Field Length: 3 numeric characters
Required Field: Yes
Default Value: 000
Data Name: CPF.EMVCPF.EMV-CV-DATA

UPDATE COUNTERS

The following three fields specify how a card should update the offline counters when processing a response message received from the issuer. These fields are used only for M/Chip 4.0 and CCD cards.

DECLINED TRANSACTION — Specifies how the card should update the offline counters when processing a response message for a declined transaction. Valid values are as follows:

- 0 = Do not update the offline counter.
- 1 = Set the counter to the upper offline limit.
- 2 = Reset the counter to zero.
- 3 = Add the transaction to the counter.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: CPF.EMVCPF.EMV-UPDATE-CNTRS-DCLN-TXN

APPROVED FINANCIAL TRANSACTION — Specifies how the card should update the offline counters when processing a response message for an approved financial transaction. Valid values are as follows:

- 0 = Do not update the offline counter.
- 1 = Set the counter to the upper offline limit.
- 2 = Reset the counter to zero.
- 3 = Add the transaction to the counter.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: CPF.EMVCPF.EMV-UPDATE-CNTRS-APPRV-FNCL-TXN

APPROVED NON-FINANCIAL TRANSACTION — Specifies how the card should update the offline counters when processing a response message for an approved non-financial transaction. Valid values are as follows:

- 0 = Do not update the offline counter.
- 1 = Set the counter to the upper offline limit.
- 2 = Reset the counter to zero.
- 3 = Add the transaction to the counter.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: CPF.EMVCPF.EMV-UPDATE-CNTRS-APPRV-NON-FNCL-TXN

CPF Screen 12

For EMV authorization processing, screen 12 holds parameters specific to BASE24-atm processing. This section describes the fields present on CPF screen 12.

```
BASE24-EMV      CARD PREFIX      LLLL      YY/MM/DD  HH:MM  12 OF 22
PREFIX:                PAN LENGTH: 00      FIID:

                        ATM EMV INFORMATION
                        FALLBACK ACTION: 0 (CONTINUE)
                        BAD OFFLINE PIN ACTION: 0 (CONTINUE)
                        OFFLINE PIN MANAGEMENT ACTION: 0 (CONTINUE)
                        NEW PIN TRANSPORT KEY LOCATOR: 0000000000000000

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                        F12-HELP
```

ATM EMV INFORMATION

The following fields contain BASE24-atm EMV processing parameters for the card prefix.

FALLBACK ACTION — Determines what action the Authorization process takes for ATM fallback transactions. Valid values are as follows:

- 0 = Continue processing the transaction
- 1 = Decline the transaction and return the card

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: CPF.EMVCPF.FALLBACK-ACT-ATM

BAD OFFLINE PIN ACTION — Specifies the processing required when a PIN-based EMV transaction contains data indicating that the offline bad PIN try limit has been exceeded. Valid values are as follows:

- 0 = Continue processing the transaction without performing EMV-specific processing
- 1 = Decline the transaction and return the card
- 3 = Generate and return a PIN Unblock script command

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: CPF.EMVCPF.OFFL-PIN-BAD-EXCD-ACT

OFFLINE PIN MANAGEMENT ACTION — Specifies the processing required when an EMV PIN Change or PIN Unblock transaction is received. Valid values are as follows:

- 0 = Continue processing the transaction without performing EMV-specific processing
- 1 = Generate and return a script command and do not log new PIN data
- 2 = Generate and return a script command and log new PIN data

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: CPF.EMVCPF.OFFL-PIN-MGMT-ACT

NEW PIN TRANSPORT KEY LOCATOR — Specifies a locator to a key in the Interchange Security File (ICHG_SECURITY) for the Transaction Security Services process. Valid values are any 16 hexadecimal characters.

If you need to notify a card management system of offline PIN changes, set this field to a PIN block encryption key locator value and the OFFLINE PIN MANAGEMENT ACTION field on CPF screen 11 to a value of 2. This PIN block encryption key locator value must correspond to a primary key value (interface name) for a record in the ICHG_SECURITY. This record must specify a PIN encryption key in the Outbound Keys field on the PIN information tab of the Interface Security Configuration window. The Transaction Security Services process translates the new PIN block to encryption under this key. The translated PIN block is then written to the PIN Change token in the Transaction Log File

(TLF). When the card management system receives the new PIN block (e.g., in a TLF extract), it uses the same key as stored in the ICHG_SECURITY to decrypt the new PIN block.

The ICHG_SECURITY is located in the Transaction Security Services database. Refer to the ***BASE24 Transaction Security Services Processing Guide*** for additional information.

Field Length:	16 alphanumeric characters
Required Field:	Yes
Default Value:	0000000000000000
Data Name:	CPF.EMVCPF.TRANSPORT-KEY

CPF Screen 13

Screen 13 holds BASE24-pos EMV processing parameters for the card prefix. All fields on CPF screen 13 are used only by BASE24-pos. Refer to the ***BASE24-pos EMV Support Manual*** for the screen layout and field descriptions.

CAF Screen 7

CAF Screen 7 enables institutions to set the expiration date and card status for the second card. This screen also displays the second card ATC for EMV transactions tracked in the Base segment of the CAF, which may or may not be used, depending on how the CPF is configured. Many institutions reissue cards, and a period exists when two physical cards with the same card number are in circulation. CAF screen 7 is shown below, followed by a description of the SECOND CARD ATC NUMBER field.

```
BASE24-BASE  CARDHOLDER FILE          LLLL          YY/MM/DD  HH:MM  07 OF 21
```

```
PAN:
```

```
MEMBER: 000
```

```
FIID:
```

```
SECOND CARD USAGE CONTROL
```

```
EXPIRATION DATE (YYMM) : 0000
```

```
CARD STATUS: 6      (CARD NOT USED)
```

```
SECOND CARD ATC NUMBER:
```

```
***** BASE24 *****
```

```
NEW PAGE:
```

```
FILE DESTINATION:
```

```
NEW LOGICAL NETWORK ID:
```

```
F12 - HELP
```

SECOND CARD ATC NUMBER — Stores the last application transaction counter (ATC) received by BASE24 from the second card. This field is used to record the number of transactions performed on the second card when the ATC is contained in the Base segment instead of the EMV segment.

Field Length: System protected

Data Name: CAF.CAFBASE.ATC-SCND-CRD

CAF Screen 13

Screen 13 is specific to EMV authorization processing, and holds the CAM authentication control data for the ICC. CAF screen 13 is shown below, followed by descriptions of its fields.

```

BASE24-EMV   CARDHOLDER FILE           LLLL           YY/MM/DD  HH:MM  13 OF 21

      PAN:                                     MEMBER: 000      FIID:

                                EMV INFORMATION
      ATC LIMIT: 9999                      BAD CAP TOKEN OVERRIDE FLAG: N
      EMV IAD FORMAT: 0 (USE CPF VALUE)    STATUS CHECK ACTION INDEX: 1 (1-4)
      SCRIPT TEMPLATE TAG: 0 (USE CPF VALUE)  SCRIPT MAC LENGTH: 0
      SEND CARD BLOCK: N                      SEND PUT DATA: N
                                LOWER CONSECUTIVE OFFLINE LIMIT: 0
      SEND PIN UNBLOCK: 0 (NO ACTION)
      SEND PIN CHANGE: 0 (NO ACTION)
      ONLINE/OFFLINE PIN SYNC: 0 (NO ACTION)

      ATC      CAP ATC      BAD CAP
      NUMBER   NUMBER      TOKEN COUNTER   CAP APSN   CAP DKI
PRIMARY CARD:    0          0              0          00
SECONDARY CARD:  0          0              0          00

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12 - HELP

```

EMV INFORMATION

The following fields contain information for EMV authorization.

ATC LIMIT — Specifies the acceptable number of transactions that can be performed on the card before issuer authorization is required. The range of valid values is 0–9999. A value of zero disables the limit check.

Field Length:	4 numeric characters
Required Field:	Yes
Default Value:	9999
Data Name:	CAF.EMVCAF.ATC-LMT

BAD CAP TOKEN OVERRIDE FLAG — A flag controlling whether additional CAP Token validation attempts can be performed if the number of consecutive failed CAP Token validation errors specified in the BAD CAP TOKEN VALIDATION LIMIT field on Card Prefix File (CPF) screen 13 has been reached. Valid values are as follows:

- Y = Yes, additional validation attempts can be performed.
- N = No, additional validation attempts cannot be performed.

If the validation is successful, the values in the BAD CAP TOKEN COUNTER fields on this CAF screen are reset to zero. This mechanism is the only way that the values in the BAD CAP TOKEN COUNTER fields can be reset once the limit has been reached without deleting and re-adding or replacing the CAF record.

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: N
Data Name: CAF.EMVCAF.BAD-CAP-TKN-OVRRD-FLG

Note: This field is not currently supported by the BASE24-atm EMV product. For more information about this field, refer to the ***BASE24-pos EMV Support Manual***.

EMV IAD FORMAT — Specifies the format of the EMV Issuer Application Data (IAD) that will be used for the card.

Note: The values in the EMV IAD FORMAT and ALTERNATE IAD FORMAT fields on CPF screen 11 identify which format is being used on a card. Prior to the ALTERNATE IAD FORMAT field on CPF screen 11 being available, the EMV IAD FORMAT field on CAF screen 13 enabled a card issuer to use the M/Chip 4 format for specific accounts within a card prefix that was still using the M/Chip 2.1 format.

Valid values are as follows:

- 0 = Use the format identified by the setting in the EMV IAD FORMAT field on CPF screen 11.
- 1 = The format as defined by Visa in the Visa Integrated Circuit Card (ICC) specification. This format is also used for UKIS cards.

The format is defined by the VISA-APPL-DATA data elements of the EMV Request Data token.

- 2 = The format recommended for MasterCard MCPA (M/Chip 2.1) cards.
The format is defined by the MCPA-APPL-DATA data elements of the EMV Request Data token.
- 3 = Use the IAD format recommended for MasterCard MCPA M/Chip 4 cards.
The format is defined by the MCHIP4-APPL-DATA data elements of the EMV Request Data token.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: CAF.EMVCAF.ISS-APPL-DATA-FRMT

SCRIPT TEMPLATE TAG — Specifies the template tag to be used when an issuer script is sent to the card. Valid values are as follows:

- 0 = Use the value from the SCRIPT TEMPLATE TAG field on CPF screen 11.
1 = Use a value of 71.
2 = Use a value of 72.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 1
Data Name: CAF.EMVCAF.SCRIPT-TPLT-TAG

SCRIPT MAC LENGTH — Specifies the MAC length to be used when an issuer script is sent to the card. Valid values are as follows:

- 0 = Use the value from the SCRIPT MAC LENGTH field on CPF screen 11.
4 = Use a length of 4 bytes.
6 = Use a length of 6 bytes.
8 = Use a length of 8 bytes.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: CAF.EMVCAF.SCRIPT-MAC-LGTH

STATUS CHECK ACTION INDEX — Identifies which of the rules for status field processing are to be used for the card. This field is used when M/Chip 2.1 and M/Chip 4 cards are issued with the same card prefix.

When the EMV IAD FORMAT field on CAF screen 13 is set to a value of 0 (use CPF value), the value in the STATUS CHECK ACTION INDEX field defined on CPF screen 11 (at the Prefix level) is used instead of the value in the STATUS CHECK ACTION INDEX field on CAF screen 13 (this field).

These rules are defined in the CVR and TVR tables. For more information on these tables, see the topic “Status Field Verification” in section 4. The range of valid values is 1–4.

Field Length: 1 numeric character
Required Field: Yes
Default Value: 1
Data Name: CAF.EMVCAF.ACTION-TABLE-INDEX

SEND CARD BLOCK — Determines whether a CARD BLOCK script command should be sent to the terminal to disable the microchip. Valid values are as follows:

Y = Yes, send CARD BLOCK script command.
N = No, do not send CARD BLOCK script command.

Note that this field is not automatically reset once an attempt to send the script has been made.

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: N
Data Name: CAF.EMVCAF.SEND-CARD-BLK

SEND PUT DATA — Determines whether a PUT DATA script command should be sent to the terminal to update the microchip data (e.g., new limits data). Valid values are as follows:

Y = Yes, send PUT DATA script command.
N = No, do not send PUT DATA script command.

This field is automatically reset once an attempt to send the script has been made.

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: N
Data Name: CAF.EMVCAF.SEND-PUT-DATA

LOWER CONSECUTIVE OFFLINE LIMIT — Specifies the maximum number of consecutive transactions that can be performed on the card before an attempt must be made to obtain authorization from the issuer. This field is used by BASE24-atm to generate the variable component of the PUT DATA script command so that a card can have a new limit downloaded. The range of valid values is 0–255.

Field Length: 4 numeric characters
Required Field: Yes
Default Value: 0
Data Name: CAF.EMVCAF.VEL-LMTS.LWR-CONSEC-LMT

SEND PIN UNBLOCK — Specifies the circumstances under which a PIN Unblock script command is generated and returned by the BASE24-atm system. Valid values are as follows:

- 0 = No action required (no EMV processing)
- 1 = Implicit (an EMV transaction indicates that the offline bad PIN try limit has been exceeded)
- 2 = Explicit (an EMV PIN Unblock transaction is received)
- 3 = Implicit and Explicit (an EMV transaction indicates that the offline bad PIN try limit has been exceeded or an EMV PIN Unblock transaction is received)

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: CAF.EMVCAF.SEND-PIN-UNBLK

SEND PIN CHANGE — Specifies the circumstances under which a PIN Change script command is generated and returned by the BASE24-atm system. Valid values are as follows:

- 0 = Do not send script (no EMV processing)
- 1 = An EMV PIN Unblock transaction is received
- 2 = An EMV PIN Change transaction is received
- 3 = An EMV PIN Unblock transaction or and EMV PIN Change transaction is received

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: CAF.EMVCAF.SEND-PIN-CHNG

ONLINE/OFFLINE PIN SYNC — Specifies whether synchronization of the online and offline PIN is required for the card. Valid values are as follows:

- 0 = PIN synchronization not required (no EMV processing)
- 1 = PIN synchronization required

Field Length: 1 numeric character
Required Field: Yes
Default Value: 0
Data Name: CAF.EMVCAF.PIN-CHNG-ACT

PRIMARY CARD DATA

The following five fields contain information about the primary card.

ATC NUMBER — The last application transaction counter (ATC) received by BASE24 from the primary card. This field is used to record the number of transactions performed on the primary card when the ATC is contained in the EMV segment instead of the Base segment.

Field Length: System Protected
Data Name: CAF.EMVCAF.ATC

CAP ATC NUMBER — The ATC received by BASE24 in the last approved Chip Authentication Program (CAP) token validation transaction performed on the primary card.

Field Length: System Protected
Data Name: CAF.EMVCAF.CAP-DATA.ATC

Note: This field is not currently supported by the BASE24-atm EMV product. For more information about this field, refer to the ***BASE24-pos EMV Support Manual***.

BAD CAP TOKEN COUNTER — The number of consecutive failed CAP token validation attempts performed on the card. This value is reset to zero when a CAP token is validated successfully for the primary card.

Field Length: System Protected
Data Name: CAF.EMVCAF.CAP-DATA.BAD-CAP-TKN-CHK-ACCUM

Note: This field is not currently supported by the BASE24-atm EMV product. For more information about this field, refer to the ***BASE24-pos EMV Support Manual***.

CAP APSN — Specifies the CAP application PAN sequence number (APSN) for the card identified by the values in the PAN and MEMBER fields on each CAF screen. The APSN identifies and differentiates cards with the same PAN. In EMV transactions, this field is generally sent to BASE24 as part of the transaction data. However, for CAP transactions, it may not be present, which means the value has to be configured on the database.

Field Length: 1–2 numeric characters
Required Field: No
Default Value: No default value
Data Name: CAF.EMVCAF.CAP-DATA.APSN

Note: This field is not currently supported by the BASE24-atm EMV product. For more information about this field, refer to the ***BASE24-pos EMV Support Manual***.

CAP DKI — Specifies the derivation key index (DKI) for the card identified by the values in the PAN and MEMBER fields on each CAF screen. The DKI identifies which of a set of master keys is to be used in the derivation of the authorization cryptogram for the card. In EMV transactions, this field is generally sent to BASE24 as part of the transaction data. However, for CAP transactions, it may not be present, which means the value has to be configured on the database.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: CAF.EMVCAF.CAP-DATA.DKI

Note: This field is not currently supported by the BASE24-atm EMV product. For more information about this field, refer to the ***BASE24-pos EMV Support Manual***.

SECONDARY CARD DATA

The following five fields contain information about the secondary card.

ATC NUMBER — The last application transaction counter (ATC) received by BASE24 from the secondary card. This field is used to record the number of transactions performed on the secondary card when the ATC is contained in the EMV segment instead of the Base segment.

Field Length: System Protected
Data Name: CAF.EMVCAF.SCND-CRD-DATA.ATC-2

CAP ATC NUMBER — The ATC received by BASE24 in the last approved Chip Authentication Program (CAP) token validation transaction performed on the secondary card.

Field Length: System Protected
Data Name: CAF.EMVCAF.SCND-CRD-DATA.CAP-DATA-2.ATC

Note: This field is not currently supported by the BASE24-atm EMV product. For more information about this field, refer to the ***BASE24-pos EMV Support Manual***.

BAD CAP TOKEN COUNTER — The number of consecutive failed CAP token validation attempts performed on the secondary card. This value is reset to zero when a CAP token is validated successfully for the card.

Field Length: System Protected
Data Name: CAF.EMVCAF.SCND-CRD-DATA.CAP-DATA-2.BAD-CAP-TKN-CHK-ACCUM

Note: This field is not currently supported by the BASE24-atm EMV product. For more information about this field, refer to the ***BASE24-pos EMV Support Manual***.

CAP APSN — Specifies the CAP application PAN sequence number (APSN) for the secondary card identified by the values in the PAN and MEMBER fields on each CAF screen. The APSN identifies and differentiates cards with the same PAN. In EMV transactions, this field is generally sent to BASE24 as part of the transaction data. However, for CAP transactions, it may not be present, which means the value has to be configured on the database.

Field Length: 1–2 numeric characters
Required Field: No
Default Value: No default value
Data Name: CAF.EMVCAF.SCND-CRD-DATA.CAP-DATA-2.APSN

Note: This field is not currently supported by the BASE24-atm EMV product. For more information about this field, refer to the ***BASE24-pos EMV Support Manual***.

CAP DKI — Specifies the derivation key index (DKI) used for the secondary card identified by the values in the PAN and MEMBER fields on each CAF screen. The DKI identifies which of a set of master keys is to be used in the derivation of the authorization cryptogram for the card. In EMV transactions, this field is generally sent to BASE24 as part of the transaction data. However, for CAP transactions, it may not be present, which means the value has to be configured on the database.

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00
Data Name: CAF.EMVCAF.SCND-CRD-DATA.CAP-DATA-2.DKI

Note: This field is not currently supported by the BASE24-atm EMV product. For more information about this field, refer to the ***BASE24-pos EMV Support Manual***.

ICC KEY File (KEYI)

The ICC KEY File (KEYI) supports the key storage requirements for the BASE24-atm EMV product. The KEYI is not used for customers who have implemented BASE24 Transaction Security Services (TSS). The key to the KEYI is the KEYI GROUP, BEGIN DATE, and END DATE fields.

The following keys are stored in the KEYI:

- A set of up to 24 double length DES keys used for authorization request cryptogram verification and authorization response cryptogram generation. These keys are indexed by the derivation key index obtained from the transaction request.
- A double length DES key used for secure messaging, the generation of the issuer script message authentication code (MAC).

The KEYI supports the key information required for ICC transactions. The KEYI holds multiple authentication keys (up to a maximum of 24) for an individual KEYI group.

A master key is held for use by the secure messaging facility for scripts. Secure messaging keys are different from those used by the card in request and response cryptogram processing.

KEYI Screen 1

KEYI screen 1 is shown below, followed by descriptions of its fields:

```

BASE24-EMV    EMV KEY FILE                LLLL      YY/MM/DD  HH:MM  01 OF 01
      KEYI GROUP:
BEGIN DATE (YYYYMM): *****    END DATE (YYYYMM): *****    FIID:
                        EMV SCHEME: 0 (UKIS)

CRYPTOGRAPHIC KEYS:
1  0000000000000000 0000000000000000 2  0000000000000000 0000000000000000
3  0000000000000000 0000000000000000 4  0000000000000000 0000000000000000
5  0000000000000000 0000000000000000 6  0000000000000000 0000000000000000
7  0000000000000000 0000000000000000 8  0000000000000000 0000000000000000
9  0000000000000000 0000000000000000 10 0000000000000000 0000000000000000
11 0000000000000000 0000000000000000 12 0000000000000000 0000000000000000
13 0000000000000000 0000000000000000 14 0000000000000000 0000000000000000
15 0000000000000000 0000000000000000 16 0000000000000000 0000000000000000
17 0000000000000000 0000000000000000 18 0000000000000000 0000000000000000
19 0000000000000000 0000000000000000 20 0000000000000000 0000000000000000
21 0000000000000000 0000000000000000 22 0000000000000000 0000000000000000
23 0000000000000000 0000000000000000 24 0000000000000000 0000000000000000

SECURE MESSAGING KEYS      FOR INTEGRITY: 0000000000000000 0000000000000000
                        FOR CONFIDENTIALITY: 0000000000000000 0000000000000000

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                        F12-HELP

```

KEYI GROUP — The primary key to the KEYI for this prefix. The value in this field must be the same as the value in the KEYI GROUP field on CPF screen 11.

Field Length: 32 alphanumeric characters
 Required Field: Yes
 Default Value: No default value
 Data Name: KEYI.KEYICC.KEYI-GRP

BEGIN DATE — The first date in the range of expiration dates to which information in this KEYI record applies. If you do not want to use this functionality, place asterisks in the BEGIN DATE and END DATE fields. Asterisks are stored as WWWWWW and cause the same KEYI record to be used regardless of card expiration date. Valid values are as follows:

000000 = Any card expiration date that is less than or equal to the specified value in the END DATE field uses this record.

yyyymm = Any card expiration date must be greater than or equal to the beginning date, but less than or equal to the end date.

***** = Any card whose expiration date has not found a match on a date in a KEYI record uses this as a wild card match.

Field Length: 6 alphanumeric characters
Required Field: Yes
Default Value: *****
Data Name: KEYI.KEYICC.BEG-DAT

END DATE — The last date in the range of card expiration dates to which information in this KEYI record applies. If you do not want to use this functionality, place asterisks in the BEGIN DATE and END DATE fields. Asterisks are stored as WWWWWW and cause the same KEYI record to be used regardless of card expiration date. Valid values are as follows:

yyyymm = Any card expiration date must be greater than or equal to the beginning date, but less than or equal to the end date.

***** = Any card whose expiration date has not found a match on a date in a KEYI record uses this as a wild card match.

Field Length: 6 alphanumeric characters
Required Field: Yes
Default Value: *****
Data Name: KEYI.KEYICC.END-DAT

FIID — The financial institution identifier of the institution to which this record applies.

Field Length: 4 alphanumeric characters
Required Field: No
Default Value: No default value
Data Name: KEYI.KEYICC.FIID

EMV SCHEME — The specific method used to perform the EMV security processing. EMV specifications define the high level EMV security requirements, but the individual payment schemes support different implementations. This field

identifies the specific implementation associated with this KEYI record and is used by the security utilities to determine the specific security module calls and security functions to be performed. Valid values are as follows:

- 0 = UKIS security scheme
- 1 = Visa security scheme as defined in Visa ICC Specifications
- 2 = MCPA (M/Chip) security scheme as defined in MasterCard specifications

Field Length: 1 numeric character
Required Field: No
Default Value: 0
Data Name: KEYI.KEYICC.EMV-SEC-SCHEME

CRYPTOGRAPHIC KEYS — A set of up to 24 issuer master keys used to generate a unique EMV master key for generation of application cryptograms. The specific master key used for an individual chip card is indicated by the derivation key index that is personalized in the card and passed in the request message. Some issuer security schemes utilize a single master key. In this case a derivation key index of zero is supplied and the first entry in the array is used.

Field Length: 16 alphanumeric characters
Occurs Up to 24 times
Required Field: No
Default Value: 0000000000000000
Data Name: KEYI.KEYICC.CRYPTO-KEYS

SECURE MESSAGING KEYS

The following two fields identify the secure messaging keys for integrity and confidentiality:

FOR INTEGRITY — The double length secure messaging key for message integrity. This is an issuer master key used to generate a unique EMV master key that is used in derivation of a session key used to authenticate script commands (using a MAC) for message integrity.

It is assumed that the secure messaging key retrieved from the KEYI and passed to the CAM utilities from the Authorization process is the appropriate key for the call to either the Thales or Atalla security module.

Field Length:	16 alphanumeric characters
Required Field:	No
Default Value:	0000000000000000
Data Name:	KEYI.KEYICC.SM-KEY

FOR CONFIDENTIALITY — The double length secure messaging key for message confidentiality.

Note: The EMV product supports the confidentiality aspect of secure messaging, (e.g., encrypting PIN data in issuer scripts) by communicating with the Transaction Security Services (TSS). Therefore, the Secure Messaging Key for Confidentiality is maintained on the TSS database, and this KEYI field is not used.

LCONF Assigns and Params

The BASE24 EMV product modules use several standard Logical Network Configuration File (LCONF) assigns and params, together with the EMV-specific assigns and params listed below.

One assign is specific to the EMV product: The KEYI assign identifies the location of the ICC Key file (KEYI).

The following params are specific to the EMV product:

- The EMV-CARD-BLOCK-SCRIPT param specifies the hexadecimal representation of binary data that is used to block an EMV card.
- The EMV-PUT-DATA-SCRIPT param specifies the hexadecimal representation of binary data that is used to reset the lower consecutive offline limit on an EMV card.
- The EMV-PIN-UNBLOCK-SCRIPT param specifies the script template for the PIN UNBLOCK script command, excluding the MAC.

Refer to “Issuer Script Commands” in section 4 for a description and examples of how these params are used in processing.

Valid Values

The LCONF params EMV-CARD-BLOCK-SCRIPT, EMV-PUT-DATA-SCRIPT, and EMV-PIN-UNBLOCK-SCRIPT must contain the values specified in the table below.

The last two bytes of each of these params specifies the length of the MAC to be appended to the end of the script command. For example, a value of 04 in the last two bytes of the EMV-CARD-BLOCK-SCRIPT param indicates a 4-byte MAC, while a value of 07 in the last two bytes of the EMV-PUT-DATA-SCRIPT param indicates a 6-byte MAC (actually a 1-byte parameter value followed by the 6-byte MAC).

Format 1 secure messaging, which is used by the CCD EMV scheme, uses only a 4-byte MAC. Format 2 secure messaging, which is used by the other EMV schemes, uses 4-byte, 6-byte, and 8-byte MACs.

All lengths in the following table are specified in hexadecimal.

Byte Description	CARD BLOCK	PUT DATA	PIN UNBLOCK
Template Tag	71 or 72	71 or 72	71 or 72
Template Length	0E or 12 10 or 14 12 or 16	0F or 13 11 or 15 13 or 17	0E or 12 10 or 14 12 or 16
Script ID Tag (2 bytes)	9F18	9F18	9F18
Script ID Length	00 or 04	00 or 04	00 or 04
Script ID Value (4 bytes) ¹	XXXXXXXX ²	XXXXXXXX ²	XXXXXXXX ²
Script Tag	86	86	86
Script Length	09, 0B, or 0D	0A, 0C, or 0E	09, 0B, or 0D
CLA	84	04	84
INS	16 or 1E	DA	24
P1	00	Data Element Tag (first byte)	00
P2	00	Data Element Tag (second byte)	00
Data Length	04, 06, or 08	05, 07, or 09	04, 06, or 08
Command Data	Not present	FF	Not present
MAC Length	04, 06, or 08 bytes	04, 06, or 08 bytes	04, 06, or 08 bytes

¹ The script ID value is optional. The script ID length identifies whether the script ID is present.

² The value X represents any hexadecimal digit in the script ID value.

Examples

The following are some examples for each of the params.

All lengths in the following table are specified in hexadecimal.

Script	Template		Script ID			Script Command ^{1, 2}				
	T	L	T	L	V	T	L	V		
Card Block	71	0E	9F18	00		86	09	84 16 00 00	04	M4
	71	12	9F18	04	12 34 56 78	86	09	84 16 00 00	04	M4
	71	10	9F18	00		86	0B	84 16 00 00	06	M6
	71	14	9F18	04	12 34 56 78	86	0B	84 16 00 00	06	M6
	71	12	9F18	00		86	0D	84 16 00 00	08	M8
	71	16	9F18	04	12 34 56 78	86	0D	84 16 00 00	08	M8
Put Data	72	0F	9F18	00		86	0A	04 DA 9F 58	05	M4
	72	13	9F18	04	12 34 56 78	86	0A	04 DA 9F 58	05	M4
	72	11	9F18	00		86	0C	04 DA 9F 58	07	M6
	72	15	9F18	04	12 34 56 78	86	0C	04 DA 9F 58	07	M6
	72	13	9F18	00		86	0E	04 DA 9F 58	09	M8
	72	17	9F18	04	12 34 56 78	86	0E	04 DA 9F 58	09	M8
PIN Unblock	71	0E	9F18	00		86	09	84 24 00 00	04	M4
	71	12	9F18	04	12 34 56 78	86	09	84 24 00 00	04	M4
	71	10	9F18	00		86	0B	84 24 00 00	06	M6
	71	14	9F18	04	12 34 56 78	86	0B	84 24 00 00	06	M6
	71	12	9F18	00		86	0D	84 24 00 00	08	M8
	71	16	9F18	04	12 34 56 78	86	0D	84 24 00 00	08	M8

See notes on the following page.

- ¹ To support an Application Block script instead of a Card Block script, the second binary byte of the V portion of the Script Command can be set to a value of 1E instead of 16.
- ² To support an update of a 1-byte EMV data element other than the LCOL in the Put Data script, the last two binary bytes of the V portion of the Script Command can be set to the EMV tag of the data element to be updated instead of 9F58 (LCOL).

ATM Terminal Data (ATD) File

BASE24-atm Terminal Data (ATD) files define the characteristics of the ATM device, including the transaction security parameters, sharing parameters, transaction totals, terminal owner, and device-dependent data. They are also used to preserve transaction context and are updated every time a transaction is processed for the terminal.

Two ATD screens contain fields that are specific to BASE24-atm EMV processing. ATD screen 21 is documented only in this manual. See the ***BASE24-atm Files Maintenance Manual*** for additional information about other ATD screens.

ATD Screen 1

ATD screen 1 contains information about the Device Handler and Authorization processes with which the terminal is associated. ATD screen 1 is shown below, followed by a description of the field used by BASE24-atm EMV modules to identify the format of the native message used by an ATM.

```

BASE24-ATM   ATM TERMINAL DATA   LLLL   YY/MM/DD HH:MM   01 OF 22
TERM ID:                FIID:        REGION:        BRANCH:        LN: LLLL

DEVICE TYPE:      (***** )
PHONE:
HARDWARE: 0 (***** )
OWNER NAME:
CITY:
COUNTRY:
DH PROCESS:
INST ID NUM: 000000000000  ALT ROUTING GROUP: 000000000000
                                NCD ROUTING GROUP: 000000000000

SHARING INFORMATION
COUNTY CODE: 0          STATE CODE: 0          COUNTRY CODE: 840
GROUPS:
SURCHARGE PROFILE:

POSTING DATE:          TIME OFFSET: 0  MULTIPLE ACCT SELECT: N (Y/N)
USER:

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                                F12-HELP

```

NATIVE MESSAGE FORMAT — A code indicating the EMV message format for requests. Valid values are as follows:

Value	Message Format	Device Handler Message Formats	
0	Standard	Diebold 10XX/478X	NCR 5XXX
1	Non-standard	Wincor D91X (DDC)	Wincor NDC+ (NDC)

Field Length: 1 numeric character
 Required Field: No
 Default Value: 0
 Data Name: ATDS1-CORE.NATV-MSG-FRMT

ATD Screen 21

ATD screen 21 is specific to the EMV product. This screen allows you to indicate the EMV capabilities of each terminal. ATD screen 21 is shown below, followed by descriptions of its fields.

```

BASE24-ATM  ATM TERMINAL DATA  LLLL  YY/MM/DD  HH:MM  01 OF 22
TERM ID:      FIID:      REGION:      BRANCH:      LN: LLLL

      EMV TERMINAL CAPABILITIES

      KEY ENTRY:  N      MAGNETIC STRIPE:  N
      IC/CONTACTS:  N      RESERVED FOR FUTURE USE:  N
      RESERVED FOR FUTURE USE:  N      RESERVED FOR FUTURE USE:  N

PLAINTEXT PIN/ICC VERIFICATION:  N  ENCIPHERED PIN/ONLINE VERIFICATION:  N
      SIGNATURE:  N  ENCIPHERED PIN/OFFLINE VERIFICATION:  N
      NO CVM REQUIRED:  N      RESERVED FOR FUTURE USE:  N

      STATIC DATA AUTHENTICATION:  N      DYNAMIC DATA AUTHENTICATION:  N
      CARD CAPTURE:  N      RESERVED FOR FUTURE USE:  N
      COMBINED DATA AUTHENTICATION:  N      RESERVED FOR FUTURE USE:  N

      ENTER 'Y' FOR EACH FACILITY WHICH IS SUPPORTED BY THIS TERMINAL

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
              F12-HELP
  
```

EMV TERMINAL CAPABILITIES

The following three fields indicate the methods supported by the terminal for entering the information from the card into the terminal:

KEY ENTRY
MAGNETIC STRIPE
IC/CONTACTS

Valid values for each of the three fields are as follows:

Y = Yes, the terminal has this capability.
N = No, the terminal does not have this capability.

The following five fields indicate the methods supported by the terminal for verifying the identity of the cardholder at the terminal:

PLAINTEXT PIN/ICC VERIFICATION
ENCIPHERED PIN/ONLINE VERIFICATION
SIGNATURE
ENCIPHERED PIN/OFFLINE VERIFICATION
NO CVM REQUIRED

Valid values for each of the five fields are as follows:

Y = Yes, the terminal has this capability.
N = No, the terminal does not have this capability.

The following four fields indicate the methods supported by the terminal for authenticating the card at the terminal and whether the terminal has the ability to capture a card.

STATIC DATA AUTHENTICATION
DYNAMIC DATA AUTHENTICATION
CARD CAPTURE
COMBINED DATA AUTHENTICATION

Valid values for each of the four fields are as follows:

Y = Yes, the terminal has this capability.
N = No, the terminal does not have this capability.

3: Transaction Processing

When an integrated circuit card is presented at the terminal, the terminal determines which applications are supported by both the card and the terminal. The terminal displays the supported applications to the cardholder and the cardholder selects the appropriate application.

This section describes the transaction paths of the BASE24-atm EMV add-on product. Only the additional steps required by an EMV transaction are described. These diagrams do not include the standard BASE24-atm processing.

Token Usage

BASE24 uses seven tokens to carry EMV-specific data during transaction processing.

The EMV Request Data token (token ID B2) contains the thirteen minimum request data elements required for inclusion in request messages, as defined by EMV. The Device Handler process or the Interchange Interface process creates this token and adds it to the transaction message before sending it to the Authorization process.

The EMV Discretionary Request Data token (token ID B3) consists of EMV-related data that is not required for authorization. However, each data element is supported by more than one EMV-compliant interface and, therefore, can be mapped between interfaces by BASE24.

The EMV Status token (token ID B4) holds data identifying the status of a transaction. Device Handler and Interchange Interface processes create this token and add it to the STM before sending it to the Authorization process. The acquiring endpoint adds the token when the transaction originates from an EMV-capable terminal, regardless of whether or not the data relates to an EMV transaction.

The EMV Response Data token (token ID B5) contains the response cryptogram, data required to generate the response cryptogram, and flags used to identify the scripts to be returned to the acquirer. If authorization is performed on BASE24, the BASE24 Authorization process creates this token. If the transaction is routed to an interchange or a host for authorization, the BASE24 Interchange Interface process or the BASE24 ISO Host Interface process creates the token.

The EMV Script Data token (token ID B6) holds EMV script data. The issuer process creates this token. In the context of EMV transactions, the issuer process can be an Interchange Interface process if the issuer is external to BASE24, or the Authorization process if BASE24 is configured for offline or online/offline authorization. The token is added to the STM before returning the message to the acquiring process. This token is present only if the transaction response contains script data.

The EMV Issuer Script Results token (token ID BJ) holds information about the processing of EMV Script data. This token is created by the acquirer interface process (e.g., Device Handler or Interchange Interface) or sent by the acquirer. It contains information about the results of EMV script processing.

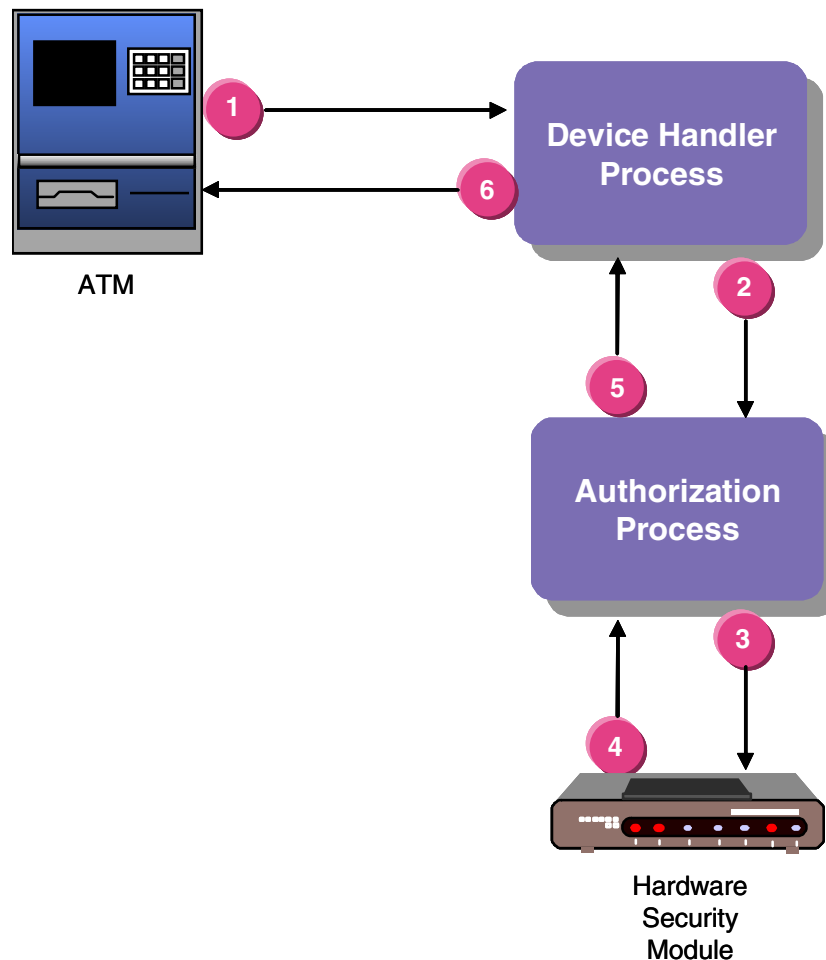
The EMV Supplementary Data token (token ID S4) carries supplementary data required by Visa in data element 55 in a contactless magnetic stripe transaction. This token is created by the acquirer interface process (e.g., Interchange Interface) or sent by the acquirer. It contains information about the Form Factor Indicator (tag 9F6E), Customer Exclusive Data (tag 9F7C), and Terminal Transaction Qualifiers (tag 9F66) data elements.

Refer to the ***BASE24 Tokens Manual*** for token formats, including the corresponding EMV tags, for each field in these tokens. All EMV tokens are considered Base message tokens, and are presented in the BASE24 Base Tokens section of the manual.

For more information about the EMV data elements refer to the MasterCard M/Chip or Visa Smart Debit Credit (VSDC) documentation sets, or the EMVCo specification.

BASE24 is the Acquirer and the Issuer

The following steps identify the flow of an EMV transaction in which BASE24 is both the acquirer and the issuer.



Steps

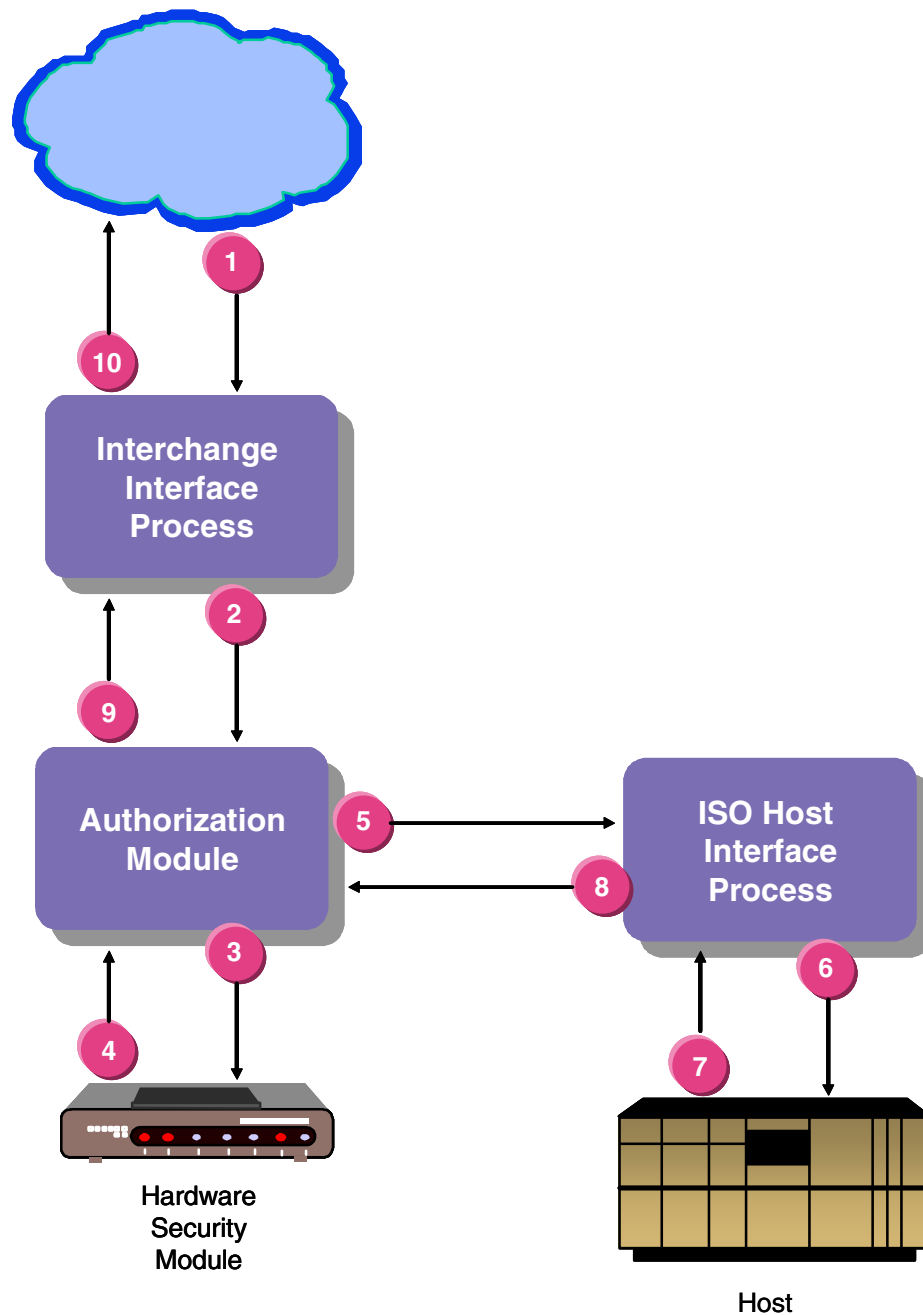
Processing

- 1 The card generates the authorization request cryptogram. The terminal transmits the request cryptogram and the data used to generate it in the request message and sends the message to the Device Handler process.

Steps	Processing
2	The Device Handler process checks the terminal capabilities indicators in the device-dependent data area of the TDF to determine which EMV tokens to send in the authorization message. For an EMV transaction, the Device Handler process formats the EMV Request Data token, the EMV Status token, and the EMV Discretionary Data token and adds them to the BASE24-atm Standard Internal Message (STM). The Device Handler process then sends the STM with the appended EMV tokens to the Authorization process.
3, 4	The presence of the EMV Request Data token and the appropriate configuration data in the IDF, CPF, CAF, and KEYI files initiates EMV authorization processing. This processing includes card and issuer authentication, using the request cryptogram and the KEYI check, generation of the authorization response cryptogram, ATC limit checking, EMV Status field checking (CVR and TVR), and issuer script handling. The Authorization process performs CAM authentication using the hardware security module.
5	The Authorization process formats an EMV Response Data token and, if applicable, an EMV Script Data token for the response message to the Device Handler process. If the transaction is not approved, a second call is made to the hardware security module using the response code to generate a new response cryptogram to be sent to the terminal.
6	The Device Handler process reformats the response message to the native format of the device and sends the message to the terminal. The card uses the information in the EMV Response Data token to authenticate the transaction authorizer by verifying the response cryptogram.

BASE24 Performs Prescreen Checks

The following steps identify the flow of an EMV transaction in which an interchange acquires the transaction, BASE24 performs prescreen checks, and sends it to the host for authorization.



Steps	Processing
1	The interchange sends a request message with EMV-specific data in the ISO data elements to the BASE24 Interchange Interface process.
2	The Interchange Interface process formats the EMV Request Data token, the EMV Status token, and the EMV Discretionary Data token (according to the interchange EMV specifications) and adds them to the BASE24-atm Standard Internal Message (STM). The Interchange Interface process sends the STM with the appended EMV tokens to the Authorization process.
3, 4	The presence of the EMV Request Data token and the appropriate configuration data in the IDF, CPF, CAF, and KEYI files initiates EMV authorization processing. Prescreen processing can be configured to include card and issuer authentication, using the request cryptogram and the KEYI check, generation of the authorization response cryptogram, and EMV Status field checking (CVR and TVR). Issuer script handling can be performed after the transaction approval is received. The Authorization process performs CAM authentication using the hardware security module.
5	The Authorization process sends the transaction request to the ISO Host Interface process.
6	The ISO Host Interface process reformats the message into the external message (SEM) format and sends it to the host.
7	The host authorizes the transaction and returns a response to the ISO Host Interface process. The host also can return a script in the Script Data token or indicate that BASE24 is to generate a script by setting one of the script flags in the EMV Response Data token.
8	The ISO Host Interface process reformats the message into STM format and returns it to the Authorization process.

Steps	Processing
9	The Authorization process formats the EMV Response Data token and, if applicable, the EMV Script Data token. If the authorization level is level 3 (online/offline) and the authorization method is Positive or Positive Balance, the Authorization process updates the ATC value in the CAF. If the transaction is not approved, a second call is made to the hardware security module using the response code to generate a new response cryptogram to be sent to the interchange. The Authorization process sends the response message to the Interchange Interface process.
10	The Interchange Interface process reformats the response message into the external message format and sends the message to the interchange.

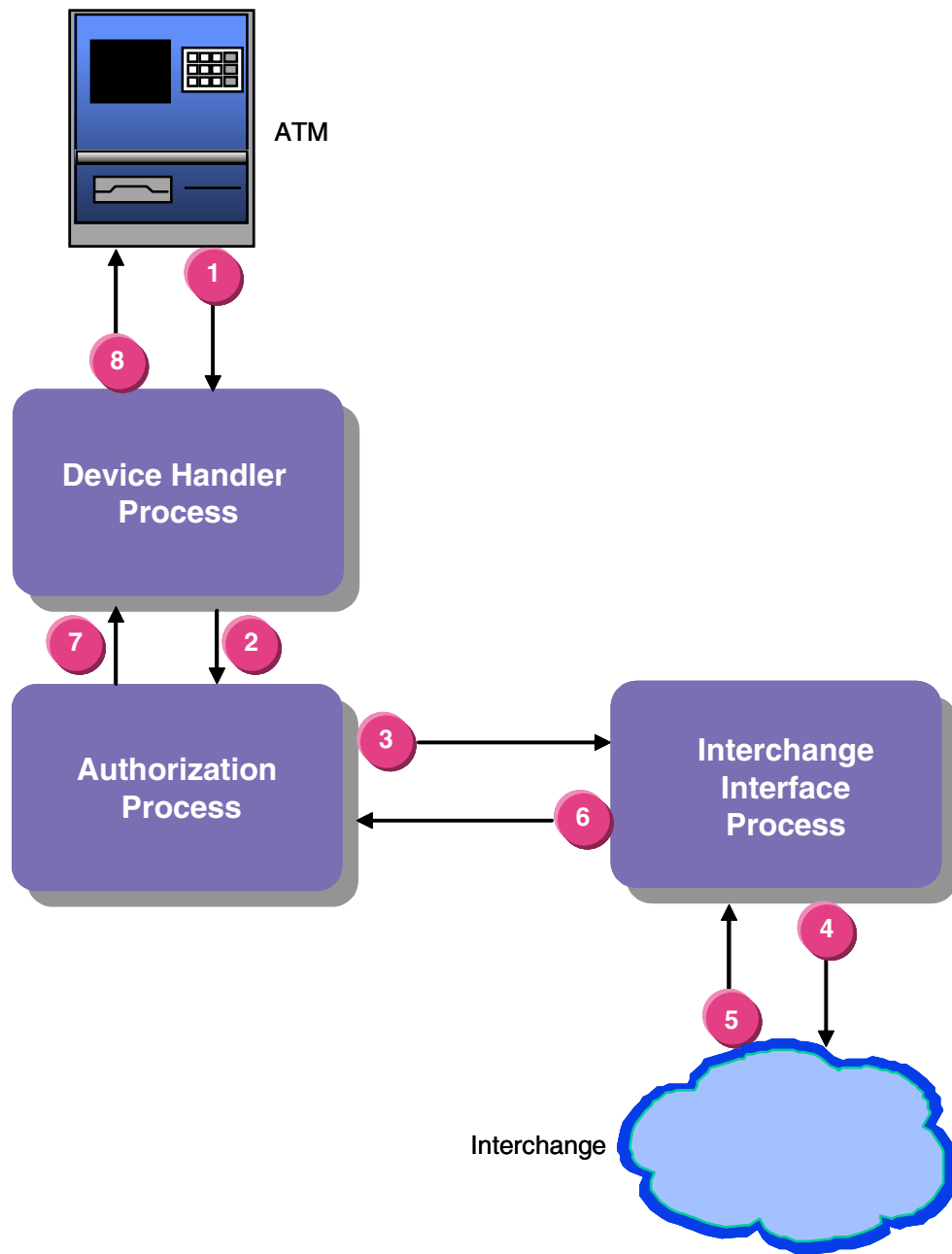
BASE24 is the Acquirer Only

In the following diagram, BASE24 is the acquirer only, sending the transaction to an interchange for authorization.

Both the terminal and the card may perform offline risk management (e.g., transaction velocity checking, floor limit checking) to determine whether the transaction should be approved, declined, or transmitted online for authorization.

Traditionally, transaction security has been performed in security zones along the transaction path for magnetic-stripe card-based transactions. For example, a different MAC would be used for each message passed between systems, where each system is considered a separate security zone, and the PIN block would be translated every time a request message leaves a security zone. With EMV cards, the security is end-to-end (i.e., between the card and the issuer), rather than between various zones of the transaction path. Therefore, the function of an acquirer in an EMV transaction is basically to ensure that the data that comes in at one end is the same as the data that goes out at the other end.

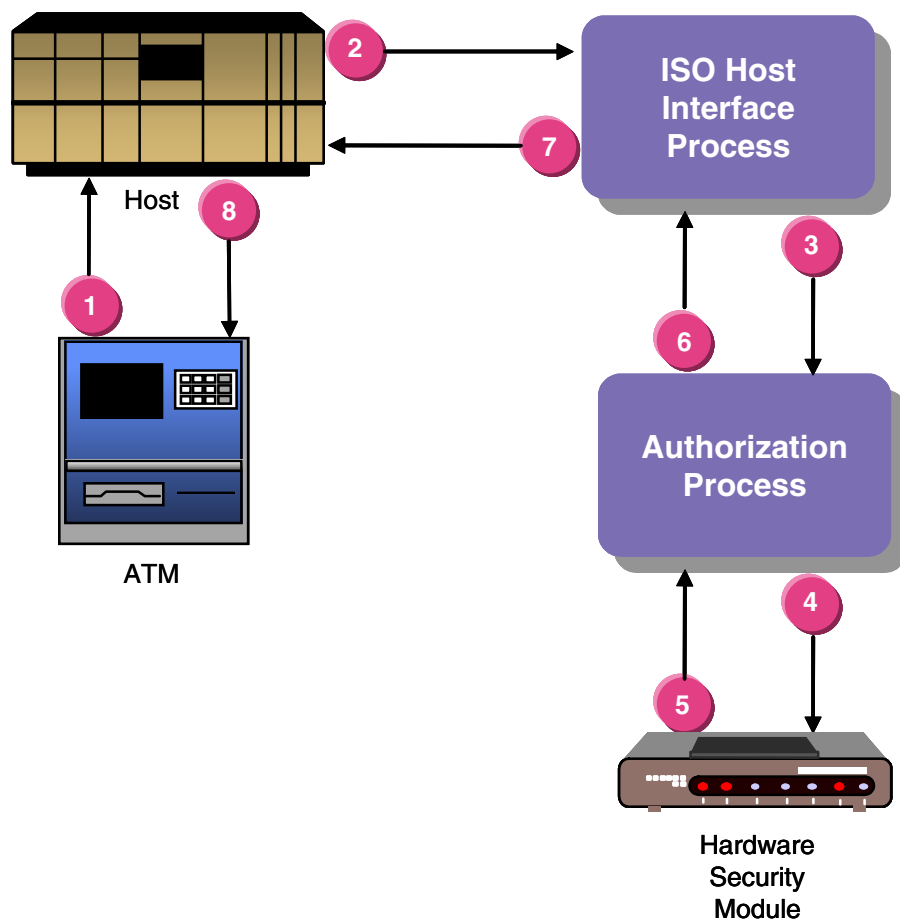
If both the card and the terminal indicate that the transaction satisfied the criteria for offline authorization, the transaction approved offline, and the card generates a cryptogram, called the transaction certificate (TC), based on data related to the card, to the terminal, and to the transaction. This certificate and the data used to generate it are transmitted in the clearing message for future cardholder disputes and chargeback purposes. BASE24 does not verify the transaction certificate.



Steps	Processing
1	If the criteria for offline authorization are not satisfied, the terminal generates an online request message indicating why the transaction was transmitted online, and the card generates an authorization request cryptogram. The terminal transmits the cryptogram and the data used to generate it in the request message.
2	If the terminal capabilities are not sent in the request message, the Device Handler process checks the terminal capabilities in the TDF to determine which EMV tokens to send. The Device Handler process formats the EMV Request Data token, the EMV Status token, and the EMV Discretionary Data token and generates the BASE24-atm Standard Internal Message (STM). The Device Handler process sends the STM with the appended EMV tokens to the Authorization process.
3	The Authorization process sends the request message to the Interchange Interface process.
4	The Interchange Interface process reformats the message into the external message (SEM) format and sends it to the interchange for authorization. Message reformatting includes moving data from the EMV tokens to the external message.
5	The interchange authorizes the transaction and returns a response to the Interchange Interface process.
6	The Interchange Interface process adds the EMV Response Data token and, if applicable, an EMV Script Data token to the internal message, reformats the response and script from the Interchange into the BASE24-atm Standard Internal Message format, and sends it to the Authorization process.
7	The Authorization process sends the response message with the appended tokens to the Device Handler process.
8	The Device Handler process reformats the response into the native mode format of the device and sends the message to the terminal. The terminal uses the information in the EMV Response Data token to authenticate the issuer by verifying the authorization response cryptogram, and uses the information in the EMV Script Data token to perform issuer scripts.

The Host is the Acquirer; BASE24 is the Issuer

The following steps identify the flow of an EMV transaction in which the host is the acquirer and BASE24 is the issuer.



Steps

Processing

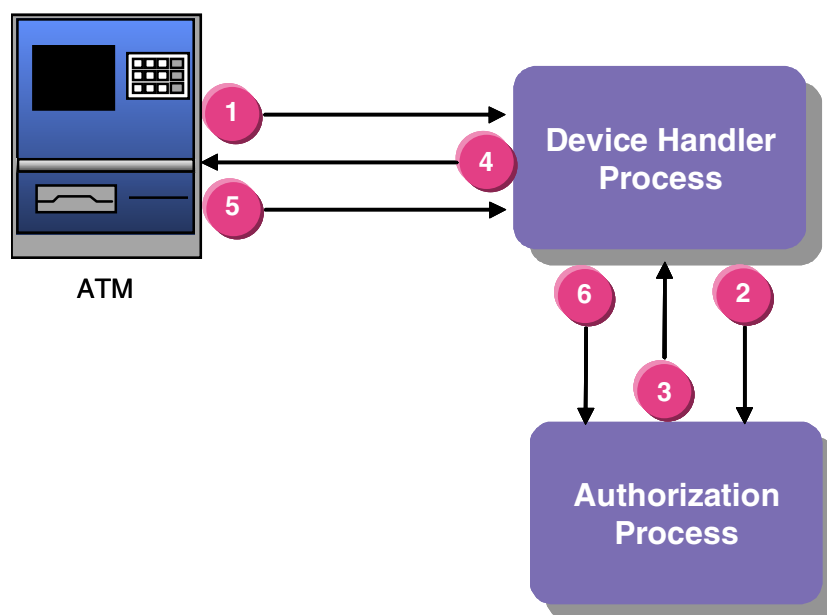
- 1 The card generates the authorization request cryptogram. The terminal transmits the request cryptogram and the data used to generate it in the request message and sends the message to the host.
- 2 The host reformats the message to the external message (SEM) format, adds EMV-specific data in the ISO data elements, and passes the message to the ISO Host Interface process.

Steps	Processing
3	The ISO Host Interface process formats the EMV Request Data token, the EMV Status token, and the EMV Discretionary Data token and adds them to the BASE24-atm Standard Internal Message (STM). The ISO Host Interface process sends the STM with the appended EMV tokens to the Authorization process.
4, 5	The presence of the EMV Request Data token and the appropriate configuration data in the IDF, CPF, CAF, and KEYI files initiates EMV authorization processing. This processing includes card and issuer authentication, using the request cryptogram and the KEYI check, generation of the authorization response cryptogram, ATC limit checking, EMV Status field checking (CVR and TVR), and issuer script handling. The Authorization process performs CAM authentication using the hardware security module.
6	The Authorization process formats an EMV Response Data token and, if applicable, an EMV Script Data token for the response message to the ISO Host Interface process. If the transaction is not approved, a second call is made to the hardware security module using the response code to generate a new response cryptogram to be sent to the terminal.
7	The ISO Host Interface process reformats the response message into the external message (SEM) format and sends the message to the host.
8	The host reformats the response message to the native format of the device and sends the message to the terminal. The card uses the information in the message to authenticate the transaction authorizer by verifying the response cryptogram.

PIN Management; BASE24 is the Authorizer

The following steps identify the flow of a PIN Management transaction in which BASE24-atm is the authorizer.

These steps illustrate processing for an ATM and a Device Handler process. The native message from the ATM identifies a PIN change or PIN unblock transaction. A transaction request can also be received by the BASE24 system from an acquirer host system, using a HISO or BIC ISO Interface process, with the first two digits of the ISO processing code set to the value 96 (PIN change) or the value 97 (PIN unblock). A transaction request can also be received by the BASE24 system from an interchange (e.g., VisaNet).



Steps

Processing

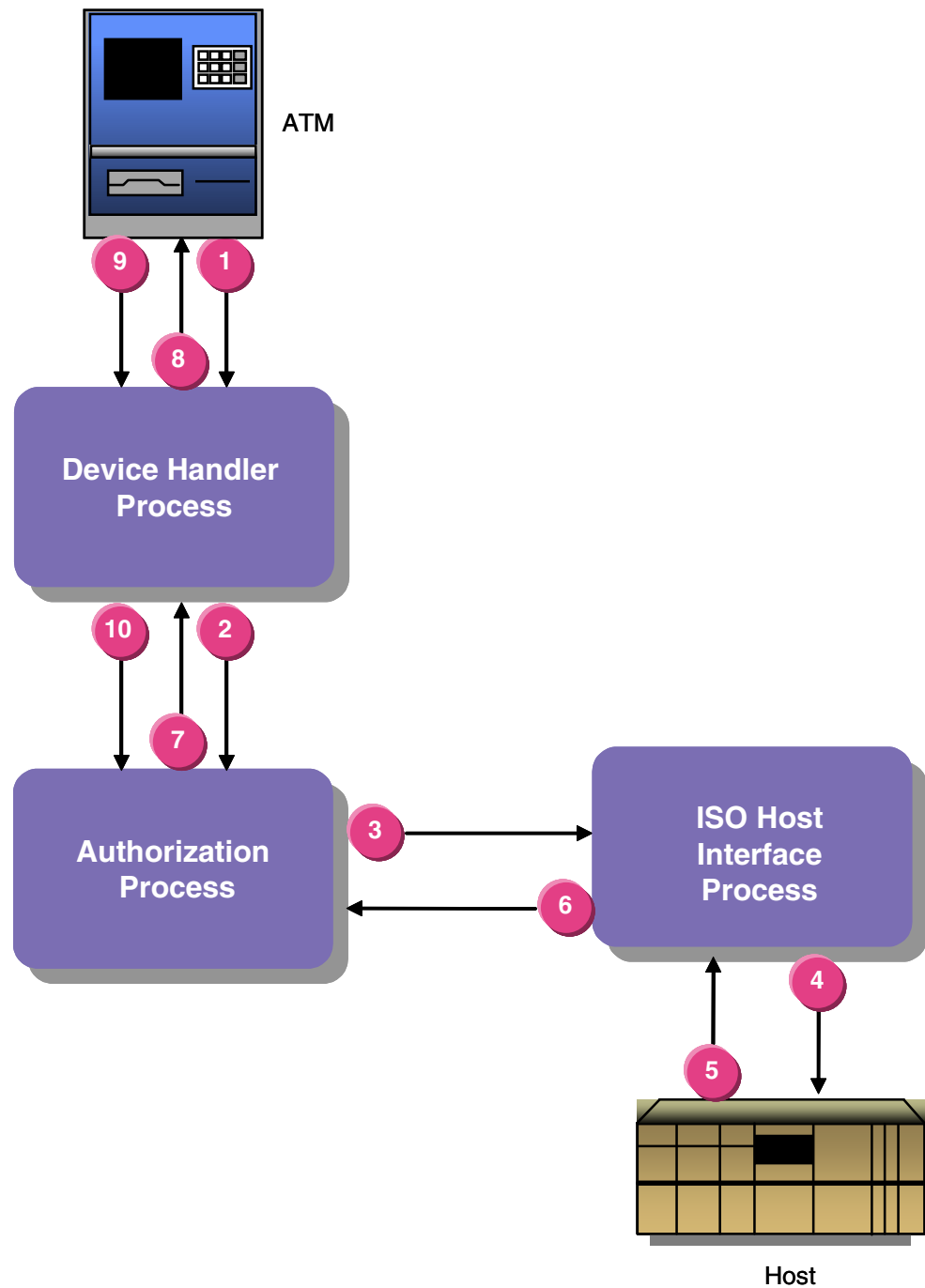
- 1 The ATM sends a PIN change or PIN unblock transaction to BASE24.
- 2 The Device Handler process sets the transaction code to a value of 81 (PIN change) or 82 (PIN unblock) and routes the transaction to the BASE24-atm Authorization process, which is configured to authorize the transaction.
- 3 The Authorization process approves the transaction and returns the response with a PIN unblock script command or PIN change script command.

Steps	Processing
4	The Device Handler process returns the response message (including the PIN unblock or PIN change script) to the ATM.
5	The ATM can subsequently indicate that the issuer script processing either was not performed or failed.
6	The Device Handler process sends a reversal message to the BASE24-atm Authorization process to negate the effect of the PIN change or unblock transaction.

PIN Management; a Host is the Authorizer

The following steps identify the flow of a PIN management transaction in which a host system is the authorizer.

These steps illustrate processing for an ATM and a Device Handler process. The native message from the ATM identifies a PIN change or PIN unblock transaction. A transaction request can also be received by the BASE24 system from an acquirer host system, using a HISO or BIC ISO Interface process, with the first two digits of the ISO processing code set to the value 96 (PIN change) or the value 97 (PIN unblock). A transaction request can also be received by the BASE24 system from an interchange (e.g., VisaNet).



Steps

Processing

- 1 The ATM sends a PIN change or PIN unblock transaction to BASE24.

Steps	Processing
2	The Device Handler process sets the transaction code to a value of 81 (PIN change) or 82 (PIN unblock) and routes the transaction to the BASE24-atm Authorization process, which can be configured to perform some prescreen checks on the transaction.
3	The Authorization process routes the transaction request to an issuer Host Interface process.
4	The issuer Host Interface process sets the first two digits of the ISO processing code set to the value 96 (PIN change) or the value 97 (PIN unblock) and passes the transaction request to the issuer host system for authorization.
5	The issuer host system authorizes the transaction and generates a PIN unblock script command or PIN change script command, places the script command in the response message with a flag in the EMV Response token to indicate the type of script command.
6	The issuer Host Interface process returns the response and script command to the Authorization process.
7	The Authorization process passes the response message and script command to the Device Handler process.
8	The Device Handler process returns the response message (including the PIN unblock or PIN change script) to the ATM.
9	The ATM can subsequently indicate that the issuer script processing either was not performed or failed.
10	The Device Handler process sends a reversal message to the BASE24-atm Authorization process to negate the effect of the PIN change or unblock transaction.

4: Authorization Processing

This section describes additional authorization processing that is performed on EMV transactions. This discussion does not include BASE24-atm standard processing. The following topics are described:

- Offline authorization
- Prescreen checking
- Issuer application data
- Application transaction counter checking
- Status field verification
- Fallback checking
- Issuer script commands
- Prefix routing
- Processing EMV card transactions as standard Track 2 card transactions
- Authorization request and response cryptograms
- Cryptographic support
- PIN management processing

Authorization Summary

The main areas of ATM authorization impacted by EMV processing are summarized in the table below. The processing functions are described in this section.

Authorization Level	Description	Processing Functions					
		CAM	ATC	CVR/TVR	FB	IS	OP
1	Prescreen processing	✓		✓	✓		
1	Response processing					✓	
2	Level 2 processing (Negative Authorization method)	✓		✓	✓		
2	Level 2 processing (Positive Authorization method)	✓	✓	✓	✓	✓	✓
3	Prescreen processing (Negative Authorization method)	✓		✓	✓		
3	Prescreen processing (Positive Authorization method)	✓	✓	✓	✓		✓
3	Response processing (Negative Authorization method)					✓	
3	Response processing (Positive Authorization method)		✓ ¹			✓	✓ ¹
3	Stand-in processing (Negative Authorization method)	✓		✓	✓		
3	Stand-in processing (Positive Authorization method)	✓	✓	✓	✓	✓	✓

Key

CAM	Card and issuer authentication (CAM check)
ATC	ATC limit checking
CVR/TVR	EMV status field checking (CVR/TVR)

Key

FB	EMV fallback checking
IS	Issuer script handling
OP	Offline PIN management

- ¹ PIN unblock support only. BASE24-atm cannot generate a PIN change script on a response message from a host.

Response Codes

If a transaction is denied, a response message is generated using the following EMV-specific response codes.

BASE24-atm Response Code	Description
81	Security module parameter error
82	Security module failure
83	KEYI record not found
84	ATC check failure
85	CVR decline
86	TVR decline
87	Request cryptogram failure
88	Fallback decline

Offline Authorization

Offline authorization is defined as the EMV-specific processing that is performed by the BASE24-atm Authorization process in an offline situation.

For both Level 2 authorization (offline only) and Level 3 stand-in authorization, EMV-specific authorization processing provides the following:

- Authorization request cryptogram verification
- Application transaction counter checking, for Positive Authorization methods (CAF and CAF/PBF)
- Fallback transaction checking
- Status field (CVR and TVR) checking (if the EMV Request Data token is present)
- Script formatting, for Positive Authorization methods (CAF and CAF/PBF)
- Offline PIN management, for Positive Authorization methods (CAF and CAF/PBF)

Each of these is described in more detail later in this section.

Prescreen Checking

Prescreen checks are checks that are performed in situations where BASE24 shares authorization responsibilities with a host. In such situations, the Authorization process can make preliminary standard checks on the transactions and decline any that do not meet preliminary requirements.

EMV prescreen checking is currently invoked from level 1 online authorization and level 3 online/offline authorization. The following prescreen checks are available.

- Authorization request cryptogram verification, based on the value of the CAM field on Card Prefix File (CPF) screen 11.
- Fallback transaction checking, based on the value of the FALLBACK field on CPF screen 11.
- Status field checking, based on the value of the TVR/CVR field on CPF screen 11.

Note: The value in the HOST PIN CHANGE OPTION field on Institution Definition File (IDF) screen 13 identifies whether or not PIN change transactions are sent to the host. If a PIN change is supposed to be sent to the host, the value in this field also identifies the action that BASE24 takes if the host is down.

Issuer Application Data

Issuer application data contains proprietary issuer data for transmission to the issuer in an online transaction. Typically this data contains information that is used in the derivation of the authorization cryptogram. The following data elements that are included in both the Visa and MasterCard definitions are used in EMV authorization processing and generation of the authorization cryptograms:

- Derivation key index
- Cryptogram version number
- Card verification results

The following fields are extracted from the issuer application data in the EMV Request Data token:

CRD-VERIF-RSLTS	Used as input when calculating the authorization cryptogram. This field is extracted for CVR status checks.
DERIV-KEY-INDEX	Used to identify which of a set of master keys is to be used in the derivation of the authorization cryptogram for the card.
CRYPTO-VER-NUM	Indicates the version of the method or algorithm used to generate the authorization cryptogram. The version defines the set of data fields to be used in generation of the cryptogram.

The BASE24-atm EMV product supports different layouts for issuer application data, depending on the card type. MasterCard and Europay specify different definitions of issuer application data than Visa does. The layout of the issuer application data for each of the currently supported formats is specified by redefinitions of the issuer application data field in the EMV Request Data token.

Valid values for the EMV IAD FORMAT and ALTERNATE IAD FORMAT fields on CPF screen 11 are as follows:

- 0 = Unspecified. The format is unknown. The contents are sent as passthrough data, but no cryptographic checks are performed on BASE24.
- 1 = The format as defined by Visa in the Visa Integrated Circuit Card (ICC) specification. This format is also used for UKIS cards.
The format is defined by the VISA-APPL-DATA data elements of the EMV Request Data token.
- 2 = The format recommended for MasterCard MCPA (M/Chip 2.1) cards.
The format is defined by the MCPA-APPL-DATA data elements of the EMV Request Data token.
- 3 = The format recommended for MasterCard MCPA (M/Chip 4) cards.
The format is defined by the MCHIP4-APPL-DATA data elements of the EMV Request Data token.
- 4 = The format mandated for CCD (IAD Format A) cards.
The format is defined by the CCD-A-APPL-DATA data elements of the EMV Request Data token.

Configuring Alternate Formats

The BASE24-atm product supports an alternate layout so cards with the same prefix can be migrated from one layout to another as they are reissued. The primary format of the data on a card for a given prefix is configurable using the EMV IAD FORMAT field on CPF screen 11. The alternate format of the data on a card for a given prefix is configurable using the ALTERNATE IAD FORMAT field on CPF screen 11.

Prior to the ALTERNATE IAD FORMAT field being available, the EMV IAD FORMAT field on CAF screen 13 enabled a card issuer to use the M/Chip 4 format for specific accounts within a card prefix that was still using the M/Chip 2.1 format.

Valid values for the EMV IAD FORMAT field on CAF screen 13 are as follows:

- 0 = Use the format specified in the EMV IAD FORMAT field on CPF screen 11. This is the default value.
 - 1 = The format as defined by Visa in the Visa Integrated Circuit Card (ICC) specification. This format is also used for UKIS cards.
The format is defined by the VISA-APPL-DATA data elements of the EMV Request Data token.
 - 2 = The format recommended for MasterCard MCPA (M/Chip 2.1) cards.
The format is defined by the MCPA-APPL-DATA data elements of the EMV Request Data token.
 - 3 = The format recommended for MasterCard MCPA (M/Chip 4) cards.
The format is defined by the MCHIP4-APPL-DATA data elements of the EMV Request Data token.
- Note:** Once the EMV IAD FORMAT field on CPF screen 11 contains a value of 3 (M/Chip 4 format), both MasterCard values in the EMV IAD FORMAT field on CAF screen 13 have the same effect.

Determining the Issuer Application Data Format

During a migration period, cards with the same prefix can have more than one application (e.g., cards with a VIS or M/Chip format migrating to a Common Core Definitions (CCD) format).

BASE24-atm uses three components to determine the issuer application data (IAD) format.

- Transaction data
- EMV IAD FORMAT field on CPF screen 11
- ALTERNATE IAD FORMAT field on CPF screen 11

BASE24-atm determines the IAD format from the transaction data, then uses that IAD format and the two IAD formats defined on CPF screen 11 to determine the IAD format for the transaction.

Transaction Data

The following table shows how the IAD format on cards that comply with specific applications can be identified from the transaction data.

IAD Length	IAD Byte 1	IAD Byte 2	IAD Format from Transaction Data
Greater than 6	06	Not applicable	VIS
Greater than 7	Not applicable	0x ¹	M/Chip 2.1
Greater than 17	Not applicable	1y ¹	M/Chip 4.0
32	0F	Az ²	CCD

¹ x and y can be any hexadecimal digit

² z can be a hexadecimal digit in the range of 5 through F

Issuer Application Data Format for a Specific Transaction

The following table shows the checks performed to derive the Issuer Application Data format for a specific transaction, using the two IAD formats defined on CPF screen 11 and the IAD format determined from the transaction data.

The first check must be on the two IAD format fields in the CPF record, as shown in the first row of the table. If these values match, the specified format is assumed, regardless of the format of the Issuer Application Data format that is determined from the transaction data.

It is possible for the IAD in a transaction to match more than one defined format. For example, a 10-byte IAD beginning %h06 %h00 in the first two bytes could indicate either a VIS or an M/Chip 2.1 format. If both IAD formats are configured on CPF screen 11, BASE24 uses the value in the EMV IAD FORMAT field rather than the value in the ALTERNATE IAD FORMAT field.

CPF Screen 11 Fields		IAD Format from Transaction Data	IAD Format Used for Transaction
EMV IAD FORMAT	ALTERNATE IAD FORMAT		
Any value	Matches value in EMV IAD FORMAT field	Not applicable	Value in EMV IAD FORMAT field
Non-zero	0	Matches value in EMV IAD FORMAT field	Value in EMV IAD FORMAT field
Non-zero	0	Does not match value in EMV IAD FORMAT field	0
0	Non-zero	Matches value in ALTERNATE IAD FORMAT field	Value in ALTERNATE IAD FORMAT field
0	Non-zero	Does not match value in ALTERNATE IAD FORMAT field	0
Non-zero	Non-zero	Matches value in EMV IAD FORMAT field	Value in EMV IAD FORMAT field
Non-zero	Non-zero	Matches value in ALTERNATE IAD FORMAT field	Value in ALTERNATE IAD FORMAT field
Non-zero	Non-zero	Does not match value in EMV IAD FORMAT field or ALTERNATE IAD FORMAT field	0

Application Transaction Counter Checking

An integrated circuit card (ICC) is a plastic card, usually the size of a credit card, that contains an embedded microprocessor chip. This chip is capable of storing large amounts of cardholder information and can contain multiple applications. The terms chip card and smart card are sometimes used interchangeably with integrated circuit card.

The application transaction counter (ATC) is a value maintained by the microchip and updated for each transaction performed by the card application. A single card can hold multiple ATCs, depending on the number of applications on the card. The ATC is also maintained in the BASE24 Cardholder Authorization File (CAF), as the value may be checked during transaction processing.

Application transaction counter checking is used only with the Positive and Positive Balance authorization methods.

To reflect the possibility of multiple ATCs being maintained on the card, there are multiple CAF fields that can be used to hold the ATC. The CAF field used in a particular transaction is determined by the setting of various fields on the CPF:

- ATC CHECK field on CPF screen 3
- ATC CHECK TYPE field on CPF screen 11
- CAP ATC UPDATE field on CPF screen 13

Note: BASE24-atm does not support CAP token validation transactions at this time.

You should set these fields based on which of the following you want to maintain:

- Separate ATCs for contactless magnetic stripe transactions, EMV transactions, and CAP token validation transactions.
- A single ATC for all three types of transactions.
- A single ATC for both contactless magnetic stripe and EMV transactions, and a separate ATC for CAP token validation transactions (or where CAP token validation transactions are not supported).
- A single ATC for both EMV and CAP token validation transactions, and a separate ATC for contactless magnetic stripe transactions (or where contactless magnetic stripe transactions are not supported).
- Separate ATCs for contactless magnetic stripe transactions and EMV transactions (where CAP token validation transactions are not supported).

- Separate ATCs for contactless magnetic stripe transactions and EMV transactions (where contactless magnetic stripe transactions are not supported).
- A single ATC for just contactless magnetic stripe transactions (where EMV transactions and CAP token validation transactions are not supported).
- A single ATC for just EMV transactions (where contactless magnetic stripe transactions and CAP token validation transactions are not supported).
- A single ATC for just CAP token validation transactions (where contactless magnetic stripe transactions and EMV transactions are not supported).
- No ATCs at all.

The relationship between these fields is described in more detail in the ***BASE24-pos EMV Support Manual***.

Two checks are performed on the ATC value received in the authorization request to detect fraudulent cards for EMV transactions.

- The value of the ATC in the message must be greater than the value in the appropriate ATC number field in the CAF.
- The value of the ATC in the message must not be greater than the sum of the appropriate ATC number field in the CAF and the ATC LIMIT field on CAF screen 13.

These checks are enabled through the ATC CHECK field on CPF screen 3 and ATC CHECK TYPE field on CPF screen 11. The values in these fields also specify whether ATC numbers are stored in the Base segment or EMV segment of the CAF.

Note: The ATC CHECK field on CPF screen 3 controls whether ATC numbers are stored in the Base segment of the CAF, so you must set it to 1 (perform ATC check on EMV transactions) or 3 (perform ATC check on EMV and contactless magnetic stripe transactions) if you set the ATC CHECK TYPE field on CPF screen 11 to a value of 8 or 9 (perform the ATC check with the ATC located in the Base segment). Otherwise, the ATC value being used in ATC checks will not be accurate.

The ATC limit is defined as the maximum number of transactions that can be performed since the last ATC check. Because EMV transactions can be authorized offline and the ATC value on the card can be updated without updating the ATC value in the CAF, the ATC value in the CAF can be less than the ATC value on the

card. Any ATC value received in a transaction that is greater than the sum of the ATC value and the ATC limit in the CAF could be viewed as being received from a fraudulent card.

If the ATC LIMIT field on CAF screen 13 is set to a value of zero, ATC limit checking is not performed. If the ATC from the card is less than or equal to zero, the transaction is declined. A full file CAF refresh resets the values in the ATC NUMBER field and SECOND CARD ATC NUMBER field on CAF screen 13 to zero. For the first transaction after the refresh, the ATC limit checking is bypassed and the value in the ATC NUMBER field or SECOND CARD ATC NUMBER field on CAF screen 13 is set to the value on the card.

Application transaction counter checking is used only with the Positive and Positive Balance authorization methods.

Second Card Processing

Second card processing is an optional feature that is enabled by LCONF parameter EXP-DAT-CMP and different expiration dates for the primary (original) and second (reissued) cards in the EXPIRATION DATE fields on CAF screens 2 and 7. This feature enables an institution to issue two cards with the same primary account number (PAN) and member number. The two cards are differentiated by comparing the expiration date on the card with two expiration dates held in the CAF, one for the primary card and another for the second card.

The system can maintain one or two additional ATC in the CAF for the second card. The SECOND ATC NUMBER fields are located on CAF screens 7 and 13. The values in the ATC CHECK field on CPF screen 3 and ATC CHECK TYPE field on CPF screen 11 control which SECOND ATC NUMBER fields are used. When a transaction is received from the second card and ATC checking is enabled, the ATC from the card's microchip is checked against the value in the appropriate SECOND ATC NUMBER field. Additionally, if the system is configured by LCONF parameter CAF-SCND-CRD-PROMOTE to automatically make the second card the new primary card, the second card's ATC is included in the data that is moved to the primary card's data.

Status Field Verification

The EMV data elements sent in request messages include two status fields:

- Terminal verification results (TVR)
- Card verification results (CVR)

These fields contain information about processing performed by an EMV-capable terminal and the ICC at the acquiring point.

In case a host or issuer is unavailable and BASE24 is not using the Positive or Positive Balance Authorization method, status field checking enables BASE24 to authorize or decline a transaction. You can also configure status field checking as a prescreen check using the TVR/CVR field in the CPF.

Status field checking is performed using preconfigured edit files (CVRTBL and TVRTBL) that are compiled into the Authorization process. These files identify which field information is to be checked and the required actions. Options available for each table are listed later in this section.

These edit files support a number of sets of test conditions and actions. The value of the issuer application data value used for a specific transaction determines which set of action index options is used. Refer to the “Issuer Application Data” topic earlier in this section for an explanation of how the issuer application data is determined for a specific transaction.

Issuer Application Data Used	Action Index Used
0	0
EMV IAD FORMAT field on CPF screen 11	STATUS CHECK ACTION INDEX field on CPF screen 11
ALTERNATE IAD FORMAT field on CPF screen 11	ALTERNATE ACTION INDEX field on CPF screen 11
EMV IAD FORMAT field on CAF screen 13	STATUS CHECK ACTION INDEX field on CAF screen 13

You can edit the configuration files to support your specific requirements. For example, you can preconfigure alternative tests or actions. In addition, the bits are checked in the order in which they appear in the edit file. Therefore, further flexibility is available in that you can configure the order in which the bits are checked.

The following lists examples of changes you can make in the edit files:

- Adjust the actions to be performed for each test by modifying column four (described below).
- Revise the sequence of the checks by reordering the entries within a given action index.
- Add checks to a specific set.
- Add new test sets by allocating a new action index and adding entries for all required checks to the end of the table. All checks for a single action index must be contiguous entries in the table.

You can copy entries into the actual table to build additional test sets, adjusting the action index and action fields accordingly.

Edit File Layout

The edit files contain four columns with the following information.

Action Index

The first column, action index, identifies the test set to which the test and prescribed actions belong. This index is used to select the tests and actions to be applied for a particular card scheme by matching the value in the STATUS CHECK ACTION INDEX field configured on CPF screen 11, ALTERNATE ACTION INDEX field configured on CPF screen 11, or STATUS CHECK ACTION INDEX field configured on CAF screen 13.

Byte and Bit

The second and third columns, Byte and Bit, specify which byte and bit of the CVR or TVR data element are to be tested. Note that in the numbering of the bits, bit 8 is the high order bit.

Authorization Action

The fourth column, Authorization Action, is applicable to all offline authorization methods and optionally as a prescreen check for online to host authorization methods. Valid values are as follows:

C = Continue processing (equivalent to no check)

D = Decline

K = Decline and retain (keep) card

Note that where the Authorization Action is C (i.e., the byte and bit settings have no effect), the entries can be left out of the table.

Card Verification Results Table

The Card Verification Results Table (CVRTBL) specifies the particular bits from the CVR status bytes that are to be tested and the actions to be taken when a specific bit is set on.

The CVR is a proprietary EMV data element (included in the issuer application data). It is therefore possible for different card schemes to use alternative definitions for the bit settings.

The CVRTBL contains the following CVR record templates:

- MasterCard M/Chip 2.1 (EMV 1996-compliant) cards and Visa VIS (as used for the UKIS ICC implementation) cards
- MasterCard M/Chip 4 (EMV 2000-compliant) cards
- CCD cards

M/Chip 2.1 and VIS CVR Records

M/Chip 2.1 and VIS use the same interpretations, except for a few differences that are noted in the following tables. All settings are applicable to both VIS and M/Chip 2.1 unless followed by VIS or M/Chip 2.1 in parentheses.

The CVRTBL defines two preconfigured test sets plus a group of alternative tests for M/Chip 2.1 and VIS cards.

The default test set (action index 1) for VIS or M/Chip 2.1 cards identifies that no checks are required. (The single null test is included as a placeholder.)

Action Index	Byte	Bit	Authorization Action	Description
1	4	4	C	Issuer script processing failed on last transaction

The base test set (action index 2) provides default tests for VIS or M/Chip 2.1 cards.

Action Index	Byte	Bit	Authorization Action	Description
2	3	4	D	Issuer authentication processing failed on last online transaction

The following table lists other alternative tests for VIS or M/Chip 2.1 cards.

Action Index	Byte	Bit	Authorization Action	Description
<i>n</i>	2	4	C	Issuer authentication failed
<i>n</i>	2	3	C	Offline PIN verification performed
<i>n</i>	2	2	C	Offline PIN verification failed
<i>n</i>	2	1	C	Unable to go online
<i>n</i>	3	8	C	Last online transaction not completed
<i>n</i>	3	7	C	PIN try limit exceeded
<i>n</i>	3	6	C	Exceeded velocity checking counters
<i>n</i>	3	5	C	New card
<i>n</i>	3	4	C	Issuer authentication failed on last online transaction

Action Index	Byte	Bit	Authorization Action	Description
<i>n</i>	3	3	C	Issuer authentication not performed after online transaction
<i>n</i>	3	2	C	Application blocked by card because PIN try limit exceeded
<i>n</i>	3	1	C	Static data authentication failed on last transaction and transaction declined offline
<i>n</i>	4	5	C	DDA failed on last online transaction and transaction declined offline (M/Chip 2.1)
<i>n</i>	4	4	C	Issuer script processing failed on last transaction
<i>n</i>	4	3	C	Lower consecutive offline limit or lower cumulative offline amount exceeded (M/Chip 2.1) DDA failed on last online transaction and transaction declined offline (VIS)
<i>n</i>	4	2	C	Upper consecutive offline limit or upper cumulative offline amount exceeded (M/Chip 2.1) Offline DDA performed (VIS)
<i>n</i>	4	1	C	Maximum offline transaction amount exceeded (M/Chip 2.1)

M/Chip 4 CVR Records

The CVRTBL defines two preconfigured test sets plus a group of alternative tests for M/Chip 4 cards.

The default test set (action index 1) identifies that no checks are required. (The single null test is included as a placeholder.)

Action Index	Byte	Bit	Authorization Action	Description
1	4	4	C	PIN try limit exceeded

The base test set (action index 3) provides default tests for M/Chip 4 cards.

Action Index	Byte	Bit	Authorization Action	Description
3	5	3	D	Issuer authentication failed on last online transaction

The following table lists other alternative tests for M/Chip 4 cards.

Action Index	Byte	Bit	Authorization Action	Description
<i>m</i>	4	7	C	Unable to go online
<i>m</i>	4	6	C	Offline PIN verification not performed
<i>m</i>	4	5	C	Offline PIN verification failed
<i>m</i>	4	4	C	PIN try limit exceeded
<i>m</i>	4	3	C	International transaction
<i>m</i>	4	2	C	Domestic transaction
<i>m</i>	4	1	C	Terminal erroneously considers offline PIN OK
<i>m</i>	5	8	C	Lower consecutive offline limit exceeded

Action Index	Byte	Bit	Authorization Action	Description
<i>m</i>	5	7	C	Upper consecutive offline limit exceeded
<i>m</i>	5	6	C	Lower cumulative offline limit exceeded
<i>m</i>	5	5	C	Upper cumulative offline limit exceeded
<i>m</i>	5	4	C	Go online on next transaction was set
<i>m</i>	5	3	C	Issuer authentication failed
<i>m</i>	5	2	C	Script received
<i>m</i>	5	1	C	Script failed
<i>m</i>	6	2	C	Match found in additional check table
<i>m</i>	6	1	C	No match found in additional check table

CCD (IAD Format A) CVR Records

The CVRTBL defines two preconfigured test sets plus a group of alternative tests for CCD (IAD format A) cards.

The default test set (action index 1) identifies that no checks are required. (The single null test is included as a placeholder.)

Action Index	Byte	Bit	Authorization Action	Description
1	4	4	C	Issuer script processing failed

The base test set (action index 4) provides default tests for CCD (IAD format A) cards.

Action Index	Byte	Bit	Authorization Action	Description
4	1	1	D	Issuer authentication failed

The following table lists other alternative tests for CCD (IAD format A) cards.

Action Index	Byte	Bit	Authorization Action	Description
<i>c</i>	1	4	C	CDA performed
<i>c</i>	1	3	C	Offline DDA performed
<i>c</i>	1	2	C	Issuer authentication not performed
<i>c</i>	1	1	C	Issuer authentication failed
<i>c</i>	2	4	C	Offline PIN verification performed
<i>c</i>	2	3	C	Offline PIN verification performed and PIN not successfully verified
<i>c</i>	2	2	C	PIN try limit exceeded
<i>c</i>	2	1	C	Last online transaction not completed
<i>c</i>	3	8	C	Lower offline transaction count limit exceeded
<i>c</i>	3	7	C	Upper offline transaction count limit exceeded
<i>c</i>	3	6	C	Lower cumulative offline amount limit exceeded
<i>c</i>	3	5	C	Upper cumulative offline amount limit exceeded
<i>c</i>	4	4	C	Issuer script processing failed

Action Index	Byte	Bit	Authorization Action	Description
<i>c</i>	4	3	C	Offline data authentication failed on previous transaction
<i>c</i>	4	2	C	Go online on next transaction was set
<i>c</i>	4	1	C	Unable to go online

Terminal Verification Results Table

The Terminal Verification Results Table (TVRTBL) is used to specify the particular bits from the TVR status bytes that are to be tested and the actions to be taken when a specific bit is set on.

The TVR is a standard EMV data element. The tables below document the TVR values as defined by EMV, and provide descriptions of each byte and bit that can be tested.

The TVRTBL defines two preconfigured test sets.

The default test set (action index 1) identifies that no checks are required. (The single null test is included as a placeholder.)

Action Index	Byte	Bit	Authorization Action	Description
1	1	8	C	Offline data authentication was not performed

The base test set (action index 2) provides default tests.

Action Index	Byte	Bit	Authorization Action	Description
2	2	7	D	Expired application
2	2	6	D	Application not yet effective

Action Index	Byte	Bit	Authorization Action	Description
2	2	5	D	Requested service not allowed for card product
2	3	8	D	Cardholder verification was not successful

The following table lists other alternative tests.

Action Index	Byte	Bit	Authorization Action	Description
<i>n</i>	1	8	C	Offline data authentication was not performed
<i>n</i>	1	7	C	Offline static data authentication failed
<i>n</i>	1	6	C	ICC data missing
<i>n</i>	1	5	C	Card appears on terminal exception file
<i>n</i>	1	4	C	Offline DDA authentication failed
<i>n</i>	2	8	C	ICC and terminal have different application versions
<i>n</i>	2	7	C	Expired application
<i>n</i>	2	6	C	Application not yet effective
<i>n</i>	2	5	C	Requested service not allowed for card product
<i>n</i>	2	4	C	New card
<i>n</i>	3	8	C	Cardholder verification was not successful
<i>n</i>	3	7	C	Unrecognized CVM (cardholder verification method)

Action Index	Byte	Bit	Authorization Action	Description
<i>n</i>	3	6	C	PIN try limit exceeded
<i>n</i>	3	5	C	PIN entry required and PIN PAD not present or not working
<i>n</i>	3	4	C	PIN entry required, PIN PAD present, but PIN was not entered
<i>n</i>	3	3	C	Online PIN entered
<i>n</i>	4	8	C	Transaction exceeds floor limit
<i>n</i>	4	7	C	Lower consecutive offline limit exceeded
<i>n</i>	4	6	C	Upper consecutive offline limit exceeded
<i>n</i>	4	5	C	Transaction selected randomly for online processing
<i>n</i>	4	4	C	Merchant forced transaction online
<i>n</i>	5	8	C	Default TDOL used
<i>n</i>	5	7	C	Issuer authentication was unsuccessful
<i>n</i>	5	6	C	Script processing failed before final GENERATE AC command
<i>n</i>	5	5	C	Issuer script processing failed after final GENERATE AC command

Fallback Checking

A fallback transaction is defined as a transaction performed on an ICC at an EMV-capable terminal but the microchip is not read. This can occur as a result of a faulty chip reader or a faulty chip, but could also be an indication of a fraudulent card. The fallback, or default, is to read the card information from the magnetic stripe. The FALLBACK field in the CPF allows you to configure whether to include fallback checking as a prescreen check. You can also specify whether the Authorization process should continue processing the transaction or decline the transaction and return the card.

If the Authorization process declines the transaction, a response code of 88 is sent in the response message.

Issuer Script Commands

Issuer script commands allow the issuer to alter the data or functionality of cards already in circulation. The BASE24-atm EMV product supports scripts originated by BASE24 as well as the issuer. These script commands are transmitted in the response message, and the card uses the script data to reconfigure itself. The supported script commands can disable the microchip or update card limits. Script data passed between the issuer and the card is secured with a message authentication code (MAC) generated by a security module.

The EMV product supports the following script commands:

- CARD BLOCK (an irreversible disabling of the microchip)
- PUT DATA (updates the LCOL field on the card)
- PIN CHANGE or PIN UNBLOCK (changes or unblocks the PIN)

BASE24 can originate script commands based on an indicator in the CAF or when requested by the issuer using a flag in the EMV Response Data token. BASE24 supports the generation of only one script command in a response message.

Issuer script data is supported either in pass-through mode or as response scripts generated by BASE24. For pass-through mode, BASE24 passes any script data received from an external authorizer through to the acquirer without validation or modification. If the EMV Script Data token is received from an issuer in a response message, the token is passed on to the acquirer end-point.

A PUT DATA script command is originated by BASE24 only if the card is configured for a positive authorization method. When a PUT DATA script is generated by BASE24, the SEND PUT DATA flag on the appropriate CAF record is reset to N only if the transaction is approved. This means that PUT DATA scripts are resent to the card on every subsequent EMV transaction, until an EMV transaction performed on the card is approved.

Configuring Issuer Script Commands

The issuer script commands are configured by BASE24 at three levels—system, card prefix, and cardholder. The system level configuration uses the LCONF params EMV-CARD-BLOCK-SCRIPT, EMV-PUT-DATA-SCRIPT, and EMV-PIN-UNBLOCK-SCRIPT, respectively. BASE24 uses the EMV-PIN-UNBLOCK-SCRIPT param for both the PIN CHANGE and PIN UNBLOCK issuer script commands.

An issuer can use the **SCRIPT TEMPLATE TAG** and **SCRIPT MAC LENGTH** fields on CPF screen 11 to configure card prefix level overrides of the MAC length specified in the LCONF params for script commands. An issuer can also use the **SCRIPT TEMPLATE TAG** and **SCRIPT MAC LENGTH** fields on CAF screen 13 to configure card level overrides of the MAC length specified in the CPF for script commands.

The MAC lengths specified in the CPF cannot exceed those specified in the LCONF. The MAC lengths specified in the CAF can exceed those specified in the CPF, but they cannot exceed those specified in the LCONF. The MAC lengths configured in the CPF and CAF do not apply to Common Core Definition cards because those cards require a MAC length of 4 bytes.

Issuer Script Formats

The hexadecimal representations of binary data contained in the three LCONF params specific to the EMV product are EMV issuer scripts. EMV issuer scripts are executed on the card itself by the EMV terminal.

Each EMV issuer script consists of a script header followed by a sequence of one or more issuer script command application protocol data units (APDUs) to be delivered serially to the ICC. The script header and each script command are composed of a tag (T), length (L), and value (V). The value for the script header is composed from the script ID, which in itself, is composed of a tag, length, and value. This structure is depicted in the following table where two commands are included in the script. All lengths must be provided in hexadecimal format.

Script Header					Command 1			Command 2		
T	L	V (script ID)			T	L	V	T	L	V
T	L	T	L	V	T	L	V	T	L	V

Script Header

The tag for the script header is always set to 71 or 72. These tags indicate the sequence of execution for processing the script by the EMV terminal. EMV terminals process scripts with a tag of 71 before issuing the final **GENERATE AC** command. EMV terminals process scripts with a tag of 72 after issuing the final **GENERATE AC** command. The first length field in the script header represents the length, in bytes, of the optional script ID plus the length of the first length field

itself. If the optional script ID is not included in the script header, the length in the second length field is set to 00. The tag field for the script ID is set to always set to the tag (9F18) for the Issuer Script Identifier data element. If the optional script ID is included in the script header, the second length field is set to the length of the script ID value plus the length of the second length field itself. The script ID is optional and is not interpreted by the EMV terminal. If provided, the script ID value consists of 4 bytes.

Script Command

Each script command provided in the script is composed of a tag set to 86 for the Issuer Script Command data element, the length of the command, and the command value. The length of the command represents the length, in bytes, of the command value plus the length of the command length field itself. The command value is provided in Application Protocol Data Unit (APDU) format, which consists of a mandatory header of four bytes followed by a conditional body of variable length, as shown in the following table.

CLA	INS	P1	P2	Lc	Data	Le
<-- Mandatory Header -->				<-- Conditional Body -->		

The mandatory header consists of the class byte of the command message (CLA), the instruction byte of the command message (INS), and two parameter bytes (P1 and P2). The conditional body contains the length of command data field (Lc), variable-length command data (Data), and the length of expected data field (Le).

Class Byte. The most significant nibble of the class byte indicates the type of command as shown in the following table.

Hexadecimal Value	Description
0	Inter-industry command
8	Proprietary to EMV specifications
Any other value	Outside the scope of the EMV specifications

A command proprietary to the EMV specification is introduced by the most significant nibble of the class byte set to 8; in other words, the structure of the command and response messages is according to the ISO/IEC 7816-4 standard,

Information technology -- Identification cards -- Integrated circuit(s) cards with contacts -- Part 4: Interindustry commands for interchange, and the coding of the messages is defined within the context of these specifications.

The least significant nibble of the class byte indicates secure messaging and logical channel mechanisms, according to ISO/IEC 7816-4. Valid values are as follows:

- C = Format 1 secure messaging used with the CCD EMV scheme
- 4 = Format 2 secure messaging used with other EMV schemes

Instruction Byte. The INS byte of a command indicates the action to be performed on the ICC. The following table lists the valid INS byte values used by BASE24, their meanings, and the relationship of INS byte values to CLA byte values. For more details on other possible values for the INS byte, refer to section 9.4.1 of Book 1 and section 6 of Book 3 in the EMV specifications.

CLA	INS	Command Instruction
8x	16	CARD BLOCK
8x	24	PIN UNBLOCK
04	DA	PUT DATA

Parameter Bytes. The parameter bytes (P1 and P2) can have any value. If they are not used, a parameter byte has a value of 00.

Length of Command Data. The number of data bytes sent in the command APDU (C-APDU) is denoted by Lc (length of command data field). If no command data is sent, this field is set to 00.

Command Data. When present, a command APDU message data field consists of a string of data elements in Basic Encoding Rules - Tag Length Value (BER-TLV) format. Up to 256 bytes of data can be sent for a command.

Length of Expected Data. The maximum number of data bytes expected in the response APDU is denoted by Le (length of expected data). When Le is present and contains the value zero, the maximum number of available data bytes (up to 256) is expected. When required in a command message, Le must always be set to 00.

Issuer Script Examples

The following tables illustrate the script data actually sent by the issuer (i.e., the contents of the EMV Script Token) in Card Block, Put Data, PIN Unblock, and PIN Change scripts.

Note: The contents of the EMV-PUT-DATA-SCRIPT param in the LCONF are used to create the PIN Change script.

Format 1 Issuer Scripts

Format 1 issuer scripts are used with the CCD card scheme. When supporting secure messaging according to Format 1, the script data actually sent by the issuer (i.e., the contents of the EMV Script Token), must contain the values specified in the table below. Note that the lower-order nibble of the CLA byte must be set to a value of C to indicate secure messaging according to Format 1. While MAC lengths of 04, 06, and 08 bytes are available, BASE24 supports only 4-byte MACs for Format 1 secure messaging.

Byte Description	CARD BLOCK	PUT DATA	PIN UNBLOCK	PIN CHANGE
Template Tag	71 or 72	71 or 72	71 or 72	71 or 72
Template Length	10 or 14	13 or 17	10 or 14	23 or 27
Script ID Tag (2 bytes)	9F18	9F18	9F18	9F18
Script ID Length	00 or 04	00 or 04	00 or 04	00 or 04
Script ID Value (4 bytes) ¹	XXXXXXXX ²	XXXXXXXX ²	XXXXXXXX ²	XXXXXXXX ²
Script Tag	86	86	86	86
Script Length	0B	0E	0B	1E
CLA	8C	0C	8C	8C
INS	16	DA	24	24

Byte Description	CARD BLOCK	PUT DATA	PIN UNBLOCK	PIN CHANGE
P1	00	Data Element Tag (first byte)	00	00
P2	00	Data Element Tag (second byte)	00	02
Data and MAC Length	06	09	06	19
Command Tag	Not present	81	Not present	87
Command Length	Not present	01	Not present	11
Command Data	Not present	XX	Not present	01 followed by 32 hexadecimal digits
MAC Tag	8E	8E	8E	8E
MAC Length ³	04	04	04	04
MAC Value ³ (4 bytes)	XXXX ²	XXXX ²	XXXX ²	XXXX ²

¹ The script ID value is optional. The script ID length identifies whether the script ID is present.

² The value X represents any hexadecimal digit.

Format 2 Issuer Scripts

Format 2 issuer scripts are used with card schemes other than CCD. When supporting secure messaging according to Format 2, the script data actually sent by the issuer (i.e., the contents of the EMV Script Token) must contain the values specified in the table on the following page. Note that the lower-order nibble of the CLA byte must be set to a value of 4 to indicate secure messaging according to Format 2. MAC lengths of 04, 06, and 08 bytes are available for Format 2 secure messaging.

Byte Description	CARD BLOCK	PUT DATA	PIN UNBLOCK	PIN CHANGE
Template Tag	71 or 72	71 or 72	71 or 72	71 or 72
Template Length	0E or 12 10 or 14 12 or 16	0F or 13 11 or 15 13 or 17	0E or 12 10 or 14 12 or 16	16 or 1A 18 or 1C 1A or 1E 1E or 22 20 or 24 22 or 26
Script ID Tag (2 bytes)	9F18	9F18	9F18	9F18
Script ID Length	00 or 04	00 or 04	00 or 04	00 or 04
Script ID Value (4 bytes) ¹	XXXXXXXX ²	XXXXXXXX ²	XXXXXXXX ²	XXXXXXXX ²
Script Tag	86	86	86	86
Script Length	09, 0B, or 0D	0A, 0C, or 0E	09, 0B, or 0D	11, 13, 15, 19, 1B, or 1D
CLA	84	04	84	84
INS	16	DA	24	24
P1	00	Data Element Tag (first byte)	00	00
P2	00	Data Element Tag (second byte)	00	02
Data and MAC Length	04, 06, or 08	05, 07, or 09	04, 06, or 08	0C, 0E, 10, 14, 16, or 18
Command Data	Not present	FF	Not present	Encrypted PIN block
MAC Value (4, 6, or 8 bytes)	XXXXXXXX ²	XXXXXXXX ²	XXXXXXXX ²	XXXXXXXX ²

- ¹ The script ID value is optional. The script ID length identifies whether the script ID is present.
- ² The value X represents any hexadecimal digit.

Prefix Routing for EMV Transactions

EMV processing allows routing of EMV transactions by prefix using a prefix route field in the EMV segment of the CPF. For EMV transactions, the prefix route code in the EMV segment of the CPF overrides the prefix route code in the base segment of the CPF. When an IDF record is used, this prefix route code allows EMV transactions to be routed and authorized separately from non-EMV transactions. When an IDF record is not used, the EMV Issuer code provides the same functionality. The EMV Issuer code overrides the value of the ISSUER field in the ATM segment of the CPF.

Processing EMV Card Transactions as Standard Track 2 Card Transactions

Some card issuers start distributing integrated circuit cards before being able to support transactions initiated by users with those cards. The BASE24-atm Authorization process can handle transactions initiated with these cards at EMV-enabled terminals by processing them as if they are initiated with standard Track 2 cards. This processing is also known as downgrading a transaction from EMV to magnetic stripe.

The CAM CHECK TYPE field in the CPF identifies how the BASE24-atm Authorization process authenticates integrated circuit cards for each card prefix. You can set this field to instruct the BASE24-atm Authorization process to process transactions initiated with integrated circuit cards at EMV-enabled terminals as if they are initiated with standard Track 2 cards instead of attempting to verify the authorization request cryptogram (ARQC).

The ARQC-VRFY field in the EMV Status token identifies integrated circuit card transactions processed as standard Track 2 card transactions without verifying the ARQC.

Authorization Request and Response Cryptograms

EMV integrated circuit cards are produced with both a magnetic stripe and a microchip. The chip also includes an electronic version of the magnetic stripe data as well as other data used for authentication and risk management. When used in an EMV device, the card and terminal perform offline risk management to determine whether the transaction should be approved or declined offline, or sent online by the acquirer for authorization. If sent online, an authorization request cryptogram, also known as an ARQC, is calculated using card and terminal data. The request cryptogram and component cryptogram data are sent to the issuer with the authorization request.

The issuer checks the request cryptogram, using the cryptogram data and a secure DES key, and calculates an authorization response cryptogram, also known as an ARPC, using response data. This response cryptogram is returned to the card by the acquirer, which can then confirm that the issuer is genuine. One call is made to the security module to both verify the request cryptogram and generate the response cryptogram. If the request cryptogram is not successfully verified, standard CVV/CVC checking is applied.

On successful completion of a transaction, another cryptogram, the transaction certificate, is calculated for use in verifying the offline clearing message. BASE24 does not process completion messages from an ATM, so BASE24 does not verify or log the transaction certificate.

A set of EMV-specific security utilities, called CAM utilities, must be used in conjunction with the Base security utilities to support this cryptogram verification. The CAM utilities support the Thales and Atalla security modules. For more information on these security modules, see the appropriate vendor documentation.

All EMV authorization processing is initiated by the presence of the EMV Request Data token and the appropriate configuration data in the CPF, CAF, and KEYI. Support for a prevalidated CAM (e.g., by Visa on the issuer's behalf) is also provided.

CAM checking is configurable at a prefix level, and is performed only on known prefixes configured in the CPF. If a card has a CAF record, application transaction counter (ATC) checking can be performed, as described later in this section.

When processing as an acquirer, the EMV product modules operate in pass-through mode, routing the CAM data in the EMV Request Data token to the issuer. The CAM data is logged to the TLF in the appropriate tokens (i.e., EMV Request

Data, EMV Response Data, and EMV Script Data tokens). The BASE24-atm Device Handler, Host Interface, or Interchange Interface process adds the EMV Request Data token.

In the event of an issuer being unavailable and BASE24 is not using the Positive or Positive Balance Authorization method, status field verification enables BASE24 to authorize or decline an EMV transaction. Status field verification is described in more detail later in this section.

For BASE24 processing as an issuer, the Authorization process optionally performs CAM authentication using either a Thales or Atalla security module.

Offline Update Counters

BASE24 includes configuration parameters at the card prefix level on CPF screen 11 that allow issuers to specify how the card should process its offline counters when an EMV response message is received from the issuer. There is one parameter for approved financial transactions, one parameter for approved non-financial transactions, and one parameter for non-approved transactions.

Check guarantee, check verification, balance inquiry, PIN change, and EMV PIN unblock transaction are non-financial transactions. All other BASE24-atm transactions are treated as financial transactions.

The values supported for each parameter are the same and include do not update offline counters, set counters to upper offline limits, reset counters to zero, and add the transaction to the appropriate counter.

BASE24 uses the appropriate value when formatting the data used in generating the Authorization Response Cryptogram (ARPC) in the Issuer Authentication Data.

Offline counter processing applies only to M/Chip 4 and Common Core Definition (CCD) cards, as these are the only EMV applications supported by BASE24 that define these bit settings.

EMV Card Verification

BASE24-atm allows issuers to validate the card verification digits (CVD) during EMV authorization processing, where the CVD on the Track 2 equivalent data is different from the CVD on the magnetic stripe.

An EMV transaction typically contains an authorization request cryptogram (ARQC), which is used by the issuer to authenticate the chip card. The ARQC verification is based on more powerful cryptography than the CVD validation, and is more likely to be able to distinguish between genuine and counterfeit EMV cards.

However it is possible for EMV issuers to receive chip-read transactions that do not contain an ARQC (e.g., MasterCard *partial grade* transactions). It is also possible for issuers to configure BASE24 to continue processing EMV transactions when the ARQC has not been verified successfully. Under these circumstances, BASE24-atm allows issuers optionally to validate the CVD during EMV authorization processing.

As described above, validating the CVD in an EMV transaction where ARQC verification has been performed successfully serves little purpose, because the validation authenticates only the Track 2 equivalent data. However, to provide complete flexibility, BASE24-atm allows issuers optionally to validate the CVD during EMV authorization processing, even when the ARQC has been verified successfully.

The EMV extension modules to ATM Authorization use four fields in the EMV segment of the Card Prefix File (CPF) to determine whether to perform card verification on EMV transactions.

- An EMV card verification check type field specifies the circumstances (if any) under which card verification is to be performed on EMV transactions.
- An EMV card verification method field specifies the algorithm and data used when card verification is performed on EMV transactions.
- An EMV card verification effective date field specifies the earliest card expiration date for which card verification is to be performed on EMV transactions.
- An EMV card verification data field contains any additional data required when card verification is performed on EMV transactions.

Cryptographic Support

EMV cryptographic support is provided for the following security functions:

- Request and response cryptogram processing
- Secure messaging with integrity for issuer script data

BASE24-atm provides cryptographic support for VIS, M/Chip, and CCD card schemes. All of the required cryptographic support can be provided by both the BASE24-atm Authorization process (using the CAM Utilities module) and the BASE24 Transaction Security Services process, except that only the Transaction Security Services process can provide cryptographic support for the generation of PIN Change scripts or the EMV 2000 method of session key derivation and CCD (IAD format A) cards.

The value in the KEYI GROUP field on CPF screen 11 is used to access the security data (e.g., master keys) required for cryptographic processing. The value in the KEYI GROUP field is used to access either the KEYI (when the Authorization process provides cryptographic support) or the TSS database (when the Transaction Security Services process is used). The KEYI GROUP fields on KEYI screen 1 and CPF screen 11 is limited to 16 characters in length. The expiration date of the card is also used to access the data, to allow different EMV security profiles to be configured as cards are reissued.

Both the KEYI and the TSS database support up to 24 authentication (cryptogram) master keys and one secure messaging (integrity) key for each EMV security profile. When requesting a specific security function, the Authorization process retrieves the derivation key index passed in requests, and passes it on to the appropriate CAM utilities procedure or TSS procedure to locate a specific cryptogram key or secure messaging key.

To support different security implementations for individual payment schemes, the EMV SCHEME field in the KEYI or the TSS database identifies the specific implementation associated with an EMV profile (i.e., authentication data). This field, in conjunction with the cryptogram version number passed in requests, identifies the specific implementation for the card, and is used by the Authorization process or the Transaction Security Services process to determine the specific security module calls and security functions to be performed.

The following subtopics discuss BASE24 cryptographic support without the BASE24 Transaction Security Services process. For information on cryptographic support provided with the Transaction Security Services process, including EMV 2000 cryptographic support, refer to the *BASE24 Transaction Security Services Processing Guide*.

EMV Scheme and Cryptogram Version Number

To support different security implementations for individual payment schemes, the EMV SCHEME field on KEYI screen 1 identifies the specific EMV implementation associated with an EMV profile (i.e., authentication data). This field, in conjunction with the cryptogram version number passed in requests, identifies the specific implementation associated with a KEYI record and is used by the Authorization process to determine the specific security module calls and security functions to be performed. Each of the valid EMV schemes are explained below.

- VIS—Defined by Visa in the *Visa Integrated Circuit Card Specification*. (Note this format is used for UKIS cards).
- M/Chip—Defined for MasterCard MCPA (M/Chip 2.1) cards.
- CCD—Mandated for CCD (IAD Format A) cards.

Note: The CCD scheme does not use the KEYI because it requires the TSS.

Cryptogram Version Numbers in EMV

The cryptogram version number is used (in conjunction with the EMV scheme) to identify the security requirements in EMV online transaction processing. In particular, the cryptogram version number defines the required processing in the following areas:

- Card key derivation method
- Session key derivation method for ARQC calculation
- Input data in ARQC/TC calculation
- Method used to calculate ARQC
- Session key derivation method for ARPC calculation
- Method used to calculate ARPC
- Session key derivation method for secure messaging

- Data format in secure messaging MAC calculation
- Padding of encrypted data for secure messaging
- Method used to encrypt data in secure messaging

The tables below indicate the security processing options, data formats, and methods used for each supported combination of EMV scheme and cryptogram version number. You can obtain further details on each of these security processing options, data formats, and methods in the EMV and card scheme specifications.

Throughout these tables, *x* represents the characters 0–9 and A–F in cryptogram version number %h0*x*, and *y* represents the characters A–F in cryptogram version number %h*y*5.

The processing performed within security devices for these functions is depicted in several diagrams at the end of this topic.

Card Key Derivation Method

Security Processing Options	EMV Application	
	EMV Scheme	CVN
Option A – Format a 16-byte data block by concatenating the PAN and application PAN sequence number (APSN), using padding or truncation if required. The left half of the key is computed by 3DES encrypting the PAN/APSN with the master key. The right half of the key is computed by 3DES encrypting the inverted (bitwise) PAN/APSN with the master key.	VIS	%h0A
	VIS	%h11 ¹
	VIS	%h0E
	M/Chip	%h0 <i>x</i>
	M/Chip	%h10
	M/Chip	%h11
	M/Chip	%h12
	M/Chip	%h13
	M/Chip	%h14
	M/Chip	%h15

Security Processing Options	EMV Application	
	EMV Scheme	CVN
Option B – Use the SHA-1 hashing algorithm on the PAN and APSN, and then create a 16-byte data block by selecting decimal digits from the output of the hash result. The left half of the key is computed by 3DES encrypting the PAN/APSN with the master key. The right half of the key is computed by 3DES encrypting the inverted (bitwise) PAN/APSN with the master key.	CCD	%hy5

- ¹ Cryptogram version number 17 for Visa cards is used to support quick Visa Smart Debit/Credit card (qVSDC) transactions. These transactions use a minimum of data to ensure quick transactions over a contactless interface, such as one employed for a low value payment toll booth operation.

Session Key Derivation Method For ARQC Calculation

Security Processing Options	EMV Application	
	EMV Scheme	CVN
None – Use card key.	VIS	%h0A
	VIS	%h11
EMV 2000 – Use a specified function that contains a branch factor, a <i>parent</i> key, and a <i>grandparent</i> key. Use the function to take bits from the ATC to derive a session key that is unique for each value of the ATC.	VIS	%h0E
	M/Chip	%h12
	M/Chip	%h13
MasterCard-proprietary – Format an 8-byte data block, and use it to generate the left and right halves of the session key. The left half is computed by replacing the third byte of the data block with %hF0, and 3DES encrypting the result with the card key. The right half is computed by replacing the third byte of the data block with %h0F, and 3DES encrypting the result with the card key.	M/Chip	%h0x
	M/Chip	%h10
	M/Chip	%h11

Security Processing Options	EMV Application	
	EMV Scheme	CVN
Common – Format an 8-byte <i>diversification value</i> , which is the 2-byte ATC concatenated with 6 bytes of %h00. Use the diversification value to generate the left and right halves of the session key. The left half is computed by replacing the third byte of the data block with %hF0, and 3DES encrypting the result with the card key. The right half is computed by replacing the third byte of the data block with %h0F, and 3DES encrypting the result with the card key.	M/Chip	%h14
	M/Chip	%h15
	CCD	%hy5

Input Data in ARQC/TC Calculation

Input Data Format	EMV Application	
	EMV Scheme	CVN
Amount, Auth Amount, Other Terminal Country Code Terminal Verification Results (TVR) Transaction Currency Code Transaction Date Transaction Type Unpredictable Number Application Interchange Profile (AIP) Application Transaction Counter (ATC) Card Verification Results (4 bytes)	VIS	%h0A
	VIS	%h0E
	M/Chip	%h0x
Amount, Auth Transaction Currency Code Unpredictable Number	VIS	%h11

Input Data Format	EMV Application	
	EMV Scheme	CVN
Amount, Auth Amount, Other Terminal Country Code Terminal Verification Results (TVR) Transaction Currency Code Transaction Date Transaction Type Unpredictable Number Application Interchange Profile (AIP) Application Transaction Counter (ATC) Card Verification Results (6 bytes)	M/Chip	%h10
	M/Chip	%h12
	M/Chip	%h14
Amount, Auth Amount, Other Terminal Country Code Terminal Verification Results (TVR) Transaction Currency Code Transaction Date Transaction Type Unpredictable Number Application Interchange Profile (AIP) Application Transaction Counter (ATC) Card Verification Results (6 bytes) Counters (8 bytes)	M/Chip	%h11
	M/Chip	%h13
	M/Chip	%h15
Amount, Auth Amount, Other Terminal Country Code Terminal Verification Results (TVR) Transaction Currency Code Transaction Date Transaction Type Unpredictable Number Application Interchange Profile (AIP) Application Transaction Counter (ATC) Issuer Application Data	CCD	%hy5

Method Used to Calculate ARQC

Security Processing Options	EMV Application	
	EMV Scheme	CVN
Visa-proprietary – Format a data block, comprising the input data padded to the right with bytes of %h00 until the data block is a multiple of 8 bytes in length. Calculate an 8-byte ARQC by using the key to generate a MAC over the data block.	VIS	%h0A
	VIS	%h11
EMV – Format a data block, comprising the ARQC concatenated with the input data and one byte containing %h80. If necessary, append bytes of %h00 until the data block is a multiple of 8 bytes in length. Calculate an 8-byte ARQC by using the key to generate a MAC over the data block.	VIS	%h0E
	M/Chip	%h0x
	M/Chip	%h10
	M/Chip	%h11
	M/Chip	%h12
	M/Chip	%h13
	M/Chip	%h14
	M/Chip	%h15
	CCD	%hy5

Session Key Derivation Method For ARPC Calculation

Derivation Method	EMV Application	
	EMV Scheme	CVN
None – Use card key.	VIS	%h0A
	M/Chip	%h0x
	M/Chip	%h10
	M/Chip	%h11
EMV 2000 – Use a specified function that contains a branch factor, a <i>parent</i> key, and a <i>grandparent</i> key. Use the function to take bits from the ATC to derive a session key that is unique for each value of the ATC.	VIS	%h0E
	M/Chip	%h12
	M/Chip	%h13
Common – Format an 8-byte <i>diversification value</i> , which is the 2-byte ATC concatenated with 6 bytes of %h00. Use the diversification value to generate the left and right halves of the session key. The left half is computed by replacing the third byte of the data block with %hF0, and 3DES encrypting the result with the card key. The right half is computed by replacing the third byte of the data block with %h0F, and 3DES encrypting the result with the card key.	M/Chip	%h14
	M/Chip	%h15
	CCD	%hy5

Input Data and Data Length in ARPC Calculation

Input Data and Data Length	EMV Application	
	EMV Scheme	CVN
ISO 8583 (1987) Response Code (2 bytes). ARPC is 8 bytes.	VIS	%h0A
	VIS	%h0E
	M/Chip	%h0x

Input Data and Data Length	EMV Application	
	EMV Scheme	CVN
16-bit ARPC Response Code (2 bytes). ARPC is 8 bytes.	M/Chip	%h10
	M/Chip	%h11
	M/Chip	%h12
	M/Chip	%h13
	M/Chip	%h14
	M/Chip	%h15
Card Status Updates (4 bytes) plus Proprietary Authentication Data (up to 8 bytes). ARPC is 4 bytes (truncated from 8 bytes).	CCD	%hy5

Method Used To Calculate ARPC

Calculation Method	EMV Application	
	EMV Scheme	CVN
Method 1 - Format an 8-byte data block, comprising the input data padded to the right with bytes of %h00. Calculate an 8-byte ARPC by XORing the data block with the ARQC, and 3DES encrypting the result with the session key. Note: This method cannot support more than 8 bytes of input data.	VIS	%h0A
	VIS	%h0E
	M/Chip	%h0x
	M/Chip	%h10
	M/Chip	%h11
	M/Chip	%h12
	M/Chip	%h13
	M/Chip	%h14
	M/Chip	%h15
Method 2 – Format a data block, comprising the ARQC concatenated with the input data and one byte containing %h80. If necessary, append bytes of %h00 until the data block is a multiple of 8 bytes in length. Calculate an 8-byte ARPC by using the key to generate a MAC over the data block. Note: This method is also used for all schemes and CVNs to calculate the MAC required for secure messaging with integrity.	CCD	%hy5

Session Key Derivation Method For Secure Messaging

Derivation Method	EMV Application	
	EMV Scheme	CVN
Visa-proprietary. Format an 8-byte <i>diversification value</i> , which is the 2-byte ATC right justified and padded to the left with 6 bytes of %h00. Use the diversification value to generate the left and right halves of the session key. The left half is computed by XORing the data block with the left half of the card key. The right half is computed by performing a bitwise inversion and XORing the result with the right half of the card key.	VIS	%h0A
EMV 2000 – Use a specified function that contains a branch factor, a <i>parent</i> key, and a <i>grandparent</i> key. Use the function to take bits from the ATC to derive a session key that is unique for each value of the ATC.	VIS	%h0E
	M/Chip	%h12
	M/Chip	%h13
Common – Format an 8-byte <i>diversification value</i> , which is the ARQC. Use the diversification value to generate the left and right halves of the session key. The left half is computed by replacing the third byte of the data block with %hF0, and 3DES encrypting the result with the card key. The right half is computed by replacing the third byte of the data block with %h0F, and 3DES encrypting the result with the card key.	M/Chip	%h0x
	M/Chip	%h10
	M/Chip	%h11
	M/Chip	%h14
	M/Chip	%h15
	CCD	%hy5

Data Format In Secure Messaging MAC Calculation

Data Format	EMV Application	
	EMV Scheme	CVN
Format 2 - The data elements (including the MAC) contained in the data field of the secured command are not composed of TLV data objects. Transaction-specific data, (e.g., the ATC), is inserted into the data field (immediately before any command data) to ensure that the MAC value generated is unique per transaction.	VIS	%h0A
	VIS	%h0E
	M/Chip	%h0x
	M/Chip	%h10
	M/Chip	%h11
	M/Chip	%h12
	M/Chip	%h13
	M/Chip	%h14
	M/Chip	%h15
Format 1 - The data field of the secured command is composed of TLV data objects. If the command to be secured has command data, then the command data is carried in the first data object, and is itself encapsulated under an EMV tag. The second data object is the MAC, which is encapsulated under tag 8E.	CCD	%hy5

Padding Of Encrypted Data For Secure Messaging

Padding Method	EMV Application	
	EMV Scheme	CVN
Visa-proprietary - Prepend encrypted data with one byte containing the length of the encrypted data. Append encrypted data with one byte containing %h80. If necessary, append bytes of %h00 until data block is a multiple of 8 bytes in length.	VIS	%h0A
	VIS	%h0E

Padding Method	EMV Application	
	EMV Scheme	CVN
MasterCard-proprietary - If the enciphered data is not a multiple of 8 bytes in length, append one byte containing %h80. If necessary, append bytes of %h00 until data block is a multiple of 8 bytes in length.	M/Chip	%h0x
	M/Chip	%h10
	M/Chip	%h11
	M/Chip	%h12
	M/Chip	%h13
	M/Chip	%h14
	M/Chip	%h15
EMV - Append enciphered data with one byte containing %h80. If necessary, append bytes of %h00 until data block is a multiple of 8 bytes in length.	CCD	%hy5

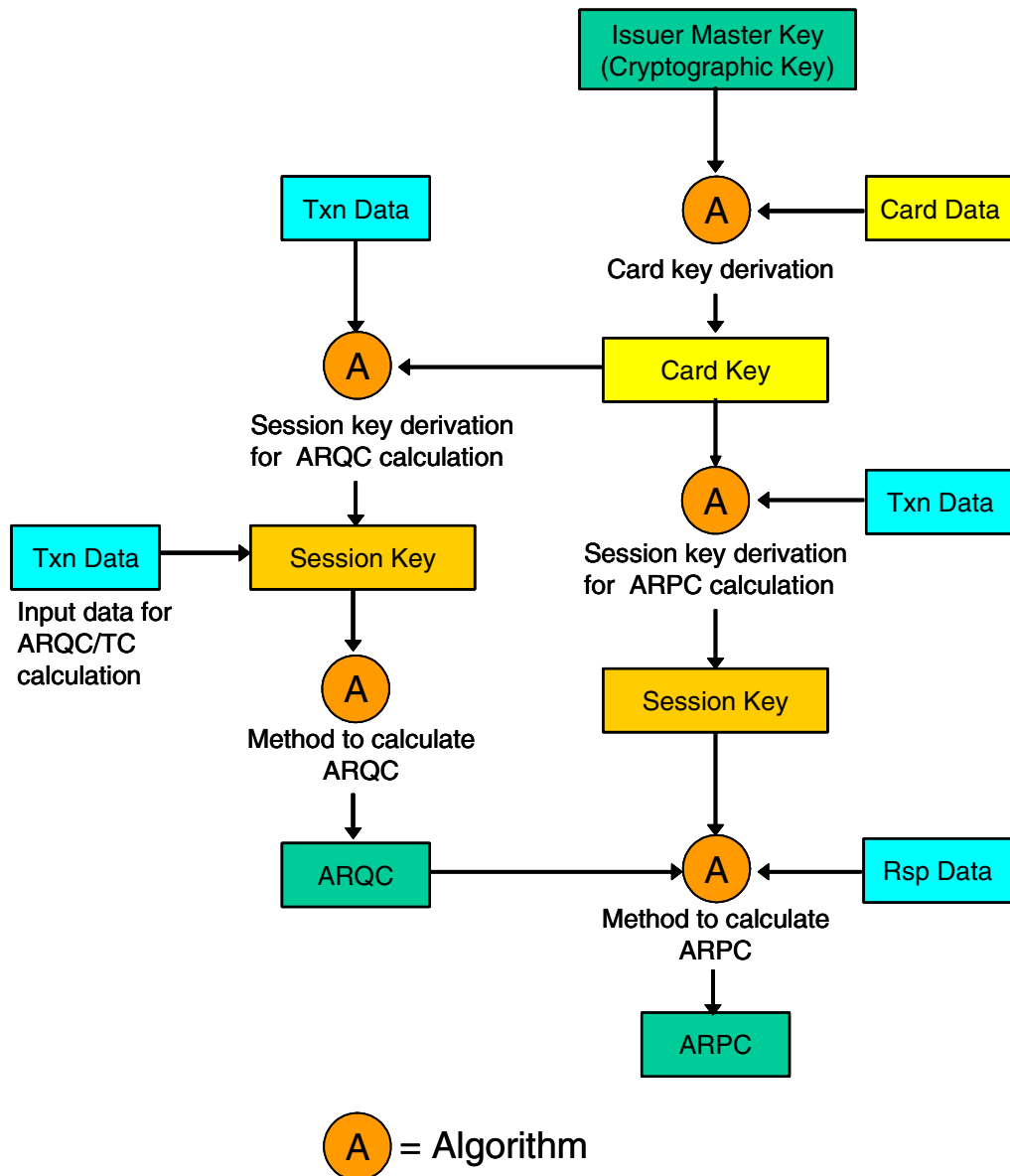
Method Used To Encrypt Data In Secure Messaging

Encryption Method	EMV Application	
	EMV Scheme	CVN
Electronic Code Book (ECB).	VIS	%h0A
	VIS	%h0E
Cipher Block Chaining (CBC).	M/Chip	%h0x
	M/Chip	%h10
	M/Chip	%h11
	M/Chip	%h12
	M/Chip	%h13
	M/Chip	%h14
	M/Chip	%h15
	CCD	%hy5

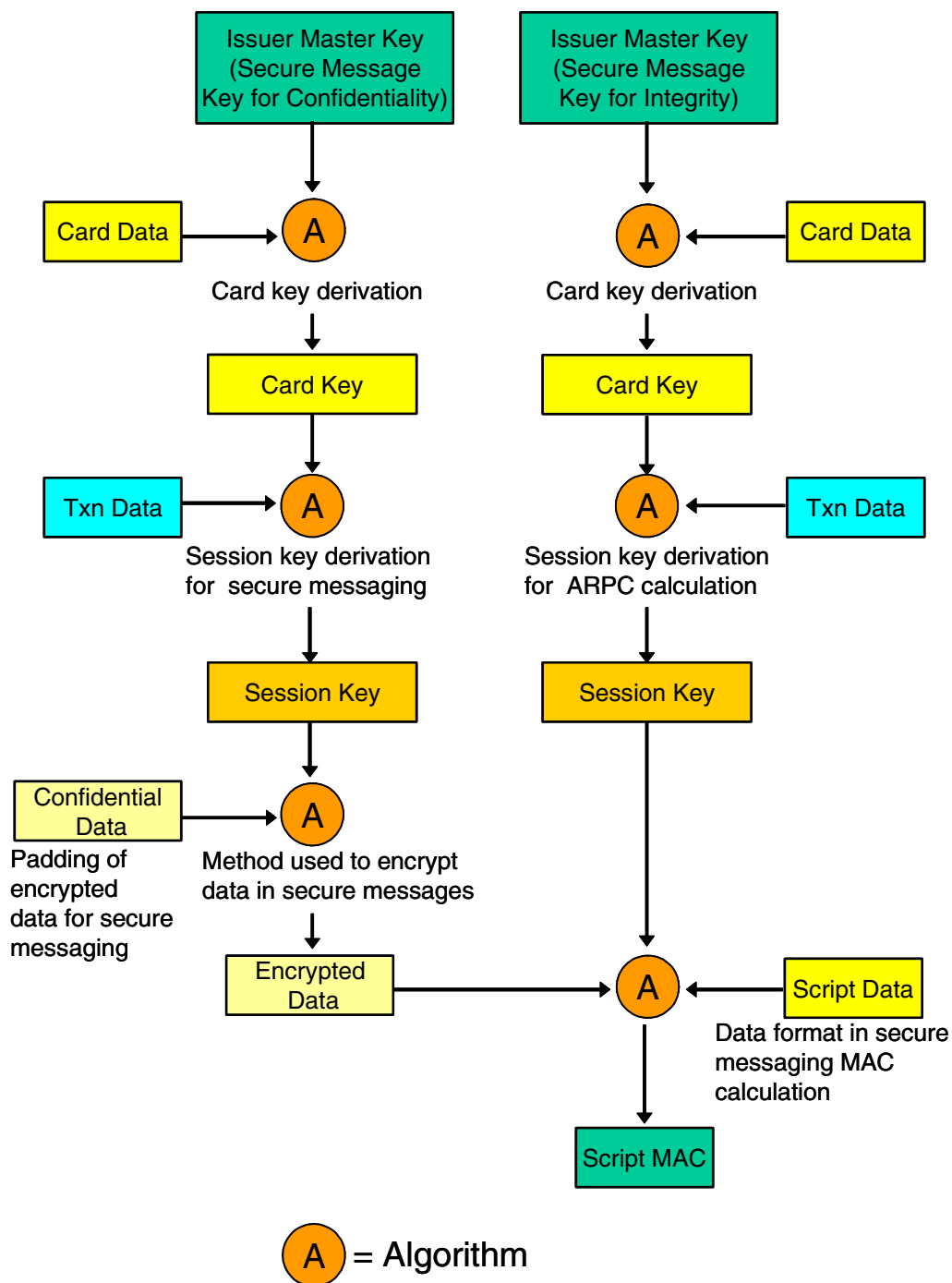
Cryptographic Processing Flows

The following diagrams depict the flow of cryptographic functions for authorization request and response cryptogram processing and for secure message processing. All of these functions are performed within a hardware security device.

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Secure Message Processing



PIN Management Processing

In order to process EMV offline PIN cardholder verification method (CVM) transactions, the BASE24-atm system supports the following functions:

- Unblock offline PIN
- Synchronized change of online and offline PIN
- Offline PIN change notification
- Script failure notification
- PIN management parameters

Unblock Offline PIN

An offline PIN can be unblocked either implicitly or explicitly.

- An EMV transaction request initiated at an ATM contains data retrieved from the EMV chip card. This data includes the Card Verification Results (CVR), which contains an indicator of whether the cardholder has exceeded the maximum number of PIN tries allowed for offline transactions. The indicator is carried in byte 3, bit 7 for VIS and M/Chip 2.1. The same indicator is carried in byte 4, bit 5 for M/Chip 4. Checking this indicator on all EMV transactions allows the BASE24 system to provide an implicit mechanism to unblock the offline PIN.
- A PIN unblock transaction request is initiated by an EMV card at an EMV-capable ATM, providing an explicit mechanism to unblock the offline PIN.

A host system can reset the bad PIN try counter on an EMV card by including a PIN unblock script command in any EMV transaction response message. In the BASE24 system, issuers can specify whether offline PIN unblocking is permitted.

- The value in the BAD OFFLINE PIN ACTION field on CPF screen 11 allows an issuer to specify whether implicit offline PIN unblocking is permitted. A setting of this parameter results in the transaction request being declined, meaning that the cardholder must perform an explicit PIN unblock transaction or PIN change transaction to reset the bad PIN try counter on the card.
- The value in the OFFLINE PIN MANAGEMENT ACTION field on CPF screen 11 allows an issuer to specify whether explicit offline PIN unblocking is permitted.

The BASE24 system must generate a PIN unblock script command, either implicitly (if the PIN try indicator of the CVR is set to the value 1) or explicitly (in response to a PIN unblock transaction request). The value in the SEND PIN UNBLOCK field on CAF screen 13 allows an issuer to specify the conditions under which a PIN unblock script is to be generated. A PIN unblock script command can be returned to the card only if the cardholder's online PIN is verified successfully and the transaction's authorization request cryptogram (ARQC) is verified successfully.

The BASE24 system must also generate a PIN change script command in response to a PIN unblock transaction request. If the cardholder forgets the PIN, a new PIN can be issued through the terminal. This processing allows a cardholder to resynchronize the online and offline PINs without actually changing the online PIN. A PIN change script command unblocks the offline PIN in addition to changing the offline PIN. The value in the SEND PIN CHANGE field on CAF screen 13 allows an issuer to specify the conditions under which a PIN change script can be generated. A PIN change script command can be returned to the card only if the cardholder's online PIN is verified successfully and the transaction's ARQC is verified successfully.

BASE24 processing does not support the HSM functionality required to encrypt the PIN block returned in the script command.

Note: You must use the Transaction Security Services process with the BASE24-atm EMV add-on product to support the generation of PIN change scripts. For more information on the offline PIN management function support provided by the Transaction Security Services process, refer to the ***BASE24 Transaction Security Services Processing Guide***.

Synchronized Change of Online and Offline PIN

An offline PIN can be changed using an explicit mechanism only. A PIN change transaction request (and optionally a PIN unblock transaction request), that can be initiated by an EMV card at an EMV-capable ATM, provides the required functionality.

A host system can change the PIN on an EMV card by including a PIN change script command in a transaction response message. The value in the OFFLINE PIN MANAGEMENT ACTION field on CPF screen 11 allows an issuer to specify whether an offline PIN change is permitted.

The value in the OFFLINE PIN MANAGEMENT ACTION field on CPF screen 11 also indicates whether the new PIN data is to be logged to the BASE24 database, as this information can be required if the card subsequently must be reissued with the new PIN. PIN change transaction logic allows the new PIN verification digits to be written to the BASE24-atm Transaction Log File (TLF). This information can be extracted and provided to a card management system, which can require the new PIN verification digits if the card subsequently must be reissued. For Visa PVV PIN verification, the new PIN cannot be derived from the new PVKI/PVV; therefore, additional information (e.g., the new PIN data itself) is required.

The BASE24 system must generate a PIN change script command, if the PIN change request is processed successfully (i.e., the cardholder's PIN verification digits are updated successfully) in the BASE24 database. The value in the SEND PIN CHANGE field on CAF screen 13 allows an issuer to specify the conditions under which a PIN change script is to be generated. A PIN change script command can be returned to the card only if the cardholder's online PIN is verified successfully and the transaction's authorization request cryptogram (ARQC) is verified successfully.

For an EMV card that does not contain an active offline PIN, it is inappropriate to return a PIN change script command, as the card may not be able to process this type of script command. Conversely, for an EMV card that does contain an active offline PIN, it is inappropriate to authorize non-EMV PIN change transactions, because a PIN change script command cannot be returned in the transaction response (resulting in nonsynchronization of the online PIN and the offline PIN). The value in the ONLINE/OFFLINE PIN SYNC field on CAF screen 13 allows an issuer to specify whether synchronization of the online and offline PIN is required for the card. The parameter specifies the action to take when a PIN Change request (either EMV or non-EMV) is performed on the EMV card.

Note: You must use the Transaction Security Services process with the BASE24-atm EMV add-on product to support the generation of PIN CHANGE scripts because the actual PIN block sent in the script has to be encrypted, and this processing is supported only in the Transaction Security Services process. No encryption is required for PIN unblock scripts because the PIN block itself is not included in the script. For more information on the offline PIN management function support provided by the Transaction Security Services process, refer to the *BASE24 Transaction Security Services Processing Guide*.

Offline PIN Change Notification

Card management systems can require knowledge of the new PIN when a PIN change transaction is performed successfully. These systems can include the new PIN on the card if and when the card is reissued, (e.g., if it is lost or when it expires). The BASE24 system allows the new PIN to be written (in encrypted form) to the BASE24-atm Transaction Log File (TLF) with the key locator to the PIN block encryption key on the Transaction Security Services (TSS) database. To decrypt this PIN block, the card management system must have access to the TSS database, or store the same PIN block encryption keys as TSS.

A PIN block encryption key locator in the ATD identifies the key in the TSS database that acts as a transport key in the zone between the BASE24-atm system and the card management system. Before logging the new PIN to the TLF, BASE24-atm and the Transaction Security Services process translate the PIN block to encryption under the new key. The card management system stores the new key on its database. Therefore, when it receives the new PIN block (in a TLF extract, for example), the card management system is able to decrypt the new PIN block.

Script Failure Notification

When authorized by the BASE24 system or the issuer host system, it is possible for EMV script processing to fail at the ATM.

If the BASE24 system is notified that the card failed to unblock the offline PIN following implicit PIN unblock processing, the BASE24 system must record this information so that appropriate personnel can be notified.

If the BASE24 system is notified that the card failed to change or unblock the offline PIN (following a PIN change transaction or an explicit PIN unblock transaction), the PIN change or unblock transaction must be reversed. This has the effect of either restoring the cardholder's original PIN verification digits to the BASE24 database or notifying the issuer that the offline PIN has not been unblocked. Similarly, the reversal must be forwarded to an issuer host system (if external to the BASE24 system), if the BASE24 system did not authorize the transaction.

PIN Management Parameters

PIN management authorization processing is controlled by the values of the PIN management parameters on the EMV segments of the CPF and the CAF.

The effect of each possible combination of parameter settings is shown in the following tables. The tables that follow show the action taken under the three conditions listed below:

- Offline PIN tries accumulator has been exceeded (implicit PIN unblock)
- PIN unblock transaction is received (explicit PIN unblock)
- PIN change transaction is received

The following table shows the processing when the offline bad PIN tries threshold has been exceeded.

Offline Bad PIN Tries Exceeded		
CPF Offline Bad PIN Tries Action Field	CAF Send PIN Unblock Field	Action
0	0	Standard processing
0	1	Standard processing
0	2	Standard processing
0	3	Standard processing
1	0	Decline the transaction
1	1	Decline the transaction
1	2	Decline the transaction
1	3	Decline the transaction
2	0	Standard processing
2	1	Standard processing
2	2	Standard processing
2	3	Standard processing
3	0	Standard processing

Offline Bad PIN Tries Exceeded		
CPF Offline Bad PIN Tries Action Field	CAF Send PIN Unblock Field	Action
3	1	Generate PIN unblock script
3	2	Standard processing
3	3	Generate PIN unblock script

The following table shows the processing when a PIN unblock transaction is received.

PIN Unblock Transaction Received			
CPF Offline PIN Management Field	CAF Send PIN Unblock Field	CAF Send PIN Change Field	Action
0	0	0	Decline the transaction
0	0	1	Decline the transaction
0	0	2	Decline the transaction
0	0	3	Decline the transaction
0	1	0	Decline the transaction
0	1	1	Decline the transaction
0	1	2	Decline the transaction
0	1	3	Decline the transaction
0	2	0	Decline the transaction
0	2	1	Decline the transaction
0	2	2	Decline the transaction
0	2	3	Decline the transaction
0	3	0	Decline the transaction
0	3	1	Decline the transaction
0	3	2	Decline the transaction
0	3	3	Decline the transaction

PIN Unblock Transaction Received			
CPF Offline PIN Management Field	CAF Send PIN Unblock Field	CAF Send PIN Change Field	Action
1	0	0	Decline the transaction
1	0	1	Generate PIN change script, do not log new PIN data
1	0	2	Decline the transaction
1	0	3	Generate PIN change script, do not log new PIN data
1	1	0	Decline the transaction
1	1	1	Generate PIN change script, do not log new PIN data
1	1	2	Decline the transaction
1	1	3	Generate PIN change script, do not log new PIN data
1	2	0	Generate PIN unblock script
1	2	1	Generate PIN unblock script
1	2	2	Generate PIN unblock script
1	2	3	Generate PIN unblock script
1	3	0	Generate PIN unblock script
1	3	1	Generate PIN unblock script
1	3	2	Generate PIN unblock script
1	3	3	Generate PIN unblock script
2	0	0	Decline the transaction
2	0	1	Generate PIN change script, do not log new PIN data
2	0	2	Decline the transaction
2	0	3	Generate PIN change script, do not log new PIN data
2	1	0	Decline the transaction

PIN Unblock Transaction Received			
CPF Offline PIN Management Field	CAF Send PIN Unblock Field	CAF Send PIN Change Field	Action
2	1	1	Generate PIN change script, do not log new PIN data
2	1	2	Decline the transaction
2	1	3	Generate PIN change script, do not log new PIN data
2	2	0	Generate PIN unblock script
2	2	1	Generate PIN unblock script
2	2	2	Generate PIN unblock script
2	2	3	Generate PIN unblock script
2	3	0	Generate PIN unblock script
2	3	1	Generate PIN unblock script
2	3	2	Generate PIN unblock script
2	3	3	Generate PIN unblock script

The following table shows the processing when a PIN change transaction is received.

PIN Change Transaction Received			
CPF Offline PIN Management Field	CAF PIN Synchronization Field	CAF Send PIN Change Field	Action
0	0	0	Standard processing
0	0	1	Standard processing
0	0	2	Standard processing
0	0	3	Standard processing
0	1	0	Standard processing
0	1	1	Standard processing
0	1	2	Standard processing

PIN Change Transaction Received			
CPF Offline PIN Management Field	CAF PIN Synchronization Field	CAF Send PIN Change Field	Action
0	1	3	Standard processing
1	0	0	Standard processing
1	0	1	Standard processing
1	0	2	Standard processing
1	0	3	Standard processing
1	1	0	Decline the transaction
1	1	1	Decline the transaction
1	1	2	Generate PIN change script, do not log new PIN data
1	1	3	Generate PIN change script, do not log new PIN data
2	0	0	Standard processing
2	0	1	Standard processing
2	0	2	Standard processing
2	0	3	Standard processing
2	1	0	Decline the transaction
2	1	1	Decline the transaction
2	1	2	Generate PIN change script, log new PIN data
2	1	3	Generate PIN change script, log new PIN data

Considerations

The following paragraphs discuss various offline PIN management considerations to take into account when configuring your BASE24-atm system.

PIN Verification

For PIN change processing, it is assumed that the system that authorizes the online PIN Change transaction (i.e., BASE24 or an issuer host) also generates the PIN Change issuer script command. (Note that the original PIN must be verified before the PIN Change transaction can be authorized.) Similarly, for explicit PIN unblock processing, it is assumed that the system that authorizes the transaction (i.e., BASE24 or an issuer Host system) will also generate the PIN Change or PIN Unblock script command. (Note that the original PIN must be verified before the PIN Change/Unblock script command can be generated.) For implicit PIN unblock processing, however, it is conceivable that the system that authorizes the transaction (i.e., BASE24 or a host) does not generate the PIN Unblock script command. For example, BASE24 could verify the online PIN, but the host could generate the PIN Unblock script command, or vice versa.

Issuer Script Commands

If a PIN Change script command or a PIN Unblock script command is generated by BASE24-atm, then it is the only command contained in the script returned to the EMV card. If multiple script commands are used (e.g., returned from an issuer host), BASE24-atm may not be able to report a script failure accurately.

Offline Bad PIN Tries

BASE24-atm does not check whether the cardholder's offline bad PIN try limit has been exceeded when an explicit PIN Unblock transaction is received. This enables cardholders to reset their offline bad PIN try counter, even when the limit has not been exceeded.

ARQC Verification

In order for BASE24-atm to generate a PIN Change script command, the Authorization Request Cryptogram (ARQC) must have already been verified. Thus, if BASE24-atm is required to authorize PIN Change transactions (and

therefore, generate the PIN Change script command), you must either configure BASE24-atm to verify the ARQC or make sure that the ARQC is verified before the transaction reaches the BASE24-atm system. A similar assumption applies if you require BASE24-atm to authorize explicit PIN Unblock transactions (and therefore, generate a PIN Change or PIN Unblock script command). For implicit PIN unblock processing, however, it is conceivable that the system that verifies the ARQC (e.g., BASE24-atm or a host) does not generate the PIN Unblock script command. For example, BASE24-atm could verify the ARQC, but the host could generate the PIN Unblock script command, or vice versa.

PIN Block Formats

When generating PIN Change issuer script commands, different destination PIN block formats are used, based on the EMV scheme selected for EMV processing. If you select the Visa scheme, the Visa Format Without using Current PIN is used for the destination PIN block. If you select the MasterCard/Europay scheme, the Standard EMV PIN Block is used for the destination PIN block. Refer to the appropriate vendor documentation for detailed information on these PIN block formats.

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5: Device Handler Processing

BASE24-atm Device Handler processes provide an interface between ATMs and the BASE24-atm Authorization process.

This section introduces the functionality provided by device-specific BASE24-atm Device Handler EMV extension modules. The EMV extension modules are additional licensable modules that, when bound with standard BASE24-atm device-specific Device Handler processes, provide the ability to process EMV transactions.

Device Handler EMV Functionality

EMV Device Handler extension modules provide the following functionality:

- Identify EMV request messages
- Validate and store EMV-specific data sent in request messages
- Accept and process card details read from a chip
- Determine whether a MAC is present in an EMV request message and, if present, verify the MAC
- Determine which EMV-specific tokens are required and add them to the STM
- Identify EMV response messages and retrieve the appropriate tokens from the STM
- Include EMV-specific data in response messages
- Determine whether a MAC is required in an EMV response message and, if required, generate the MAC

EMV data is passed between processes in a BASE24-atm system in data tokens. For requests, the EMV modules map data from the request message to tokens and appends these tokens to the STM. The EMV extension modules format the EMV Request Data token, the EMV Status token, and the EMV Discretionary Data token, as needed.

For responses, the EMV extension modules map EMV data from the EMV Response Data token and the EMV Script Data token to the response message sent back to the terminal.

EMV Processing

The Device Handler EMV extension module removes and saves the EMV data from an incoming request message and reformats the message as a device-specific message. The device-specific Device Handler process performs normal processing on the message and then the EMV extension module builds the EMV tokens that are appended to the STM before passing the message to the Authorization process. The EMV extension module performs the following:

- If EMV data is successfully retrieved, this data is formatted and loaded into the appropriate tokens and passed to the Authorization process for further processing.
- If EMV terminal capabilities are not provided in the request message, the module checks the EMV terminal capability settings for the terminal on BASE24-atm Terminal Data files (ATD) screen 21 to determine which EMV tokens to include.

Response messages from the Authorization process are reformatted into the device-specific format using data passed from the Authorization process. If the EMV Response Data token and, optionally, the EMV Script Data token are present, the following processing occurs:

- EMV response data is added to the message and returned to the terminal. The STM can contain the EMV Response Data token, which is used in formatting the response to the terminal.
- The content of the EMV Script Data token, if present, is passed to the device in the appropriate field.
- If the request contained EMV tokens but the response does not contain the EMV Response Data token, no EMV data is included in the response to the terminal.

If the Device Handler process declines the transaction, no EMV data is included in the response to the terminal. Similarly, if the issuer does not return EMV data, none is returned in the response to the device.

Device-Specific Processing

Refer to the following appendixes for device-specific EMV processing and message formats.

- Appendix A contains information for the Diebold 10XX/478X EMV module.
- Appendix B contains information for the NCR 5XXX EMV module.
- Appendix C contains information for the Shared Diebold 912 EMV module.
- Appendix D contains information for the Shared NDC+ EMV module.

Additional device-specific EMV information is not provided in this manual for the Tidel Device Handler and Triton Device Handler. Refer to the documentation provided with the source files for these device handlers.

6: Interchange Interface Processing

This section describes how the following interchange interface processes handle EMV transactions.

- American Express Global Network (AMEX GNS)
- MasterCard (BankNet and MDS)
- Visa

EMV Support in the BASE24 External Message

The BIC ISO Interchange Interface and BASE24 ISO Host Interface processes handle EMV transactions using tokens to carry EMV information internally and placing these tokens in data element S-126 of the BASE24 external message.

The BIC ISO Interchange Interface and BASE24 ISO Host Interface processes use the standard technique for configuring the EMV-specific tokens to be included in external messages.

Enhanced EMV Response Code Mapping

BASE24 maps response codes carried internally with ISO response codes used in the BASE24 external message. Two types of mapping are supported: standard mapping and enhanced EMV mapping.

The difference between standard mapping and enhanced EMV mapping is how EMV-related response codes are handled. Using standard mapping, EMV-related response codes are mapped to more general response codes which can obscure the reason an EMV transaction is declined. Using enhanced EMV mapping, however, unique ISO response codes are used that allow EMV-related response codes to retain their EMV characteristics. For example, an outbound BASE24-atm EMV response code of 084 would map to 05 under standard mapping and U4 under enhanced EMV mapping as shown in the table below. Where the former does not retain the specific reason for the response (due to an ATC check failure), the latter does.

BASE24-atm	Standard Mapping	Enhanced EMV Mapping
084 ATC check failure	05 Do not honor	U4 ATC check failure

For a complete list of BASE24-atm response code mappings, refer to the ***BASE24 External Message Manual***.

The type of mapping used for BASE24-atm and BASE24-pos EMV messages is controlled by the ENHANCED-EMV-RC-MAPPING param in the LCONF. Param values control the type of mapping as follows:

Param Value	Response Code Mapping	
	BASE24-atm	BASE24-pos
0	Standard Mapping	Standard Mapping
1	Enhanced EMV Mapping	Standard Mapping
2	Standard Mapping	Enhanced EMV Mapping
3	Enhanced EMV Mapping	Enhanced EMV Mapping

Token Impacts

The following table describes how token processing is performed when an interchange is involved.

EMV Request Data token	The interchange EMV module adds the EMV Request Data token on receiving inbound EMV requests, and uses the token in formatting outbound EMV-format requests.
EMV Discretionary Data token	The interchange EMV module adds the EMV Discretionary Data token at the same time that it adds the EMV Request Data token. The EMV Discretionary Data token provides information about the capabilities of the terminal. None of this data is required for authorization.
EMV Status token	The interchange EMV module adds the EMV Status token on receiving EMV messages, or magnetic stripe transactions from an EMV-capable terminal.
EMV Response Data token	The interchange EMV module adds the EMV Response Data token on receiving an inbound response with EMV information, and uses the token in formatting outbound EMV-format responses.
EMV Script Data token	The interchange EMV module adds the EMV Script Data token on receiving inbound responses with EMV information, and with the Issuer Script bit set in the EMV Response token. It uses the token in formatting outbound EMV-format responses.
EMV Issuer Script Results token	The interchange EMV module adds the EMV Issuer Script Results token on receiving inbound reversals with EMV information, and uses the token in formatting outbound EMV-format reversals. The token contains information about the results of EMV script processing.

Refer to the *BASE24 Tokens Manual* for configuring the tokens used by interface processes.

AMEX GNS Interchange Interface

The American Express Global Network (AMEX GNS) interchange uses data element P-55, ICC System Related Data, to pass chip card information from the acquirer to the issuer. This information is passed in the ATM Financial Transaction Request (1200) and ATM Financial Transaction Response (1210) messages. The data includes encrypted elements that only the issuer or issuer agent is able to use. The passing of this data provides for Online Mutual Authentication (OMA) as follows:

- The chip is able to authenticate itself to the issuer in the ATM Financial Transaction Request (1200) message.
- The issuer is able to authenticate itself to the chip in the ATM Financial Transaction Response (1210) message. The American Express Global Network interchange does not edit the contents of data element P-55.

The American Express Global Network interchange performs the following message edits:

- If data element P-55 is present in the 1200 ATM Financial Transaction Request message, then positions 1 (Card Data Input Capability) and 7 (Card Data Input Mode Code) of data element 22 (Point of Service Data Code) must be set to the value 5 (Integrated Circuit Card (ICC)).
- If data element P-55 is present in the 1210 ATM Financial Transaction Response message, then it must be present in the corresponding 1200 ATM Financial Transaction Request. Data element P-55 is mandatory in the 1210 ATM Financial Transaction Response message if it is present in the 1200 ATM Financial Transaction Request message.
- The American Express Global Network interchange mandates that in the 1210 ATM Financial Transaction Response messages, data element P-55 must contain the issuer authentication data and issuer script data only if the transaction is approved. Therefore, if the transaction is denied, the issuer authentication data and issuer script data is not included in the 1210 ATM Financial Transaction Response messages. Regardless of whether the transaction is approved or disapproved, the version header and version number must be echoed in the 1210 ATM Financial Transaction Response message.

BASE24 as Acquirer

For BASE24 as an acquirer, the American Express Global Network Interchange ISO Interface process receives BASE24-atm Standard Internal Messages (STMs) with one or more EMV tokens attached. The American Express Global Network EMV extension is called to complete the formatting of the SEM. The extent of this formatting is dependent upon the level of chip support determined by the contents of the transaction message.

For transactions performed at non-chip-capable terminals, no further formatting is required.

For transactions performed at chip-capable terminals where the PAN is not read from the chip, and no chip data is available, the EMV Status token is used to indicate that it is not a chip read transaction in the message sent to the American Express Global Network interchange.

For transactions performed at chip-capable terminals where the PAN is read from the chip, but no chip data is available, the EMV Status token is used to indicate that it is a chip-read transaction in the message sent to the American Express Global Network interchange.

For transactions performed at chip-capable terminals where the PAN is read from the chip and chip data is available, both the EMV Status token and the EMV Request Data token are present in the internal message.

When receiving a response from the American Express Global Network interchange containing data element P-55, the STM to be returned is formatted as a response to the Authorization process. The American Express Global Network EMV extension is called to complete the formatting of the STM. If the original request is contained in the EMV Request Data token, then this formatting can include adding the EMV Response token and the EMV Script Data token prior to sending the formatted STM.

BASE24 as Issuer

For BASE24 as issuer, the American Express Global Network Interchange ISO Interface process receives external request messages. The American Express Global Network EMV extension is called to complete the formatting. The extent of this formatting is dependent upon the level of chip support determined by the contents of the transaction message.

If the terminal is EMV capable, but the chip data elements are not available, the American Express Global Network EMV extension appends only the EMV Status token to the STM to be forwarded to the Authorization process. This token differentiates between transactions where the PAN is read from a chip and those where it is not.

If the terminal is EMV capable, the card data is read from the chip, and the chip data elements are available, the American Express Global Network EMV extension appends both the EMV Status token and the EMV Request Data token to the STM to be forwarded to the Authorization process. The STM, with the tokens, is sent to the Authorization process. The Authorization process validates the request cryptogram, generates the response cryptogram, and verifies the application transaction counter.

When a response is sent to the American Express Global Network interchange, the American Express Global Network EMV extension is called to complete the formatting. If data element P-55 is present in the original request from the American Express Global Network interchange, this formatting includes extracting data from the EMV Response token and (if present) the EMV Script Data token. If the card authentication is successful, (issuer authentication data is present in the EMV Response token), this data is mapped to the corresponding field in data element P-55 in the response message sent to the American Express Global Network interchange. If card authentication is not successful, data element P-55 is sent in the response message to the American Express Global Network interchange with only the Version Header and Version Number subfields populated.

American Express Global Network Interchange Interface EMV External Message

The American Express Global Network interchange uses data element P-55, Integrated Circuit Card System Related Data, to pass chip information from the acquirer to the issuer. This data element contains many subelements and is supported only for transactions where chip data is successfully read from the chip at a chip-capable terminal.

Most of the information in the P-55 data element is also contained in the EMV-specific tokens. Although the P-55 data element may be present in the external message, it is not logged to the Interchange Log File (ILF).

The format of EMV-specific tokens is described in the ***BASE24 Logical Network Configuration Manual***.

EMV Request Data Token

The following table shows the mapping between fields in the EMV Request Data token and the EMV tag values in data element P-55 of the request message. All data elements in this token are required.

Token Field	EMV Tag Values in Data Element P-55 of the Request Message
CRYPTO-INFO-DATA	9F27
TVR	95
ARQC	9F26
AMT-AUTH	9F02
AMT-OTHER	9F03
AIP	82
ATC	9F36
TERM-CNTRY-CDE	9F1A
TRAN-CRNCY-CDE	5F2A
TRAN-DAT	9A
TRAN-TYPE	9C
UNPREDICT-NUM	9F37
ISS-APPL-DATA	9F10

EMV Discretionary Data Token

The following table shows the mapping between fields in the EMV Discretionary Data token and the EMV tag values in data element P-55 of the request message.

Token Field	EMV Tag Values in Data Element P-55 of the Request Message
TERM-SERL-NUM	9F1E

Token Field	EMV Tag Values in Data Element P-55 of the Request Message
EMV-TERM-CAP	9F33
EMV-TERM-TYPE	9F35
APPL-VER-NUM	9F09
CVM-RSLTS	9F34
DF-NAME	84

EMV Status Token

The following table shows the mapping between fields in the EMV Status token and the interchange/interface message.

Token Field	Value
PT-SRV-ENTRY-MDE	Set to the value 0yx, where y is the value of field 22 subfield 7 (Card Data Input Mode) and x is mapped from field 22 subfield 12 (PIN Capture Capability).
TERM-ENTRY-CAP	Set to the value 5 (EMV-capable terminal) if field 22 subfield 1 (Card Data Input Capability) equals the value 5; otherwise set to the value 0.
LAST-EMV-STAT	Set to the value 1 if either of the following exists: - Field 22 subfield 7 (Card Data Input Mode) equals 5. - Field 22 subfield 7 (Card Data Input Mode) equals 2, field 35 (Track 2 Data) is present, and the first digit of the Service Code (position 21 in field 35, excluding the length subfield) equals 2. Otherwise, set to the value 0.
DATA-SUSPECT	Set to the value 0.
APPL-PAN-SEQ-NUM	Set to the value of field 55 subfield 15 (Application PAN Sequence Number), if present; otherwise set to spaces.

Token Field	Value
DEV-INFO	Set to spaces.
RSN-ONL-CDE	Set to spaces.
ARQC-VRFY	Set to the value 0.

EMV Response Data Token

The following table shows the mapping between fields in the EMV Response Data token and the EMV tag values in data element P-55 of the response message.

Token Field	EMV Tag Values in Data Element P-55 of the Response Message
ISS-AUTH-DATA	91

EMV Script Data Token

The following table shows the mapping between fields in the EMV Script Data token and the EMV tag values in data element P-55 of the response message.

Token Field	EMV Tag Values in Data Element P-55 of the Response Message
ISS-SCRIPT-DATA	71 or 72

MasterCard Interchange Interfaces (BankNet and MDS)

The BankNet EMV module supports BASE24-atm EMV transactions when bound with the BankNet Interchange Interface process. The MDS EMV module supports BASE24-atm EMV transactions when bound with the BASE24 MDS Interchange Interface process. Both of these modules are additional licensable modules and comply with the message formats and rules developed by MasterCard.

The main function of the BASE24 MasterCard Debit Switch (MDS) EMV Interchange Interface module and the BankNet EMV Interchange Interface module is the formatting of internal message token data to and from EMV-specific external message fields and subfields. They also handle the initialization, compression, and expansion of the EMV-related fields in the external message.

In this section, the BankNet interchange and the MDS interchange are collectively referred to as the MasterCard interchanges.

BASE24 as Acquirer

For BASE24 as an acquirer, the MasterCard Interchange Interface processes receive BASE24-atm Standard Internal Messages (STMs) with one or more EMV tokens attached. Standard BASE24 external message formatting is performed and then the MasterCard EMV module completes the formatting. The extent of this formatting is dependent on the level of chip support determined by the contents of the transaction message:

- For transactions performed at terminals that are not chip-capable, no further formatting is required.
- For transactions performed at chip-capable terminals where the PAN is not read from the chip and no chip data is available, the EMV Status token is used to set data element P-22 (Point of Service Entry Mode) in the message sent to the MasterCard interchanges to indicate that it is not a chip-read transaction.
- For transactions performed at chip-capable terminals where the PAN is read from the chip, but no chip data is available, the EMV Status token is used to set data element P-22 in the message sent to the MasterCard interchanges to indicate that it is a chip-read transaction.
- For transactions performed at chip-capable terminals where the PAN is read from the chip and chip data is available, both the EMV Status token and the EMV Request Data token are present in the internal message. The EMV

Discretionary Data token may also be present. In addition to the processing described above, the data in the EMV Request Data token (and optionally from the EMV Discretionary Data token) are moved to data element P-55 (ICC System-Related Data) in the message sent to the MasterCard interchanges.

When receiving a response from the MasterCard interchanges containing EMV data, existing logic handles the initial formatting of the STM. The EMV module then completes the formatting of the STM. If the original request contained the EMV Request Data token, the formatting may include adding the EMV Response Data token and possibly the EMV Script Data token prior to sending the formatted STM.

BASE24 as Issuer

For BASE24 as an issuer, the BankNet or MDS Interchange Interface process receives external request messages from the MasterCard interchanges. Standard BASE24 internal message formatting is performed, and then the EMV module completes the formatting, adding the appropriate tokens. The extent of this formatting is dependent upon the level of chip support determined by the contents of the transaction message:

- For transactions not containing data element P-55, no specific EMV processing is performed.
- If the terminal was EMV-capable, but the chip data elements were not available (i.e., data element P-55 is not present), the EMV module appends only the EMV Status token to the STM. The information in this token differentiates between transactions where the PAN is read from a chip and those where it is not.
- If the terminal was EMV-capable, the card data was read from the chip, and the chip data elements are available, the EMV module appends both the EMV Status token and the EMV Request Data token (and optionally the EMV Discretionary Data token) to the STM before sending it to the Authorization process.

When the BASE24-atm Authorization process sends a response to the BankNet or MDS Interchange Interface process, existing logic formats the external response and then the EMV module completes the formatting. If data element P-55 was present in the original request from the MasterCard interchanges, the EMV module extracts data from the EMV Response Data token and (if present) the EMV Script Data token to include in data element P-55 of the response message.

BankNet and MDS EMV External Message

The MasterCard interchanges use data element P-55, Integrated Circuit Card System Related Data, to pass chip information from the acquirer to the issuer. This data element contains many subelements and is supported only for transactions where chip data is successfully read from the chip at a chip-capable terminal.

The MDS interchange passes this information only in Financial Transaction Request (0200), Financial Transaction Request Response (0210), Financial Transaction Advice (0220), and Financial Transaction Advice Response (0230) messages. The BankNet interchange passes this information only in Authorization Request (0100), Authorization Request Response (0110), Authorization Advice (0120), and Authorization Advice Response (0130) messages. Internally, BASE24 handles messages sent from the BankNet interchange as 0200, 0210, 0220, and 0230 messages.

Most of the information in data element P-55 is also contained in the EMV-specific tokens. Although data element P-55 may be present in the external message, it is not logged to the Interchange Log File (ILF).

The format of EMV-specific tokens is described in the *BASE24 Tokens Manual*.

EMV Request Data Token

The following table shows the mapping between fields in the EMV Request Data token and the EMV tag values in data element P-55 of the request message. All data elements in this token are required.

Token Field	EMV Tag Values in Data Element P-55 of the Request Message
CRYPTO-INFO-DATA	9F27
TVR	95
ARQC	9F26
AMT-AUTH	9F02
AMT-OTHER	9F03
AIP	82
ATC	9F36

Token Field	EMV Tag Values in Data Element P-55 of the Request Message
TERM-CNTRY-CDE	9F1A
TRAN-CRNCY-CDE	5F2A
TRAN-DAT	9A
TRAN-TYPE	9C
UNPREDICT-NUM	9F37
ISS-APPL-DATA	9F10

EMV Discretionary Data Token

The following table shows the mapping between fields in the EMV Discretionary Data token and the EMV tag values in data element P-55 of the request message.

Token Field	EMV Tag Values in Data Element P-55 of the Request Message
TERM-SERL-NUM	9F1E
EMV-TERM-CAP	9F33
EMV-TERM-TYPE	9F35
APPL-VER-NUM	9F09
CVM-RSLTS	9F34
DF-NAME	84

EMV Status Token

The following table shows the mapping between fields in the EMV Status token and the interchange/interface message.

Token Field	Value
PT-SRV-ENTRY-MDE	Set to the value 02x if field 22 (Point-of-Service Entry Mode) equals the value 80x or 90x; otherwise set to the value of field 22.
TERM-ENTRY-CAP	Set to the value 5 (EMV-capable terminal) if field 61 subfield 11 (POS Card Data Terminal Input Capability Indicator) equals the value 5, 8, or 9.
LAST-EMV-STAT	Set to the value 1 if field 22 (Point-of-Service Entry Mode) equals the value 05x or 80x; otherwise set to the value 0.
DATA-SUSPECT	Set to the value 0.
APPL-PAN-SEQ-NUM	Set to the value of field 23 (Card Sequence Number), if present; otherwise set to spaces.
DEV-INFO	Set to spaces.
RSN-ONL-CDE	Set to spaces.
ARQC-VRFY	Set to the value 0.

EMV Response Data Token

The following table shows the mapping between fields in the EMV Response Data token and the EMV tag values in data element P-55 of the response message.

Token Field	EMV Tag Values in Data Element P-55 of the Response Message
ISS-AUTH-DATA	91

EMV Script Data Token

The following table shows the mapping between fields in the EMV Script Data token and the EMV tag values in data element P-55 of the response message.

Token Field	EMV Tag Values in Data Element P-55 of the Response Message
ISS-SCRIPT-DATA	71 or 72

Visa Interchange Interface

The BASE24 Visa EMV product module is an additional licensable module which, when bound with the BASE24 VisaNet ISO Interchange Interface module (or VisaNet Interface), complies with the message formats and rules developed by Visa for the Visa Smart Debit/Credit (VSDC) Service. It offers support for early and full option issuer processing, or general EMV acquirer processing, as described by Visa, and supports BASE24-atm EMV processing.

The main function of the BASE24 Visa EMV module is the formatting of internal message token data to and from EMV-specific external message fields and subfields. It also handles the initialization, compression, and expansion of the EMV-related fields in the external message.

The Visa Interchange Interface also supports the following PIN management transactions:

700000 = PIN Change processing code
720000 = PIN Unblock processing code

BASE24 as Acquirer

For EMV acquirers, the following levels of support are provided:

Full Support	A number of new EMV data elements are included in the message to Visa. The BASE24 Visa EMV module assumes this mode if the EMV Request Data token is appended to the STM.
Easy Entry	The EMV data elements are not included in the message to Visa. The BASE24 Visa EMV module assumes this mode if the EMV Request Data token is not appended to the STM, but the card details are read from the chip.

For BASE24 as an acquirer, the BASE24 VisaNet Interchange Interface process receives standard internal request messages with EMV Request Data, EMV Discretionary Data, and EMV Status tokens attached. Standard external message formatting is performed and then the BASE24 Visa EMV module completes the formatting. If only the EMV Status token is present, the transaction is deemed to be either an Easy Entry transaction or a magnetic stripe transaction from an EMV-capable terminal, and the internal message is formatted without the EMV data elements.

When receiving a response from Visa, the Interchange Interface process formats the STM to return to the Authorization process. If the response contains EMV data, the Visa EMV module formats the EMV Response Data token, and if applicable, the EMV Script Data token prior to sending the formatted STM.

BASE24 as Issuer

For issuer processing, the two levels of support offered by Visa provide varying degrees of impact on the current message structure:

- | | |
|--------------|---|
| Early Option | If issuers request this option, Visa removes all EMV-specific fields from inbound request messages. Issuers receive new values in existing fields. The BASE24 Visa EMV module uses these fields to format the EMV Status token appended to the STM for forwarding to the Authorization process. The BASE24 Authorization process does not take any specific action on this token data. This form of processing is known as pass-through. |
| Full Option | Issuers receive a number of new EMV data elements. The new data items are added to the EMV Status and EMV Request Data tokens, which contain all the information needed for the Authorization process or back-end host to perform card authentication processing. The EMV Response Data and EMV Script Data tokens returned from the Authorization process are used in formatting the EMV data elements in the response to Visa. Full Option issuers may use the VisaNet Card and Issuer Authentication Services. |

For card authentication by Visa, the BASE24 Visa EMV module sets a flag in the EMV Status token to indicate a prevalidated request cryptogram. This causes the Authorization process to suppress the card and issuer authentication processing. The processing assumes that the VisaNet Issuer Authentication Service is used, and that this service returns issuer authentication data in the response message to the acquirer.

Where BASE24 is the issuer, the BASE24 VisaNet Interchange Interface process receives external request messages with formats dependent on the level of issuer support:

Early Option	External request message processing moves the EMV-specific data into an EMV Status token.
Full Option	The STM request is formatted and the Visa EMV module completes the formatting with EMV changes, adding the EMV Status and EMV Request Data tokens. The STM, with tokens appended, is sent to the Authorization process. When the Authorization process responds, the response to Visa is formatted. The Visa EMV module completes the formatting, extracting the data required to format the EMV data elements from the EMV Response and (if present) EMV Script Data tokens.

For fallback transactions the EMV Status token is added to the STM.

On transactions acquired from VisaNet, a check is made on how the card details were entered for the transaction. If the card details are read from the chip, the Point of Service Entry Mode field in the EMV Status token must be set to 05x, where x indicates the PIN entry capability of the terminal.

VisaNet EMV External Message

The EMV external message can use data element P-55 or an extension to the standard VIP (BASE I/SMS) messages that incorporates a tertiary (or third) bit map and associated fields. Both formats are provided in this discussion.

The USE-FIELD-55-FOR-EMV-DATA param in the Logical Network Configuration File (LCONF) controls the format of outbound messages. The VisaNet Interchange Interface process uses the tertiary bit map format if this param is missing or invalid. For inbound messages, the VisaNet Interchange Interface process can accept messages containing either format or a combination of formats.

The following tables show the mapping between fields in the EMV data tokens and fields in the VisaNet external message. Note that token fields not listed have no comparable field in the VisaNet message.

EMV Request Data Token (Data Element P-55)

The following table shows the mapping between fields in the EMV Request Data token and the EMV tag values in data element P-55. All data elements in this token are required.

Token Field	EMV Tag Values in Data Element P-55
CRYPTO-INFO-DATA	9F27
TVR	95
ARQC	9F26
AMT-AUTH	9F02
AMT-OTHER	9F03
AIP	82
ATC	9F36
TERM-CNTRY-CDE	9F1A
TRAN-CRNCY-CDE	5F2A
TRAN-DAT	9A
TRAN-TYPE	9C
UNPREDICT-NUM	9F37
ISS-APPL-DATA	9F10

EMV Discretionary Data Token (Data Element P-55)

The following table shows the mapping between fields in the EMV Discretionary Data token and the EMV tag values in data element P-55.

Token Field	EMV Tag Values in Data Element P-55
TERM-SERL-NUM	9F1E
EMV-TERM-CAP	9F33

Token Field	EMV Tag Values in Data Element P-55
EMV-TERM-TYPE	9F35
APPL-VER-NUM	9F09
CVM-RSLTS	9F34
DF-NAME	84

EMV Status Token (Data Element P-55)

The following table shows the mapping between fields in the EMV Status token and the interchange/interface message using data element P-55.

Token Field	External Message Field	Visa Field Name
PT-SRV-ENTRY-MDE	Set to the value 05x if field P-22 is equal to the value 95x, set to the value 02x if field P-22 is equal to the value 90x; otherwise set to the value of field P-22.	Point of Service Entry Mode (field P-22)
TERM-ENTRY-CAP	P-60, subfield 2	Terminal Entry Capability
LAST-EMV-STAT	P-60, subfield 3	Chip Condition Code
DATA-SUSPECT	P-60, subfield 7	Chip Authentication Reliability Indicator
APPL-PAN-SEQ-NUM	P-23	Card Sequence Number
ARQC-VRFY	P-44, subfield 8	Card Authentication Results Code

EMV Response Data Token (Data Element P-55)

The following table shows the mapping between fields in the EMV Response Data token and the EMV tag values in data element P-55.

Token Field	EMV Tag Values in Data Element P-55
ISS-AUTH-DATA	91

EMV Script Data Token (Data Element P-55)

The following table shows the mapping between fields in the EMV Script Data token and the EMV tag values in data element P-55.

Token Field	EMV Tag Values in Data Element P-55
ISS-SCRIPT-DATA	71 or 72

EMV Issuer Script Results Token (Data Element P-55)

The following table shows the mapping between fields in the EMV Issuer Script Results token and the EMV tag values in data element P-55.

Token Field	EMV Tag Values in Data Element P-55
ISS-SCRIPT-RSLTS-TKN	9F5B

EMV Request Data Token (Tertiary Bit Map)

The following table shows the mapping between fields in the EMV Request Data token and the tertiary bit map of the VisaNet message.

Token Field	Visa Field	Visa Field Name
TVR	T-131	Terminal Verification Results
ARQC	T-136	Cryptogram
AMT-AUTH	T-147	Cryptogram Amount
AMT-OTHER	T-149	Cryptogram Cashback Amount
AIP	T-138	Application Interchange Profile
ATC	T-137	Application Transaction Counter
TERM-CNTRY-CDE	T-145	Terminal Country Code
TRAN-CRNCY-CDE	T-148	Cryptogram Currency Code
TRAN-DAT	T-146	Terminal Transaction Date
TRAN-TYPE	T-144	Cryptogram Transaction Type
UNPREDICT-NUM	T-132	Unpredictable Number
LGTH	T-134	Length of Field T-134
ISS-APPL-DATA	T-134	Issuer Application Data

EMV Discretionary Data Token (Tertiary Bit Map)

The following table shows the mapping between fields in the EMV Discretionary Data token and the tertiary bit map of the VisaNet message.

Token Field	Visa Field	Visa Field Name
TERM-SERL-NUM	T-133	Terminal Serial Number
EMV-TERM-CAP	T-130	Terminal Capability Profile

EMV Status Token (Tertiary Bit Map)

The following table shows the mapping between fields in the EMV Status token and the tertiary bit map of the VisaNet message.

Token Field	External Message Field	Visa Field Name
PT-SRV-ENTRY-MDE	Set to the value 05x if field P-22 is equal to the value 95x, set to the value 02x if field P-22 is equal to the value 90x; otherwise set to the value of field P-22.	Point of Service Entry Mode (field P-22)
TERM-ENTRY-CAP	P-60, subfield 2	Terminal Entry Capability
LAST-EMV-STAT	P-60, subfield 3	Chip Condition Code
DATA-SUSPECT	P-60, subfield 7	Chip Authentication Reliability Indicator
APPL-PAN-SEQ-NUM	P-23	Card Sequence Number
ARQC-VRFY	P-44, subfield 8	Card Authentication Results Code

EMV Response Data Token (Tertiary Bit Map)

The following table shows the mapping between fields in the EMV Response Data token and the tertiary bit map of the VisaNet message. The tertiary bit used for this data depends on whether BASE24 is the issuer or the acquirer.

Token Field	Visa Field		Visa Field Name
	BASE24 as Issuer	BASE24 as Acquirer	
ARPC	T-139, subfield 1	T-140, subfield 1	Authorization Response Cryptogram (ARPC)

Token Field	Visa Field		Visa Field Name
	BASE24 as Issuer	BASE24 as Acquirer	
ISS-RESP-CDE	T-139, subfield 2	T-140, subfield 2	ARPC Response Code

EMV Script Data Token (Tertiary Bit Map)

The following table shows the mapping between fields in the EMV Request Data token and the tertiary bit map of the VisaNet message.

Token Field	Visa Field	Visa Field Name
ISS-SCRIPT-DATA	T-142	Issuer Script

EMV Issuer Script Results Token (Tertiary Bit Map)

The following table shows the mapping between fields in the EMV Issuer Script Results token and the tertiary bit map of the VisaNet message.

Token Field	Visa Field	Visa Field Name
ISS-SCRIPT-RSLTS-TKN	T-143	Issuer Script Results

ACI Worldwide, Inc.

A: Diebold 10XX/478X Device Support

The Diebold 10XX/478X Device Handler is an existing BASE24-atm process that provides the interface between Diebold 10XX/478X ATMs and BASE24. The Diebold 10XX/478X EMV module is an additional licensable module which, when bound with the Diebold 10XX/478X Device Handler process, provides EMV support for Diebold 10XX/478X ATMs or ATMs that emulate the Diebold 10XX/478X message. There is also a Shared Diebold 912 EMV module that can be bound with the Diebold 10XX/478X Device Handler process for ATMs that use Wincor Nixdorf's D912 message formats. A Diebold EMV module and a Shared D912 EMV module can be bound to the same Diebold 10XX/478X Device Handler process.

The Diebold EMV module scans request messages for EMV data. If EMV data is found in the message, the EMV module processes this data (i.e., checking its validity, adding EMV tokens, etc.) before sending the message to authorization.

The Diebold EMV module also handles any EMV data received from authorization and formats the appropriate message sent to the ATM. It also handles the printing of the application identifier and the application PAN sequence number on a receipt.

Processing of EMV configuration data load messages is handled by the EMV module. The EMV module reads the EMV application and EMV scheme data records from the configuration file (CNFGIX) and formats the message sent to the device.

The processing of EMV data in solicited and unsolicited status messages is done by the EMV module. In addition it also processes any error status codes generated by the EMV Smart Card Application.

Any special reversal processing required because of a failure from the device's EMV Smart Card Application is handled by the EMV module.

This appendix describes configuration screens required to support BASE24-atm EMV transactions. For general information about DCT functions, see the ***BASE24 Device Control Manual***. For general information about the CNFGIX Configuration File, see the ***BASE24-atm Diebold 10XX/478X Device Support Manual***.

BASE24-atm Diebold Entry Programs

The Diebold entry programs access the Diebold Configuration File (CNFGIX) that controls Diebold ATM display screens, states, timers, financial institution table (FIT) data, and printer configuration. All of the data in the CNFGIX can be modified through the entry programs.

The following Diebold entry programs are supported.

ENTIBOLD — Supports all Diebold states and cardholder screens and the FIT records, miscellaneous data, rear balance functions, consumer and statement printer configuration, and other information for Diebold and Wincor ATMs. This Diebold entry program does not support Wincor-specific states or tables. The program supports operation codes 070–078 for state SC (smart card). New Diebold-specific enhancements should be added to this program.

ENTIBLD2 — Supports all Diebold states, Wincor EMV state e, Wincor EMV AID table, cardholder screens, FIT records, miscellaneous data, timers, and printer configuration. This program supports operation codes 010, 011, 095, 097, and 099 for state SC (smart card). This program is provided for backwards compatibility and should not be updated for new enhancements.

ENTWBOLD — Supports all Wincor states, Wincor tables and cardholder screens. This program supports Wincor-specific screens, states, and tables only. It does not support Diebold-specific states or tables. Any new Wincor-specific enhancements should be added to this program.

The Diebold entry programs provide the mechanism to configure Diebold ATMs or Wincor ATMs using the BASE24-atm product. For comprehensive documentation regarding the Diebold or Wincor ATMs, consult the available vendor literature.

The Diebold entry programs allow you to define the following attributes:

- ATM display screens
- State tables
- Miscellaneous data
- Financial institution table
- Rear balance function definition data
- Consumer printer-configured text definition

- Statement printer-configured text definition
- Merchant Banking Center buttons and amounts
- EMV application data
- EMV scheme data
- Wincor EMV AID table

Note that the ENTWBOLD program is used to configure Wincor ATM display screens, Wincor State screens, and the Wincor EMV AID table. All other functions for Wincor devices must be configured through the ENTIBOLD program.

Refer to section 4 of the ***BASE24-atm Diebold 10XX/478X Device Support Manual*** for ENTIBOLD, ENTIBLD2, and ENTWBOLD screens and field descriptions.

Diebold DCT

The Diebold 10XX/478X DCT requester provides the interface between the main DCT and the DCT for Diebold 10XX/478X ATMs. The Diebold 10XX/478X DCT screen and requester support a load option of LOAD EMV DATA. The LOAD EMV DATA option enables an operator to download all EMV application and EMV scheme data to an ATM.

Device Configuration

The ATM must be configured for standard Diebold EMV Smart Card processing. Applications (i.e., application identifiers) and EMV Chip Card Schemes can be configured at the ATM or they can be configured using the Diebold Entry Program and downloaded to the ATM. Any EMV data not configured at the ATM must be configured using the Diebold Entry Program and downloaded to the ATM in order to support EMV. Configuration changes that are downloaded override any existing configuration data.

Refer to section 4 of the ***BASE24-atm Diebold 10XX/478X Device Support Manual*** for ENTIBOLD, ENTIBLD2, and ENTWBOLD screens and field descriptions.

Note: Diebold does not support the clearing of entries from the applications table. Therefore, if new applications are downloaded to the ATM, they are added to the existing list. Once an application has been added to the table, it cannot be removed. Refer to Diebold's ***EMV User's Guide*** for more information regarding terminal configuration.

Extended Smart Card State Screen

The Extended Smart Card State (SC_b) gives the network programmer the ability to provide Smart Card State based-operations in the state flow of the terminal. The Enhanced Smart Card State is variable in length and is defined per operation. The basic requirements of an Enhanced Smart Card State are that it contains the State Number, Group, State Type, Operation, Good Operation Next State, Invalid Operation Next State, and Device Failure Next State. Other fields can be defined for the Enhanced Smart Card State on a per operation basis.

The Extended Smart Card State Screen is shown below followed by descriptions of its fields.

```

Extended Smart Card State

State Type ____ State Number ____ Group ____ Unit Number 001

Operation ____

Operation must be one of:
    010 -- Authorization Request Cryptogram
    011 -- ICC Transaction Termination
    095 - Update Buffer with ICC Data
    097 - Select Application/Initiate Application
    099 - ICC Session Initialization

=====
F1= ENTER OPERATION TYPE          F10=PRINT          F16=EXIT
  
```

State Type — The state type associated with the value in the State Number field. The BASE24-atm product supports state types that Diebold has defined. When a state type defined by Diebold is entered in this field, the necessary fields for that state type are listed on the screen. You can define additional state types, but the

Diebold entry program does not check the data entered in those states to ensure that they meet the prescribed parameters. These additional state types are also not listed on the cross-reference listings of states and screens, and states and states.

Field Length: 3 alphanumeric characters, followed by a system-protected text description
Required Field: Yes
Default Value: A

Refer to to section 4 of the ***BASE24-atm Diebold 10XX/478X Device Support Manual*** for the values defined by Diebold.

State Number — A number that is associated with the desired state. Valid values are 000 through 254, 256 through ZZZ.

Field Length: 3 alphanumeric characters
Required Field: Yes
Default Value: 000

Group — The group number used to identify a particular load group, such as a collection of states, screens, and so on, that is downloaded to the ATM. Each group provides certain functionality, such as Group 0 is the basic group, Group 5 provides Fast Cash support, and so on. The group number of each ATM is assigned in the TERMINAL GROUP field of the ATD. Refer to the topic “InterBold i Series/MDS Display Screen” for a complete list of groups.

Field Length: 4 numeric characters
Required Field: Yes
Default Value: 0000

Unit Number — A Diebold system-filled field, always 001.

Field Length: System protected
Required Field: N/A
Default Value: N/A

Operation — The operation to be performed.

Valid values are as follows:

010 = Authorization Request Cryptogram
011 = ICC Transaction Termination
095 = Update Buffer With ICC Data
097 = Select Application/Initiate Application
099 = ICC Session Initialization

Field Length: 3 numeric characters

Required Field: Yes

Default Value: 000

Extended Smart Card State – Authorization Request Cryptogram Operation Screen

The Authorization Request Cryptogram module retrieves the transaction type and dollar amount flag from the buffer specified in the Transaction Type Buffer ID. Processing Restrictions, CVM List Processing, Terminal Risk Management, and Terminal Action Analysis are performed, and a request for an ARQC is issued to the ICC. When the Smart Card Buffer ID is not 000, the Authorization Request Cryptogram along with supporting data are placed in the Smart Card Buffer to be sent to the Terminal Driver in the next Transaction Request message.

The Extended Smart Card State – Authorization Request Cryptogram Operation screen is shown below.

```
Extended Smart Card State ---- Authorization Request Cryptogram Operation
State Type ____      State Number ____      Group ____      Unit Number 001
Operation 010      (Authorization Request Cryptogram)
```

```
Good Operation Next State..... ____
Invalid Operation Next State..... ____
Device Failure Next State..... ____
Appl. Authentication Cryptogram Next State.. ____
Transaction Type Buffer ID..... ____
Smart Card Buffer ID..... ____
```

```
=====
F1= CHANGE STATE TYPE      F2=READ      F3=ADD      F4=DELETE
F5=UPDATE      F9=NEXT      F10=PRINT      F16=EXIT
```

Operation 010 (Authorization Request Cryptogram)

The Authorization Request Cryptogram operation begins the authorization process. Refer to the Diebold vendor documentation for descriptions of the fields in the highlighted area on this screen.

Extended Smart Card State – ICC Transaction Termination Operation Screen

The ICC Transaction Termination module issues a GENERATE AC request to the ICC for an Application Authentication Cryptogram (AAC) to complete the transaction. This operation is designed to complete a transaction with the ICC when a timeout occurs while waiting for a response from the host to an ICC Transaction.

The Extended Smart Card State – ICC Transaction Termination Operation screen is shown below.

```

Extended Smart Card State ---- ICC Transaction Termination Operation

State Type ____      State Number ____      Group ____      Unit Number 001

Operation 011      (ICC Transaction Termination)

  Good Operation Next State..... ____
  Invalid Operation Next State..... ____
  Device Failure Next State..... ____
  Smart Card Buffer ID..... ____

=====
F1= CHANGE STATE TYPE      F2=READ      F3=ADD      F4=DELETE
F5=UPDATE      F9=NEXT      F10=PRINT      F16=EXIT
  
```

Operation 011 (ICC Transaction Termination)

The ICC Transaction Termination operation allows the transaction to complete when a timeout from the host occurs. Refer to the Diebold vendor documentation for descriptions of the fields in the highlighted area on this screen.

Extended Smart Card State – Update Buffer with ICC Data Operation Screen

The Update Buffer with ICC Data module places the ICC Data specified by the ICC Data field in the TCS buffer specified by the Update Buffer ID.

The Extended Smart Card State – Update Buffer with ICC Data Operation screen is shown below.

Extended Smart Card State ---- Update Buffer with ICC Data Operation

State Type ____ State Number ____ Group ____ Unit Number 001

Operation 095 (Update Buffer with ICC Data)

```
Good Operation Next State ..... ____
Invalid Operation Next State ..... ____
Device Failure Next State ..... ____
ICC Data ..... ____
Update Buffer ID ..... ____
Include Sentinels ..... ____
```

=====

F1= CHANGE STATE TYPE	F2=READ	F3=ADD	F4=DELETE
F5=UPDATE	F9=NEXT	F10=PRINT	F16=EXIT

Operation 095 (Update Buffer with ICC Data)

The Update Buffer with ICC Data operation provides for updating the buffer with current ICC data. Refer to the Diebold vendor documentation for descriptions of the fields in the highlighted area on this screen.

Extended Smart Card State – Select Application/Initiate Application Operation Screen

The Select Application Module attempts to select an application from an internally-held list of supported applications using the Application Identifier (AID). A list of applications, supported mutually by the terminal and the Integrated Circuit Card (ICC), is determined. The application with the highest priority is selected. In the event two or more applications exist with the highest priority, the first application in the list is selected. When the Good Operation Next State has a value of 255, the next state to be performed is the Good Operation Next State downloaded to the terminal during terminal initialization. The Good Operation Next State is downloaded using the Modified 912 Write command with the A modifier.

The Extended Card State – Select Application/Initiate Application Operation screen is shown below.

```

Extended Smart Card State ---- Select Application/Initiate Application Operation
State Type ____ State Number ____ Group ____ Unit Number 001
Operation 097 (Select Application/Initiate Application)

  Good Operation Next State ..... ____
  Invalid Operation Next State ..... ____
  Device Failure Next State ..... ____
  No Application Supported Next State .... ____
  Easy Entry Next State ..... ____
  Sub Operation ..... ____

=====
F1= CHANGE STATE TYPE  F2=READ          F3=ADD          F4=DELETE
F5=UPDATE              F9=NEXT        F10=PRINT       F16=EXIT
  
```

Operation 097 (Select Application/Initiate Application)

The Select Application/Initiate Application operation selects an application from a list of prioritized applications. Refer to the Diebold vendor documentation for descriptions of the fields in the highlighted area on this screen.

Extended Smart Card State – ICC Session Initialization Operation Screen

The ICC Session Initialization module initializes the connection to the chip. If the chip is not connected, the Smart Card Reader attempts to move the Smart Card into position and connect the chip. After the Smart Card Reader connects to the chip, it applies power and resets the chip.

The Extended Smart Card State – ICC Session Initialization Operation screen is shown below.

```
Extended Smart Card State ---- ICC Session Initialization Operation

State Type ____ State Number ____ Group ____ Unit Number 001

Operation 099 (ICC Session Initialization)

  Good Operation Next State ..... ____
  Invalid Operation Next State ..... ____
  Device Failure Next State ..... ____
  Invalid Card Next State ..... ____

=====
F1= CHANGE STATE TYPE  F2=READ          F3=ADD          F4=DELETE
F5=UPDATE              F9=NEXT          F10=PRINT       F16=EXIT
```

Operation 099 ICC Session Initialization)

The ICC Session Initialization operation is used to power-up the chip on the Smart Card. Refer to the Diebold vendor documentation for descriptions of the fields in the highlighted area on this screen.

Interbold State Table Screen

The InterBold State Table screen is used to reconfigure state tables and conditions that are associated with various ATM transaction states and groups. The Interbold State Table screen is shown below.

INTERBOLD STATE TABLE

Type ___ Service Code Check Extended State
 State Number ___ Group ___

Network Failure Next State ___
Service Code 0 Next State ___
Service Code 1 Next State ___
Service Code 2 Next State ___
Service Code 3 Next State ___
Service Code 4 Next State ___
Service Code 5 Next State ___
Service Code 6 Next State ___
Service Code 7 Next State ___
Service Code 8 Next State ___
Service Code 9 Next State ___
No Service Code Next State ___
Track 2 Data Buffer ID ___
Sub-Operation ___

=====

F1= CHANGE STATE TYPE	F2=READ	F3=ADD	F4=DELETE
F5=UPDATE	F9=NEXT	F10=PRINT	F16=EXIT

Transaction processing at the ATM is controlled by the configuration of state tables. Refer to the Diebold vendor documentation for descriptions of the fields in the highlighted area on this screen.

EMV Application Data Screen

The EMV Application Data screen is used to enter application tag information. The EMV Application Data screen is shown below.

EMV APPLICATION DATA

Record ____ Group ____

Application Identifier _____

AID Flag _ (0 = Full, 1 = Partial) Next State ____ Scheme Identifier ____

Tag _____	Value _____
Tag _____	Value _____
Tag _____	Value _____
Tag _____	Value _____
Tag _____	Value _____
Tag _____	Value _____
Tag _____	Value _____
Tag _____	Value _____
Tag _____	Value _____
Tag _____	Value _____
Tag _____	Value _____

=====

F2=READ F3=ADD F4=DELETE F5=UPDATE
F9=NEXT F10=PRINT F16=EXIT

Refer to the EMV vendor documentation for descriptions of the fields in the highlighted area on this screen.

EMV Scheme Data Screen

The EMV Scheme Data screen is used to enter scheme tag information. The EMV Scheme Data screen is shown below.

EMV SCHEME DATA

Record ____ Group ____

Scheme Identifier ____

Tag _____	Value _____
Tag _____	Value _____
Tag _____	Value _____
Tag _____	Value _____
Tag _____	Value _____
Tag _____	Value _____
Tag _____	Value _____
Tag _____	Value _____
Tag _____	Value _____
Tag _____	Value _____
Tag _____	Value _____

=====

F2=READ F3=ADD F4=DELETE F5=UPDATE

F9=NEXT F10=PRINT F16=EXIT

Refer to the EMV vendor documentation for descriptions of the fields in the highlighted area on this screen.

Diebold 10XX Commands Screen

The Diebold 10XX Commands screen is used to enter commands that control the processing messages. The Diebold 10XX Commands screen is shown below.

```
BASE24-DCT   DIEBOLD 10XX COMMANDS   LNET           YY/MM/DD HH:MM   40 OF 46

  ATM TERMINAL ID: _____ FIID: _____

- LOAD GROUP           - LOAD SCREEN           - LOAD MAC KEY
- LOAD GROUP SCREEN    - LOAD STATE           - LOAD PUBLIC KEY
- LOAD GROUP STATE     - LOAD PIN KEY         - LOAD MASTER KEY
- LOAD EMV DATA      - LOAD ALL KEYS

THE FOLLOWING ARE ONLY APPLICABLE FOR 1000 TERMINALS WITH EMG
- COPY FONT           TO LIBRARY
- COPY ICON    0 TO LIBRARY
- COPY CELL DATA FROM LIBRARY    TO LIBRARY
- COPY  0 CELLS FROM LIBRARY  CELL  0 TO LIBRARY CELL  0
- RESTORE STANDARD CONFIGURATION

***** BASE24 *****
NEW PAGE:           FILE DESTINATION:           NEW LOGICAL NETWORK ID:
F1-ENTER  F3-OPEN F4-LOAD  F5-CLOSE  F7-GOTO F 9-RESET  F12-HELP  F14-PROOF LIST
```

The highlighted field is EMV specific when using the Smart Card.

B: NCR Device Support

The NCR 5XXX Device Handler is an existing BASE24-atm process that provides the interface between NCR 5XXX ATMs and BASE24. The NCR 5XXX EMV module is an additional licensable module which, when bound with the NCR 5XXX Device Handler process, provides EMV support for NCR 5XXX ATMs or ATMs emulating the NCR 5XXX message. There is also a Shared NDC EMV module that can be bound with the NCR 5XXX Device Handler process for ATMs using either or both of the following Wincor Nixdorf NDC message formats:

- The format configured according to the initial version of the Shared NDC EMV implementation. This appendix contains the configuration screens required to support BASE24-atm EMV transactions with these ATMs.
- The later version as defined in Wincor Nixdorf's *NDC/Diebold D91x Message Format Extension for EMV*. See appendix D for the configuration screens required to support BASE24-atm EMV transactions with these ATMs.

The same Wincor Nixdorf hardware can automatically support both NDC message formats, depending on whether the Wincor EMV Chip Card state or the standard NCR EMV states are downloaded to an ATM. There is no need to load special software or make changes at the ATM site. Likewise, the Shared NDC+ EMV module dynamically determines which format is being used based on the configuration data that is downloaded to the ATM.

An NCR EMV module and a Shared NDC+ EMV module can be bound to the same NCR 5XXX Device Handler process.

NCR ATMs are configured using software and files supplied by NCR and contained within the ATMs themselves. BASE24 device support emulates the NCR configuration file using the CNFG5070 Configuration File, which can be accessed using the BASE24 Device Control Terminal (DCT). With access to the CNFG5070, you can configure EMV-specific parameters and state table entries and then download these parameters to the ATM. BASE24-atm NCR 5XXX EMV support is compliant with version 2.14 of the NCR CAM 2 specification.

This appendix describes the implementation requirements associated with the NCR NDC+ EMV Dip Card Reader. This processing is required for customers using NCR's Dip Card Reader due to the format difference of the request message sent by the EMV Dip Reader. NCR's EMV Dip Reader does not send the Track 2 data buffer if the Track 2 equivalent data is first obtained from the chip. Requests are processed by the BASE24-atm NCR 5XXX EMV device handler so that Track 2 equivalent data is processed instead of Track 2.

This appendix also describes the configuration screens required to support BASE24-atm EMV transactions with the NCR 5XXX EMV module. For general information about DCT functions, see the ***BASE24 Device Control Manual***. For general information about the CNFG5070 Configuration File, see the ***BASE24-atm NCR 5XXX Device Support Manual***, and see the following vendor documents for specific information about the configuration file parameters:

- ***NCR NDC+ Programmer's Reference Manual***
- ***NCR NDC+ and Aptra Advance NDC EMV Integrated Circuit Card (ICC) Reference Manual***

CNFG5070 Configuration File

To access the configuration file from the DCT, select screen 45 (ATM NCR 5XXX COMMANDS) from the DCT Product Menu. The NCR 5XXX Commands screen appears.

- If you want to download EMV-specific configuration information, select DOWNLINE LOAD DATA and press the **F1** key. The NCR 5XXX Load Commands screen appears.
- If you want to enter configuration parameters or state table data, select ACCESS CONFIGURATION FILE and press the **F1** key. The NCR 5XXX Configuration File Menu appears.

The NCR 5XXX Load Commands screen and the NCR 5XXX Configuration File Menu are described in this appendix, as are EMV-related screens accessed from the NCR 5XXX Configuration File Menu.

NCR 5XXX Load Commands

The NCR 5XXX Load Commands screen allows you to download specified information to NCR 5XXX ATMs. One field on this screen is specific to the BASE24-atm EMV product. The NCR 5XXX Load Commands screen is shown below, followed by a description of the LOAD EMV GROUP field.

```

BASE24-DCT   NCR 5XXX LOAD COMMANDS   LLLL           YY/MM/DD  HH:MM  01 OF 01
ATM TERMINAL ID: _____  LOAD GROUP:      0      FIID: _____

  _ LOAD ALL (INCLUDING KEYS)           _ LOAD PUBLIC KEY
  _ LOAD ALL IN GROUP                   _ LOAD MASTER KEY
  _ LOAD ALL SCREENS                     _ LOAD ALL KEYS
  _ LOAD ALL STATES                       LOAD KEY ENTRY MODE 1
  _ LOAD ALL FITS
  _ LOAD DATE AND TIME
  _ LOAD MAC KEY
  _ LOAD KEY TYPE 1
  _ LOAD CUST SCREEN    0      LANG 0
  _ LOAD C-SCREEN      C  0
  _ LOAD R-SCREEN      R  0
  _ LOAD STATE         0
  _ LOAD FIT           0
  _ LOAD EMV GROUP      0

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
F1-ENTER F2-CNFG F3-CNTRS F4-OPEN F5-CLOSE F6-LOAD F7-RESET F8-CPLF F16-EXIT

```

LOAD EMV GROUP — The number of the EMV group to be downloaded to the ATM from the customized load group specified in the LOAD GROUP field. The range of valid values is 0 through 9999. After entering the group number, place the cursor to the left of the field name and press the **F6** key.

Field Length: 1–4 numeric characters
 Required Field: No
 Default Value: 0

NCR 5XXX Configuration File

Three menu items highlighted on the following NCR 5XXX Configuration File Menu contain fields that are specific to the BASE24-atm EMV product with NCR 5XXX ATMs, ATMs emulating the NCR 5XXX message, and ATMs using Wincor Nixdorf's NDC message format that is configured with the same states and tables as NCR's CAM II implementation. These menu items are described below.

```

BASE24-DCT   NCR 5XXX CONFIG FILE   LLLL           YY/MM/DD  HH:MM  01 OF 01

CONFIGURATION FILE NAME \SYS1.$DATA.PRO1DATA.CNFG5070

GROUP NUMBER      0

-  ENHANCED CONFIGURATION PARAMETERS
-  STATE
-  SCREEN
-  FINANCIAL INSTITUTION TABLES
-  SUPERVISOR CARD DATA
-  PRINTER CONTROL SCREENS (Kxx SCREENS)
-  CONFIGURATION IDENTITY NUMBER
-  HOPPER MAPPING TABLE
-  EMV CONFIGURATION MENU
-  WINCOR EMV AID TABLE

-  PRINT

***** BASE24 *****
NEW PAGE:           FILE DESTINATION:       NEW LOGICAL NETWORK ID:
F1-SELECT F7-GOTO F10-PRINT F12-HELP F16-EXIT SF16-LOGOFF

```

ENHANCED CONFIGURATION PARAMETERS

Contains information about optional features for NCR 5XXX terminals.

STATE

Lists the state screens needed to configure NCR terminals.

EMV CONFIGURATION MENU

Lists four options from which a screen can be accessed in order to configure various tables for NCR 5XXX ATMs, ATMs that emulate the NCR 5XXX message, and ATMs that use Wincor Nixdorf's NDC message format that is configured with the same states and tables as NCR's CAM II implementation.

To select one of these menu items, place the cursor next to the function and press the **F1** key.

Enhanced Configuration Parameters

Enhanced Configuration Parameters screen 1 contains information about optional features for NCR 5XXX terminals. This screen is displayed when you place the cursor to the left of the ENHANCED CONFIGURATION PARAMETERS option on the NCR 5XXX Configuration File Menu and press the **F1** key.

Enhanced Configuration Parameters screen 1 is shown below. The three EMV-specific fields are highlighted. Descriptions of these fields follow the screen example.

```

BASE24-DCT   ENHANCED CONFIG PARAM   LLLL   YY/MM/DD   HH:MM   01 OF 02

GROUP:      0

OPTIONS:
CAMERA CONTROL      1 (ON)
READY/SUPPLY        1 (READY, AMT 8 DGTS)
AUTO VOICE          1 (ON)
ROLL WIDTH          40
TRACK 1 FORMAT      1 (ISO, BCKWD SRCH)
SPEC CMD RJCT       0 (NO)
TRAN STATUS         1 (YES)
J-PRNTR BKUP TIM LMT 999 (NOT USED)
ENV DISP FLT        999 (NOT USED)
ENHANCED TI         999 (NOT USED)
KEYBOARD AUDIO ECHO 0 (NO)
RPT EJ DUAL-MD ERRS 0 (NONE)
INCLUDE PAN         0 (YES)
INCLUDE EMV ERR INFO 999 (NOT USED)
ACQUIRER ID        0
DATE FORMAT         1 (MMDDYY)
LEFT PRINT COL      1
J-PRNT BNA CNTS     999 (NOT USED)
BNA CNTS IN RQST    999 (NOT USED)
J-PRNTR BKUP PRNT LMT 999 (NOT USED)
MEI FLASH RATE      999 (NOT USED)
REPORT DAS ERRORS   0 (NO)
EJ BKUP MODE        0 (STANDARD)
CARDS ACCEPTED 0 (MAG ONLY)
EMV ACTIONED 0 (YES)

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:    NEW LOGICAL NETWORK ID:
F9-NEXT PAGE       F11-PREVIOUS PAGE     F12-HELP

```

CARDS ACCEPTED — Indicates the type of card to be accepted at the terminals. Valid values are as follows:

- 0 = Accept cards with only a magnetic stripe.
- 1 = Accept cards with both a magnetic stripe and an integrated circuit chip.
- 2 = Accept cards with an integrated circuit chip only when the Card Read state is not 0 (i.e., accept only when a card with a magnetic stripe has already started the transaction).
- 999 = Do not use this option.

Field Length: 1–3 numeric characters, followed by a system-protected text description
Required Field: No
Default Value: 0

INCLUDE EMV ERR INFO — Indicates the type of information to be included in application error messages. Valid values are as follows:

- 0 = Do not include EMV error information. EMV command and response data are not to be included in EMV application error messages, both solicited and unsolicited.
- 1 = Include EMV error information. EMV command and response data are to be included in EMV application error messages, both solicited and unsolicited.
- 999 = Do not use this option.

Field Length: 1–3 numeric characters, followed by a system-protected text description
Required Field: No
Default Value: 999

EMV ACTIONED — Determines whether EMV handling commands are actioned or not actioned. For more information on actioned and not actioned, see the *NCR NDC+ Programmer's Reference Manual* for NCR 5XXX terminals. Valid values are as follows:

- 0 = EMV handling commands are to be actioned.
- 1 = EMV handling commands are not to be actioned.

Field Length: 1–3 numeric characters, followed by a system-protected text description
Required Field: No
Default Value: 0

State Table Data Menu

The State Table Data Menu lists the state screens needed to configure the NCR terminals. This screen is displayed when you place the cursor to the left of the STATE option on the NCR 5XXX Configuration File Menu and press the **F1** key. The highlighted fields shown on the screen below are specific to the BASE24-atm EMV product for NCR 5XXX ATMs, ATMs that emulate the NCR 5XXX message, and ATMs that use Wincor Nixdorf's NDC message format that is configured with the same states and tables as NCR's CAM II implementation, and are described on the following pages. The Global Extension state screen is also described because it applies to several of the EMV-specific states.

BASE24-DCT	STATE TABLE DATA	LLLL	YY/MM/DD	HH:MM	01 OF 42
GROUP	0	STATE NUMBER	0		

2:42 (A-CARD READ)	3:42 (B-PIN ENTRY)	4:42 (C-UNLOCK ENV DISP)
5:42 (D-PRESET OP-CODE)	6:42 (E-4 FDK SELECT)	7:42 (F-AMOUNT ENTRY)
8:42 (G-AMOUNT CHECK)	9:42 (H-4 FDK INFO)	10:42 (I-TRAN REQUEST)
11:42 (J-CLOSE)	12:42 (K-FIT SWITCH)	13:42 (L-CARD WRITE)
14:42 (M-ENH PIN ENTRY)	15:42 (N-CAMERA CONTROL)	16:42 (P-DPM CAV)
17:42 (Q-DPM ACCEPT DOC)	18:42 (R-ENH AMT ENTRY)	19:42 (S-LANG SWITCH)
20:42 (T-CRD READ PIN INIT)	21:42 (V-LANG FROM CARD)	22:42 (W-FDK SWITCH)
23:42 (X-8 FDK INFO)	24:42 (Y-8 FDK SELECT)	25:42 (Z-GLOBAL EXT)
26:42 (ENH FIT SWITCH)	27:42 (i-AUDIO CONTROL)	28:42 (k-SMART FIT CHECK)
29:42 (+-BEGIN EMV INIT)	30:42 (,-COMPL EMV INIT)	31:42 (;-EMV RE-INIT)
32:42 (--AUTO LANG SELCT)	33:42 (?-SET EMV TXN DATA)	34:42 (.-BEGIN EMV SELCT)
35:42 (/--COMPL EMV SELCT)	36:42 (b-CUST SELCT PIN)	37:42 (>-CASH ACCEPT)
38:42 (w-CHECK ACCEPT)	39:42 (e-WINCOR EMV CARD)	40:42 (m-PIN & LANG SLCT)
41:42 (r-SHARED BNA)	42:42 (q-SHARED CHECK IN)	

TYPE THE NUMBER OF THE DESIRED FUNCTION IN NEW PAGE FIELD AND KEY F7

***** BASE24 *****

NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:
	F12-HELP	

To access a screen, enter the appropriate screen number (the number to the left of the colon) in the NEW PAGE field and press the **F7** key. Each of these EMV-specific screens is described on the following pages.

Smart FIT Check State

The Smart FIT Check State screen determines the next state to be performed based upon a search of the financial institution table (FIT) using the EMV data. It allows the chip to direct the transaction flow instead of the magnetic stripe. The Smart FIT Check State screen is shown below, followed by descriptions of its fields.

BASE24-DCT	STATE TABLE DATA	LLLL	YY/MM/DD	HH:MM	28 OF 42
GROUP	0	STATE NUMBER	0		
SMART FIT CHECK					
STATE TYPE	GOOD-READ NEXT-STATE	ERROR SCREEN	READ COND1	READ COND2	READ COND3
k	0	0	0	0	0
	CARD-RETAIN FLAG	NO-FIT NEXT-STATE			
	0	0			
***** BASE24 *****					
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:			
	F12-HELP				

STATE TYPE — The value k identifies the Smart FIT Check state.

Field Length: System Protected

GOOD-READ NEXT-STATE — The number of the state to which the terminal proceeds if there is a matching FIT entry. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

ERROR SCREEN — The number of the screen displayed if the read conditions are not satisfied due to a bad card or mechanism error. The card is ejected. Valid values are 0 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

READ COND1 — A number sequence that specifies which track or tracks on the card are read. If the specified track is read, the state specified in the GOOD-READ NEXT-STATE field is performed. If the track could not be read, the action specified in the READ COND2 field is attempted. This field can be zero filled to the left. Valid values are 1 through 7.

Note: For specific information on what values to enter in the READ COND1 field, refer to the *NCR NDC+ Programmer's Reference Manual* for NCR 5XXX terminals.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

READ COND2 — A number sequence that specifies which track or tracks on the card are read. If the specified track is read, the state specified in the GOOD-READ NEXT-STATE field is performed. If the track could not be read, the action specified in the READ COND3 field is attempted. This field can be zero filled to the left. Valid values are 1 through 7.

Note: For specific information on what values to enter in the READ COND2 field, refer to the *NCR NDC+ Programmer's Reference Manual* for NCR 5XXX terminals.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

READ COND3 — A number sequence that specifies which track or tracks on the card are read. If the specified track is read, the state specified in the GOOD-READ NEXT-STATE field is performed. If the track could not be read, the card is ejected and the screen specified in the ERROR SCREEN field is displayed. This field can be zero filled to the left. Valid values are 1 through 7.

Note: For specific information on what values to enter in the READ COND3 field, refer to the *NCR NDC+ Programmer's Reference Manual* for NCR 5XXX terminals.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

CARD-RETAIN FLAG — Indicates when the ATM is to return the card. This field can be zero filled to the left. Valid values are as follows:

0 = Terminal returns the card immediately.
1 = Terminal returns the card as specified by a transaction reply command message.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

NO-FIT NEXT-STATE — The number of the state to which the terminal proceeds if the financial institution number on a card does not match any FIT. If FITs are not used, this value must be 0. Valid values are 0 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Begin EMV Initialization State

The Begin EMV Initialization state is responsible for staging the ICC into the smart card reader, powering on the smart card reader, and building the ICC application candidate list. It can also automatically select an ICC application if the AUTOMATIC EMV APPL SELECTION FLAG field is configured to do so. The Begin EMV Initialization state screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT   STATE TABLE DATA          LLLL          YY/MM/DD  HH:MM  29 OF 42

      GROUP      0      STATE NUMBER      0

                        BEGIN EMV INITIALIZATION

STATE         EMV INIT STARTED      EMV INIT NOT STARTED      EMV INIT
TYPE          STATE                  STATE                      REQUIREMENT
+             0                      0                          0

      AUTOMATIC EMV APPL      APPLICATION LABEL      CARDHOLDER
      SELECTION FLAG          VALIDATION FLAG          CONFIRM FLAG
      0                      0                          0

      SCREEN TO CLEAR
      SCREEN NUMBER

      0

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
               F12-HELP

```

STATE TYPE — The value + identifies the Begin EMV Initialization state.

Field Length: System Protected

EMV INIT STARTED STATE — The number of the state to which the terminal proceeds if EMV initialization has started. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

EMV INIT NOT STARTED STATE — The number of the state to which the terminal proceeds if EMV initialization has not been started because the alternative technology byte (ATB) condition in the EMV INIT REQUIREMENT field has not been met. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EMV INIT REQUIREMENT — Indicates when EMV processing should be initiated. Valid values are as follows:

- 0 = Always begin EMV initialization.
- 1 = Begin EMV initialization only if the magnetic stripe track 2 data has been read and contained an ATB value of 2 or 6.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

AUTOMATIC EMV APPL SELECTION FLAG — Indicates if an automatic EMV application selection is required. Valid values are as follows:

- 0 = Cardholder is to select or confirm the EMV application to be used.
- 1 = EMV application is to be automatically selected.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

APPLICATION LABEL VALIDATION FLAG — Indicates action to be taken if an EMV application does not provide an Application Label during the candidate list build processing. Valid values are as follows:

- 000 = Check if the EMV application provides an Application Label. If an Application Label is not provided, use the corresponding default Application Label. If no default is available, the EMV application is not added to the candidate list.
- 001 = Do not check if the EMV application provided an Application Label.

- 002 = Check if the EMV application provides an Application Label. If an Application Label is not provided, all EMV processing terminates with a fake EMV application level error.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

CARDHOLDER CONFIRM FLAG — Specifies action to be taken if an EMV application indicates, using the Application Priority Indicator during candidate list build processing, that cardholder confirmation is required. Valid values are as follows:

- 000 = Do not check if the EMV application requires cardholder confirmation.
- 001 = Check if the EMV application requires cardholder confirmation. If cardholder confirmation is required, the EMV application is not added to the candidate list. If the Application Priority Indicator is not provided, the EMV application is treated as not requiring cardholder confirmation.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

SCREEN TO CLEAR SCREEN NUMBER — Specifies the screen to clear before displaying the selected application name. Valid values are any valid screen number 000 through 999.

Field Length: 3 numeric characters
Required Field: Yes
Default Value: 0

Complete EMV Initialization State

The Complete EMV Initialization state determines the next action to be performed based upon the results from the EMV initialization state. The Complete EMV Initialization state screen is shown below, followed by descriptions of its fields.

BASE24-DCT	STATE TABLE DATA	LLLL	YY/MM/DD	HH:MM	30 OF 42
GROUP	0	STATE NUMBER	0		
COMPLETE EMV INITIALIZATION					
STATE TYPE	PLEASE WAIT SCREEN NUMBER	EMV INIT SUCCESSFUL STATE	CARD NOT SMART STATE		
,	0	0	0		
	NO USABLE APPL STATE	EMV APPL LEVEL ERROR STATE	EMV HDWR LEVEL ERROR STATE		
	0	0	0		
***** BASE24 *****					
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:			
	F12-HELP				

STATE TYPE — The value , (comma) identifies the Complete EMV Initialization state.

Field Length: System Protected

PLEASE WAIT SCREEN NUMBER — An optional screen that can be displayed to the cardholder while waiting for the completion of EMV initialization. Valid values are 0 through 999.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

EMV INIT SUCCESSFUL STATE — The number of the state to which the terminal proceeds if EMV initialization is successful. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

CARD NOT SMART STATE — The number of the state to which the terminal proceeds if the card does not have a chip on it or if a failure condition, such as dirty ICC contacts, occurs. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

NO USABLE APPL STATE — The number of the state to which the terminal proceeds if initialization found no usable EMV applications for inclusion in the candidate list. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EMV APPL LEVEL ERROR STATE — The number of the state to which the terminal proceeds if initialization fails as a result of an EMV application-level error. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EMV HDWR LEVEL ERROR STATE — The number of the state to which the terminal proceeds if initialization fails as a result of an EMV hardware-level error. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EMV Reinitialize State

The EMV Reinitialize state enables the EMV application currently being used for processing to be reinitialized. The EMV Reinitialize state screen is shown below, followed by descriptions of its fields.

```
BASE24-DCT  STATE TABLE DATA          LLLL      YY/MM/DD  HH:MM  31 OF 42

      GROUP      0      STATE NUMBER      0

                        EMV RE-INITIALIZE

STATE        GOOD      BAD
TYPE         STATE     STATE
;            0         0

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP
```

STATE TYPE — The value ; (semicolon) identifies the EMV Reinitialize state.

Field Length: System Protected

GOOD STATE — The number of the state to which the terminal proceeds if the EMV application is able to be reinitialized. The processing identified by the Begin EMV Application Select and Initialization state is started in parallel and is performed using the EMV application currently selected. The outcome of this parallel processing is acted upon by the Complete EMV Application Selection and Initialization state, as appropriate to the final outcome of the reinitialization processing. Processing then proceeds to the state identified in this field.

Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

BAD STATE — The number of the state to which the terminal proceeds if the EMV application is not able to be reinitialized. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Automatic Language Selection State

The Automatic Language Selection state enables the language preference of the cardholder to automatically be selected if the EMV application contains the language preference data object.

Note: Automatic language selection is not currently supported.

The Automatic Language Selection state screen is shown below, followed by descriptions of its fields.

```
BASE24-DCT  STATE TABLE DATA          LLLL          YY/MM/DD  HH:MM  32 OF 42

      GROUP      0      STATE NUMBER      0

                        AUTOMATIC LANGUAGE SELECTION

STATE          LANGUAGE MATCH      NO LANGUAGE MATCH
TYPE           STATE               STATE
-              0                   0

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP
```

STATE TYPE — The value - (hyphen) identifies the Automatic Language Selection state.

Field Length: System Protected

LANGUAGE MATCH STATE — The number of the state to which the terminal proceeds if the EMV language preference data finds a match with those languages supported at the ATM. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

NO LANGUAGE MATCH STATE — The number of the state to which the terminal proceeds if the EMV language preference data does not find a match with those languages supported at the ATM. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Set EMV Transaction Data State

The Set EMV Transaction Data state enables selective transaction-specific data to be made available for EMV processing. The Set EMV Transaction Data state screen is shown below, followed by descriptions of its fields.

```
BASE24-DCT    STATE TABLE DATA          LLLL          YY/MM/DD  HH:MM  33 OF 42

      GROUP      0      STATE NUMBER      0

                        SET EMV TRANSACTION DATA

STATE  NEXT  CURRENCY  TRANSACTION  AMOUNT AUTHORIZED
TYPE   STATE TYPE      TYPE          SOURCE BUFFER
?      0      0        0              0

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP
```

STATE TYPE — The value ? (question mark) identifies the Set EMV Transaction Data state.

Field Length: System Protected

NEXT STATE — The number of the state to which the terminal proceeds after processing the remaining fields on this screen. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

CURRENCY TYPE — Indicates the reference currency type for the ICC Currency Data Objects Table. Valid values are 0 through 255. This field can be zero filled to the left. A value of 000 indicates no action should be taken.

For more information about the ICC Currency Data Objects Table, see the *NCR NDC+ Programmer's Reference Manual* for NCR 5XXX terminals.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TRANSACTION TYPE — Indicates the reference transaction type for the ICC Currency Data Objects Table, expressed as a decimal equivalent of the hexadecimal ISO value. Valid values are 0 through 255. This field can be zero filled to the left. A value of 000 indicates no action should be taken.

For more information about the ICC Currency Data Objects Table, see the *NCR NDC+ Programmer's Reference Manual* for NCR 5XXX terminals.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

AMOUNT AUTHORIZED SOURCE BUFFER — Identifies the source buffer from which the Amount Authorized data object will be derived and indicates whether the amount in the buffer requires conversion (from major or minor currency units) before being placed in the Amount Authorized data object. Valid values are as follows:

000 = No action should be taken.
010 = Use General Purpose Buffer B; no conversion is required.
011 = Use General Purpose Buffer B; conversion is required.
020 = Use General Purpose Buffer C; no conversion is required.
021 = Use General Purpose Buffer C; conversion is required.
030 = Amount Buffer; no conversion is required.
031 = Amount Buffer; conversion is required.

If conversion from major to minor currency units is required, the value held within the Transaction Currency Exponent data object is used to indicate how many trailing zeros are to be added to the amount in the buffer when placed in the Amount Authorized data object.

The CURRENCY TYPE field is processed before this field to ensure the Transaction Currency Exponent data object can be set before the conversion process requires it.

For more information about the Amount Authorized data object and the Transaction Currency Exponent data object, see the ***NCR NDC+ Programmer's Reference Manual*** for NCR 5XXX terminals.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

Begin EMV Application Selection and Initialization State

The Begin EMV Application Selection and Initialization state enables a cardholder to select which EMV application to use.

Note: The Begin EMV Application Selection and Initialization state screen has three extension states associated with it. These extension states are accessed through the Global Extension screen (state Z), which is described later in this section.

The Begin EMV Application Selection and Initialization state screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT   STATE TABLE DATA           LLLL           YY/MM/DD  HH:MM  34 OF 42

      GROUP      0      STATE NUMBER      0

      BEGIN EMV APPLICATION SELECTION & INITIALIZATION

STATE         CRDHLDR SELECTION           FDK SCREEN NUMBERS
TYPE          SCREEN NUMBER               EXTENSION STATE
.             0                           0

      ACTION KEYS                         EXIT PATHS
      EXTENSION STATE                     EXTENSION STATE
      0                                   0

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP

```

STATE TYPE — The value . (period) identifies the Begin EMV Application Selection and Initialization state.

Field Length: System Protected

CRDHLDR SELECTION SCREEN NUMBER — The number of the screen that displays the list of EMV applications from which the customer can select. This screen is displayed only when the EMV application can be selected by the customer. Valid values are 0 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

FDK SCREEN NUMBERS EXTENSION STATE — The number of the state for the extension state that identifies which screen numbers hold the text of the EMV application name for each function display key (FDK) and indirectly which of the FDKs are to be used. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

ACTION KEYS EXTENSION STATE — The number of the state for the extension state that identifies the FDK for each action key and the screen numbers which hold the text for each FDK used as an action key. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EXIT PATHS EXTENSION STATE — The number of the state for the extension state that identifies the next state numbers for the exit paths from this state. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Complete EMV Application Selection and Initialization State

The Complete EMV Application Selection and Initialization state determines the next action to be performed based on the result from the Begin EMV Application Selection and Initialization state.

Note: The Complete EMV Application Selection and Initialization state screen has one extension state associated with it. This extension state is accessed through the Global Extension screen (state Z), which is described later in this section.

The Complete EMV Application Selection and Initialization state screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT    STATE TABLE DATA    LLLL    YY/MM/DD  HH:MM  35 OF 42

      GROUP      0      STATE NUMBER    0

      COMPLETE EMV APPLICATION SELECTION & INITIALIZATION

STATE  PLEASE WAIT    EMV APPL NAME    EMV APPL NAME
TYPE  SCREEN NUMBER  TPLT SCREEN NUMBER  SCREEN NUMBER
/      0              0              0

      EXIT PATHS
      EXTENSION STATE
      0

```

```

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
               F12-HELP

```

STATE TYPE — The value / (slash) identifies the Complete EMV Application Selection and Initialization state.

Field Length: System Protected

PLEASE WAIT SCREEN NUMBER — The number of the screen to display to the cardholder while waiting for completion of the parallel processing of the EMV application selection and initialization. Valid values are 0 through 999. If the screen number is set to 000, no screen is displayed. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EMV APPL NAME TPLT SCREEN NUMBER — The number of the screen that contains the template to be used for generating the display data of the screen containing the name of the selected EMV application. Valid values are 0 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EMV APPL NAME SCREEN NUMBER — The number of the screen to store the output of processing the template screen. This screen contains the name of the selected EMV application. Valid values are 0 through 999. If the screen number is set to 000, the name of the EMV application is not stored. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EXIT PATHS EXTENSION STATE — The number of the state for the extension state that identifies the next state numbers for the exit paths from this state. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Global Extension Screen

The Global Extension screen is a generic screen that is used to access extension state table data for all states that have an extension state. An extension state is used to extend, or expand, the functions of a particular state. The state tables in general only allow eight values to be set. For those states requiring more than eight configuration values, the global extension table is used to hold the excess values.

Fields on the following EMV state screens use the Global Extension screen:

- The Begin EMV Application Selection and Initialization state screen (state .) contains the following extension states:
 - FDK Screen Numbers
 - Action Keys
 - Exit Paths
- The Complete EMV Application Selection and Initialization state screen (state /) contains the Exit Paths extension state.

The Global Extension screen is used to input extension state table data for all of the extension states mentioned above. The additional configuration values on the Global Extension screen have generic field names, and the meaning of a specific field changes depends on the original state screen that references this screen.

The table entry fields on the Global Extension screen correspond to the extension state table entries described in the *NCR NDC+ Programmer's Reference Manual* and the *NCR NDC+ and Aptra Advance NDC EMV Integrated Circuit Card (ICC) Reference Manual*.

The contents of the table entry fields for each of the extension states are described on the pages following the Global Extension screen.

The Global Extension screen is accessed from the State Table Data Menu by entering a value of 25 in the NEW PAGE field and pressing the **F7** key. The Global Extension screen is shown below. Field descriptions are provided for each extension state on the following pages.

```

BASE24-DCT      STATE TABLE DATA      LLLL      YY/MM/DD  HH:MM  25 OF 42

      GROUP      0      STATE NUMBER      0

                                GLOBAL EXTENSION

STATE           TABLE           TABLE           TABLE           TABLE           TABLE           TABLE
TYPE            ENTRY-2           ENTRY-3           ENTRY-4           ENTRY-5           ENTRY-6           ENTRY-7
  Z              0              0              0              0              0              0

                TABLE           TABLE
                ENTRY-8           ENTRY-9
                  0              0

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP

```


FDK Template Screen Numbers Extension State

The following fields are used with the FDK Template Screen Numbers Extension State field on the Begin EMV Application Selection and Initialization State screen. Each field is input as a table entry on the Global Extension screen. For example, the FDK A ICC Application Name Template Screen Number (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen.

TABLE ENTRY-2 (FDK A ICC Application Name Template Screen Number) — The screen number to use as a template to display the ICC application number against function display key (FDK) A. Valid values are 000 through 999. A value of 000 indicates that this FDK is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-3 (FDK B ICC Application Name Template Screen Number) — The screen number to use as a template to display the ICC application number against FDK B. Valid values are 000 through 999. A value of 000 indicates that this FDK is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-4 (FDK C ICC Application Name Template Screen Number) — The screen number to use as a template to display the ICC application number against FDK C. Valid values are 000 through 999. A value of 000 indicates that this FDK is not used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-5 (FDK D ICC Application Name Template Screen

Number) — The screen number to use as a template to display the ICC application number against FDK D. Valid values are 000 through 999. A value of 000 indicates that this FDK is not used.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

TABLE ENTRY-6 (FDK F ICC Application Name Template Screen

Number) — The screen number to use as a template to display the ICC application number against FDK F. Valid values are 000 through 999. A value of 000 indicates that this FDK is not used.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

TABLE ENTRY-7 (FDK G ICC Application Name Template Screen

Number) — The screen number to use as a template to display the ICC application number against FDK G. Valid values are 000 through 999. A value of 000 indicates that this FDK is not used.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

TABLE ENTRY-8 (FDK H ICC Application Name Template Screen

Number) — The screen number to use as a template to display the ICC application number against FDK H. Valid values are 000 through 999. A value of 000 indicates that this FDK is not used.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

TABLE ENTRY-9 (FDK I ICC Application Name Template Screen

Number) — The screen number to use as a template to display the ICC application number against FDK I. Valid values are 000 through 999. A value of 000 indicates that this FDK is not used.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

Action Keys Extension State

The following fields are used with the Action Keys Extension State field on the Begin EMV Application Selection and Initialization State screen. Each field is input as a table entry on the Global Extension screen. For example, the More Applications Screen Number (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen. The fields TABLE ENTRY-6 through TABLE ENTRY-9 are not used with the Action Keys Extension State and must be set to 0.

TABLE ENTRY-2 (More Applications Screen Number) — The screen number to display when the function display key (FDK) is activated, to enable the cardholder to access the next screen of ICC application names. Valid values are 000 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-3 (FDK for More Applications) — The FDK to be used to allow the cardholder to access the next screen of ICC application names. Valid values are as follows:

001 = FDK A
002 = FDK B
003 = FDK C
004 = FDK D
005 = FDK F
006 = FDK G
007 = FDK H
008 = FDK I

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-4 (Back to Start of List Screen Number) — The screen number to display when the FDK is activated, to enable the cardholder to return to the first screen of ICC application names. Valid values are 000 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-5 (FDK for Back to Start of List) — The FDK to be used to enable the cardholder to return to the first screen of ICC application names. Valid values are as follows:

001 = FDK A
002 = FDK B
003 = FDK C
004 = FDK D
005 = FDK F
006 = FDK G
007 = FDK H
008 = FDK I

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Exit Paths Extension State on the Begin EMV Application Selection and Initialization State Screen

The following fields are used with the Exit Paths Extension State field on the Begin EMV Application Selection and Initialization State screen. Each field is input as a table entry on the Global Extension screen. For example, the Time Out Next State Number (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen. The fields TABLE ENTRY-7 through TABLE ENTRY-9 are not used with the Exit Paths Extension State field on the Begin EMV Application Selection and Initialization State screen and must be set to 0.

TABLE ENTRY-2 (Time Out Next State Number) — The next state if the cardholder does not respond in time when input is requested. Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-3 (Cancel Key Next State Number) — The next state if the cardholder selects the cancel key when input is requested. Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-4 (Cardholder Selected Application Next State Number) — The next state if the cardholder selects an ICC application when input is requested. Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-5 (Application Selected Automatically Next State

Number) — The next state if the ICC application selection is being performed automatically because there is only one ICC application available. Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

TABLE ENTRY-6 (No Usable Applications Next State Number) — The next state if there are no ICC applications for the cardholder to choose from. Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

Exit Paths Extension State on the Complete EMV Application Selection and Initialization State Screen

The following fields are used with the Exit Paths Extension State field on the Complete EMV Application Selection and Initialization State screen. Each field is input as a table entry on the Global Extension screen. For example, the ICC Application Select and Initialization Successful Next State Number (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen. The TABLE ENTRY-9 field is not used with the Exit Paths Extension State field on the Complete EMV Application Selection and Initialization State screen and must be set to 0.

TABLE ENTRY-2 (ICC Application Select and Initialization Successful Next State Number) — The next state number if the ICC application was successfully selected and initialized. Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-3 (Card Not Smart Next State Number) — The next state number if the card was found not to be an ICC. Valid values are 000 through 254 and 256 through 999.

Note: This path could also be taken if a failure condition occurs which cannot be distinguished from the integrated circuit not being present on the card. An example could be dirty ICC contacts.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-4 (ICC Application Not Usable Next State Number) — The next state number if the selected ICC application was found not to be usable at the terminal. Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-5 (No Suitable ICC Applications Next State

Number) — The next state number if automatic ICC application selection and initialization was performed, but no suitable applications were found. Valid values are 000 through 254 and 256 through 999.

Note: This entry applies only when automatic ICC application selection has been performed.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-6 (ICC Application Level Error Next State

Number) — The next state number if the ICC application selection and initialization failed as a result of an ICC application level error. Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-7 (ICC Hardware Level Error Next State

Number) — The next state number if the ICC application selection and initialization failed as a result of an ICC hardware level error. Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-8 (Processing Not Performed Next State

Number) — The next state number if the ICC application selection and initialization is not being performed. Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EMV Configuration Menu

The EMV Configuration Menu contains a list of four options from which a screen can be accessed in order to configure various tables. This screen is displayed when you place the cursor to the left of the EMV CONFIGURATION MENU option on the NCR 5XXX Configuration File Menu and press the **F1** key. This screen can be displayed for NCR 5XXX ATMs, ATMs emulating the NCR 5XXX message, and ATMs using Wincor Nixdorf's NDC message format that is configured using the same state and tables as NCR's CAM II implementation. The EMV Configuration Menu is shown below, followed by a description of the options.

```
BASE24-DCT  EMV CONFIGURATION MENU  LLLL      YY/MM/DD  HH:MM  01 OF 01

      EMV GROUP      0

      _ CURRENCY DATA OBJECTS TABLE
      _ TRANSACTION DATA OBJECTS TABLE
      _ TERMINAL DATA OBJECTS TABLE
      _ APPLICATION IDENTIFIER TABLE
      _ LANGUAGE SUPPORT TABLE

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
F1-SELECT F7-GOTO F10-PRINT F12-HELP F16-EXIT SF16-LOGOFF
```

EMV GROUP — The number used to identify a particular load group that can be downloaded to the ATM. Valid values are 0 through 9999.

Field Length: 1–4 numeric characters
Required Field: No
Default Value: 0

The following table briefly describes each of the four options on this screen:

Option	Description
CURRENCY DATA OBJECTS TABLE	Configures ISO currencies
TRANSACTION DATA OBJECTS TABLE	Configures ISO transaction types
TERMINAL DATA OBJECTS TABLE	Configures terminal country code and type
APPLICATION IDENTIFIER TABLE	Configures application identifiers
LANGUAGE SUPPORT TABLE	Configures terminal language

To select one of these options, place your cursor to the left of the option name and press the **F1** key. The screens for each of these options are described on the following pages.

Currency Data Objects Table

The Currency Data Objects Table determines the ISO currencies available in the EMV environment. The Currency Data Objects Table screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT  CURRENCY DATA TABLE  LLLL  YY/MM/DD  HH:MM  01 OF 01

      EMV GROUP      0

      CURRENCY      ISO CURRENCY      CURRENCY
      TYPE          CODE              EXPONENT
      00            000  ( ***)       00
      00            000  ( ***)       00
      00            000  ( ***)       00
      00            000  ( ***)       00
      00            000  ( ***)       00
      00            000  ( ***)       00
      00            000  ( ***)       00

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:  NEW LOGICAL NETWORK ID:
                  F12-HELP

```

CURRENCY TYPE — The currency type used in the transaction. Valid value range is 00 through FF.

Field Length: 2 alphanumeric characters
 Required Field: Yes
 Default Value: 00

ISO CURRENCY CODE — The ISO code of the available currency. Valid values are 000 through 999. The ISO 4217 standard, *Codes for the Representation of Currencies and Funds*, describes ISO standard currency codes.

Field Length: 3 numeric characters, followed by a system-protected text description
 Required Field: Yes
 Default Value: 000

CURRENCY EXPONENT — The currency exponent derived from the value of the ISO CURRENCY CODE field. The range of possible values is 00 through 99.

Field Length: System Protected

Transaction Data Objects Table

The Transaction Data Objects Table determines the available ISO transaction types. The Transaction Data Objects Table screen is shown below, followed by descriptions of its fields.

```
BASE24-DCT  TRANSACTION DATA TABLE  LLLL  YY/MM/DD  HH:MM  01 OF 01
```

```
EMV GROUP      0
```

ISO TRANSACTION TYPE	CATEGORY CODE
00 (UNKNOWN)	0
00 (UNKNOWN)	0
00 (UNKNOWN)	0
00 (UNKNOWN)	0
00 (UNKNOWN)	0
00 (UNKNOWN)	0
00 (UNKNOWN)	0
00 (UNKNOWN)	0
00 (UNKNOWN)	0
00 (UNKNOWN)	0

```
***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP
```

ISO TRANSACTION TYPE — The ISO code for an available transaction type. The ISO transaction type is represented by the first two digits of the processing code from the 1987 ISO 8583 standard, *Bank Card Originated Messages—Interchange Message Specifications—Content for Financial Transactions*. Valid values are 00 through 99.

Field Length:	2 numeric characters, followed by a system-protected text description
Occurs:	Up to 10 times
Required Field:	No
Default Value:	00

CATEGORY CODE — The Transaction Category Code. Values 0 through F can be entered in this field. However, the only supported value is Z.

Field Length: 1 alphanumeric character

Required Field: Yes

Default Value: 0

Terminal Data Objects Table

The Terminal Data Objects Table is used to configure the terminal country code and terminal type. The Terminal Data Table screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT  TERMINAL DATA TABLE  LLLL  YY/MM/DD  HH:MM  01 OF 01

      EMV GROUP      0

      TERMINAL COUNTRY CODE  000  (***)

      TERMINAL TYPE          14  (ONLINE ONLY  )

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                      F12-HELP

```

TERMINAL COUNTRY CODE — A code indicating the country of the terminal, according to the ISO 3166 standard, *Codes for the Representation of Names of Countries*. Valid values are 000 through 999.

Field Length: 3 numeric characters, followed by a system-protected text description

Required Field: Yes

Default Value: 000

TERMINAL TYPE — A code indicating the communications ability of the terminal. Valid values are as follows:

14 = Online only

15 = Offline with online capability

16 = Offline only

Field Length:	2 numeric characters, followed by a system-protected text description
Required Field:	Yes
Default Value:	14

Application Identifier Table

The Application Identifier (AID) Table identifies the applications that can be processed. The Application Identifier Table screen is shown below, followed by descriptions of its fields.

```

BASE24-DCT  APPLICATION ID TABLE  LLLL  YY/MM/DD  HH:MM  01 OF 01

EMV GROUP      0      AID NUMBER      0
PRIMARY AID
LOWEST  VERSION NUMBER  0000      DEFAULT APPL LABEL
HIGHEST VERSION NUMBER  0000
NUMBER OF DATA OBJECTS      0      TERMINAL ACTION CODE
DATA OBJECTS

NUMBER OF COMPLETION DATA OBJECTS  0
COMPLETION DATA OBJECTS

SECONDARY AIDs
____
____
____
____
____

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP

```

Note: On this screen, a value of 609 in the EMV GROUP field identifies an ATM using Wincor Nixdorf's NDC message format that is configured with the same states and tables as NCR's CAM II implementation. A valid value other than 609 in the EMV GROUP field identifies an NCR 5XXX ATM or any other ATM emulating the NCR 5XXX message.

For more information about these fields, refer to the *NCR NDC+ Programmer's Reference Manual* for NCR 5XXX terminals.

AID NUMBER — The number of the application. Valid values are 0 through 255.

Field Length: 1–3 numeric characters
 Required Field: Yes
 Default Value: 0

PRIMARY AID — The ID of the primary application. This field is left-justified, cannot contain embedded spaces, and must contain an even number of characters. Valid values are 0–9 and A–F.

Field Length: 0–32 hexadecimal characters
Required Field: No
Default Value: No default value

LOWEST VERSION NUMBER — The lowest version number of the primary application. Valid values are 0000 through FFFF.

Field Length: 4 alphanumeric characters
Required Field: No
Default Value: 0

DEFAULT APPL LABEL — A label used to identify an application process. This field is left-justified and can contain embedded spaces.

Field Length: 1–16 alphanumeric characters
Required Field: Yes
Default Value: 0

HIGHEST VERSION NUMBER — The highest version number of the primary application. Valid values are 0000 through FFFF.

Field Length: 4 alphanumeric characters
Required Field: No
Default Value: 0

NUMBER OF DATA OBJECTS — The number of data objects to be contained in the transaction request. Valid values are 0 through 34.

Field Length: 1–2 numeric characters
Required Field: No
Default Value: 0

TERMINAL ACTION CODE — The Terminal Action Code. Valid value range is 0000000000 through FFFFFFFF, blank characters are not allowed.

Field Length: 10 alphanumeric characters
Required Field: Yes
Default Value: 0000000000

DATA OBJECTS — The tags of the data objects to be contained in the transaction request. This field is left-justified and cannot contain embedded spaces. Valid values are 0–9 and A–F.

Field Length: 1–68 hexadecimal characters
Required Field: No
Default Value: No default value

NUMBER OF COMPLETION DATA OBJECTS — The number of data objects to be contained in the completion status message. Valid values are 00 through 27.

Field Length: 1–2 numeric characters
Required Field: No
Default Value: 0

COMPLETION DATA OBJECTS — The tags of the data objects to be contained in the completion status message. This field is left-justified and cannot contain embedded spaces. Valid values are 0–9 and A–F.

Field Length: 1–54 hexadecimal characters
Required Field: No
Default Value: No default value

SECONDARY AIDs — The IDs of up to twelve secondary application IDs. Each entry is left-justified, cannot contain embedded spaces, and must contain an even number of characters. This field must be filled from top to bottom, left to right. That is, if you enter three application IDs, you must enter the first two on line one and the third one on the left side of line two. Valid values are 0–9 and A–F.

Field Length: 2–32 hexadecimal characters
Occurs: 1–12 times

Required Field: No
Default Value: No default value

Language Support Table

The Language Support Table identifies the languages to be used on the screen and audio at a terminal. NCR 5XXX ATMs and ATMs emulating the NCR 5XXX message support the Language Support Table. However, ATMs using Wincor Nixdorf’s NDC message format that is configured with the same states and tables as NCR’s CAM II implementation do not support it.

The Language Support Table screen is shown below, followed by descriptions of its fields.

BASE24-DCT

LANGUAGE SUPPORT TABLE

LLLL

YY/MM/DD

HH:MM

01 OF 01

EMV GROUP

0

ISO LANGUAGE CODE

—

SCREEN BASE

—

AUDIO BASE

—

OP CODE POSN

—

OP CODE VAL

—

ISO LANGUAGE CODE

—

SCREEN BASE

—

AUDIO BASE

—

OP CODE POSN

—

OP CODE VAL

—

ISO LANGUAGE CODE

—

SCREEN BASE

—

AUDIO BASE

—

OP CODE POSN

—

OP CODE VAL

—

ISO LANGUAGE CODE

—

SCREEN BASE

—

AUDIO BASE

—

OP CODE POSN

—

OP CODE VAL

—

BASE24

NEW PAGE:

FILE DESTINATION:

NEW LOGICAL NETWORK ID:

F12-HELP

ISO LANGUAGE CODE — A code that specifies the language to be used. Valid values are AA through ZZ.

Field Length: 2 alphabetic characters
Required Field: No
Default Value: No default value

<language description> — A description of the language specified by the ISO LANGUAGE CODE.

Field Length: alphabetic characters
Required Field: No
Default Value: No default value

SCREEN BASE — A code that specifies the Screen Base to be used when the language code on the smart card matches a code in the language table. Valid values are 000 through 999.

Field Length: 3 numeric characters
Required Field: Yes
Default Value: 000

AUDIO BASE — A code that specifies the Audio Directory Base to be used at the terminal when the language code on the smart card matches a code in the language code table. Valid values are 000 through 007.

Field Length: 3 numeric characters
Required Field: Yes
Default Value: 000

OP CODE POSN — A code that specifies positions in the Op Code Buffer. Valid values are 0 through 7 for each of the three bytes, duplicates are not allowed (i.e., 011 is not allowed). If the chip language preference matches one of the language codes in the table, the three op code buffer positions indicated in this field are set to the value indicated in the OP CODE VAL field.

Field Length: 3 numeric characters
Required Field: Yes
Default Value: 000

OP CODE VAL — A code that specifies the Op Code Value to be used when the language code on the smart card matches a code in the language code table. The three op code buffer positions indicated in the OP CODE POSN field correspond to the values in this field. Valid values are A through I inclusive, excluding the value E, and including the @ character.

Field Length: numeric characters
Required Field: Yes
Default Value: 000

C: Shared Diebold 912 Device Support

The Diebold 10XX/478X Device Handler is an existing BASE24-atm process that provides the interface between Diebold 10XX/478X ATMs and BASE24. The Shared Diebold 912 EMV module is an additional licensable module which, when bound with the Diebold 10XX/478X Device Handler process, provides EMV support for ATMs that use Wincor Nixdorf's Diebold 912 message formats. There is also a Diebold 10XX/478X EMV module that can be bound with the Diebold 10XX/478X Device Handler process for Diebold 10XX/478X ATMs. A Diebold EMV module and a Shared Diebold 912 EMV module can be bound to the same Diebold 10XX/478X Device Handler process.

The Shared Diebold 912 EMV module scans request messages for EMV data. If EMV data is found in the message, the EMV module processes this data (i.e., checking its validity, adding EMV tokens, etc.) before sending the message to authorization.

The Shared Diebold 912 EMV module also handles any EMV data received from authorization and formats the appropriate message sent to the ATM. It also handles the printing of the application identifier and the application PAN sequence number on a receipt.

Processing of EMV configuration data load messages is handled by the EMV module. The EMV module reads the EMV application and EMV scheme data records from the configuration file (CNFGIX) and formats the message sent to the device.

The processing of EMV data in solicited and unsolicited status messages is done by the EMV module. In addition it also processes any error status codes generated by the EMV Smart Card Application.

Any special reversal processing required because of a failure from the device's EMV Smart Card Application is handled by the EMV module.

This appendix describes configuration screens required to support BASE24-atm EMV transactions. For general information about DCT functions, see the ***BASE24 Device Control Manual***. For general information about the CNFGIX Configuration File, see the ***BASE24-atm Diebold 10XX/478X Device Support Manual***.

BASE24-atm Diebold Entry Programs

The BASE24-atm Diebold entry programs access the Diebold Configuration File (CNFGIX) controls Diebold ATM display screens, states, timers, financial institution table (FIT) data, and printer configuration. All of the data in the CNFGIX can be modified through the entry programs.

The following Diebold entry programs are supported.

ENTIBOLD — Supports all Diebold states and cardholder screens and the FIT records, miscellaneous data, rear balance functions, consumer and statement printer configuration, and other information for Diebold and Wincor ATMs. This Diebold entry program does not support Wincor-specific states or tables. The program supports operation codes 070–078 for state SC (smart card). New Diebold-specific enhancements should be added to this program.

ENTIBLD2 — Supports all Diebold states, Wincor EMV state e, Wincor EMV AID table, cardholder screens, FIT records, miscellaneous data, timers, and printer configuration. This program supports operation codes 010, 011, 095, 097, and 099 for state SC (smart card). This program is provided for backwards compatibility and should not be updated for new enhancements.

ENTWBOLD — Supports all Wincor states, Wincor tables and cardholder screens. This program supports Wincor-specific screens, states, and tables only. It does not support Diebold-specific states or tables. Any new Wincor-specific enhancements should be added to this program.

The Diebold entry programs provide the mechanism to configure Diebold ATMs or Wincor ATMs using the BASE24-atm product. For comprehensive documentation regarding the Diebold or Wincor ATMs, consult the available vendor literature.

The Diebold entry programs allow you to define the following attributes:

- ATM display screens
- State tables
- Miscellaneous data
- Financial institution table
- Rear balance function definition data
- Consumer printer-configured text definition

- Statement printer-configured text definition
- Merchant Banking Center buttons and amounts
- EMV application data
- EMV scheme data
- Wincor EMV AID table

Note that the ENTWBOLD program is used to configure Wincor ATM display screens, Wincor State screens, and the Wincor EMV AID table. All other functions for Wincor devices must be configured through the ENTIBOLD program.

Refer to section 4 of the ***BASE24-atm Diebold 10XX/478X Device Support Manual*** for ENTIBOLD, ENTIBLD2, and ENTWBOLD screens and field descriptions.

Diebold DCT

The Diebold 10XX/478X DCT requester provides the interface between the main DCT and the DCT for Diebold 10XX/478X ATMs and ATMs using Wincor Nixdorf's D912 message format. The Diebold 10XX/478X DCT screen and requester support a load option of LOAD EMV DATA. The LOAD EMV DATA option enables an operator to download all EMV application and EMV scheme data to an ATM.

Device Configuration

The ATM must be configured for standard Diebold EMV Smart Card processing. Applications (i.e., application identifiers) and EMV Chip Card Schemes can be configured at the ATM or they can be configured using the Diebold Entry Program and downloaded to the ATM. Any EMV data not configured at the ATM must be configured using the Diebold Entry Program and downloaded to the ATM in order to support EMV. Configuration changes that are downloaded override any existing configuration data.

Refer to section 4 of the ***BASE24-atm Diebold 10XX/478X Device Support Manual*** for ENTIBOLD, ENTIBLD2, and ENTWBOLD screens and field descriptions.

Shared State Tables

The InterBold State Table screen is accessed by entering function code 2 for State Tables from the ENTIBOLD or ENTIBLD2 Diebold Entry Program Menu or from the ENTWBOLD Diebold Entry Program Shared/Wincor Support Main Menu. The InterBold State Table screen allows you to configure state tables and conditions that are associated with various ATM transaction states and groups.

The Shared State Select Menu can be accessed by entering the function code 2 on the ENTWBOLD Diebold Entry Program Shared/Wincor Support Main Menu and pressing the **F1** key. The Shared State Select Menu enables you to access the InterBold State Table screen to configure Wincor Shared states e, @r, and @q. After you enter one of these values in the State Type field plus the state number and group information, press the **F2** key to display the Interbold State Table screen.

The Shared State Table Menu is shown below, followed by a description of its fields.

```

                                SHARED STATE SELECT MENU

                                State Type

                                State Number  000   Group  0000

                                Select the State Type From the Following List

                                e  (Wincor EMV Chip Card)
                                @r (Wincor Enhanced CheckIn State)
                                @q (Wincor Enhanced CashIn State)

=====
                                F1=CHANGE STATE TYPE  F2=READ  F9=NEXT          F10=PRINT  F16=EXIT

```

State Type — The state type associated with the number in the State Number field. The BASE24-atm product supports the state types that Wincor has defined. When a BASE24-supported state type defined by Wincor is entered in this field, the necessary fields for that state type are listed on the screen.

Field Length: 2 alphanumeric characters, followed by a system-protected text description

Required Field: Yes

Default Value: No default value

The presently supported values defined by Wincor are listed below.

State Type Designation	Description
e	Wincor EMV Chip Card state
@r	Wincor Enhanced Check In state
@q	Wincor Enhanced Cash In state

The entry screens associated with the Wincor EMV Chip Card state appear later in this appendix. For more information about the Enhanced Check In state, refer to the *Wincor Nixdorf NDC/Diebold D91x Message Format Extension for Cheque In, June 13, 2007, Release 1.34*. For more information about the Enhanced Cash In state, refer to the *Wincor Nixdorf NDC_Diebold D91x Message Format Extension for CashIn V1.44*.

State Number — A number that is associated with the desired state. Valid values are 000 through 254, 256 through ZZZ.

Field Length: 3 alphanumeric characters
Required Field: No
Default Value: 000

Group — The number used to identify a particular load group (i.e., a collection of states, screens, tables, and parameters) that is downloaded to the ATM. Each group provides certain functionality, for example Group 0 is the basic group, Group 5 provides Fast Cash support, and so on.

Field Length: 4 alphanumeric characters
Required Field: No
Default Value: 0000

Wincor EMV Chip Card State Operation Code Entry Screen

The Wincor EMV Chip Card State Operation Code Entry screen is used to access the other Wincor EMV Chip Card State screens. The Wincor EMV Chip Card State Operation Code Entry screen is shown below, followed by descriptions of its fields.

```

                                INTERBOLD STATE TABLE
                                Type  e    Wincor EMV Chip Card State
                                State Number 000 Group 0000

Operation 000

Operation must be one of:
    000 -- Contact Chip and Candidate List Creation
    001 -- Customer EMV Application Selection
    002 -- Start/Restart EMV Application
    003 -- Initialize EMV Transaction Data
    004 -- Generate First Application Cryptogram (ARQC)
    005 -- Generate Second Application Cryptogram (AAC)
    006 -- EMV Language Selection

=====
F1=ENTER OPERATION CODE                      F10=PRINT                      F16=EXIT

```

Type — A code that identifies the type of states being defined. A value of e identifies the Wincor EMV Chip Card State screens, and cannot be modified from the Wincor EMV Chip Card State Operation Code Entry screen.

State Number — The number of the state for which the current operation code is being defined. Valid values are 0 through 254 and 256 through ZZZ.

Group — A value that identifies a load group containing configuration, communications keys, and miscellaneous information defined for a terminal. Valid values are 0 through 9999.

Operation Code — Identifies which Wincor EMV Chip Card State screen to display. Valid values are as follows:

- 000 = Contact Chip and Candidate List creation
- 001 = Customer EMV Application selection
- 002 = Start/Restart EMV Application
- 003 = Initialize EMV Transaction Data
- 004 = Generate First Application Cryptogram (ARQC)
- 005 = Generate Second Application Cryptogram (AAC)
- 006 = EMV Language Selection

Wincor EMV Chip Card State Contact Chip Card and Candidate List Creation Screen

The Wincor EMV Chip Card State Contact Chip Card and Candidate List Creation screen is used to specify the states and screens associated with processing contact chip cards and creating candidate lists. The Wincor EMV Chip Card State Contact Chip Card and Candidate List Creation screen is shown below.

```

                                INTERBOLD STATE TABLE
                                Type e   Wincor EMV Chip Card State
                                State Number 000 Group 0000

Operation 000 (Contact Chip Card and Candidate List creation)

Good Next State..... 000
No Chip/Chip Failure Next State..... 000
No Candidate Application Next State.... 000
EMV Application Error Next State..... 000
EMV Hardware Error Next State..... 000
Please Wait Screen..... 000
Processing Not Started Next State..... 000
Service Code Check Flag..... 000
Confirmed Candidate Checking Flag..... 000

=====
F1=CHANGE STATE TYPE  F2=READ    F3=ADD    F4=DELETE  F5=UPDATE
                      F9=NEXT    F10=PRINT F16=EXIT

```

Transaction processing at the ATM is controlled by the configuration of state tables. Refer to the Wincor vendor documentation for descriptions of the fields in the highlighted area on this screen.

Wincor EMV Chip Card State Customer EMV Application Selection Screen

The Wincor EMV Chip Card State Customer EMV Application Selection screen is used to specify the states and screens associated with the selecting the customer EMV application. The Wincor EMV Chip Card State Customer EMV Application Selection screen is shown below.

```

INTERBOLD STATE TABLE
Type e      Wincor EMV Chip Card State
State Number 000 Group 0000
Operation 001 (Customer EMV Application selection)

Application Selected Next State..... 000
One Application Available Next State..... 000
No Candidate Application Available Next State... 000
Timeout Next State..... 000
Cancel Next State..... 000
Application Selection Screen..... 000
Application Screen Template FDK A...  ___ B...  ___ C...  ___ D...  ___
                               F...  ___ G...  ___ H...  ___ I...  ___
Next Page FDK.....  ___
Next Page FDK Active Screen.....  ___
First Page FDK.....  ___
First Page FDK Active Screen.....  ___
FDK Priority Order for Application 1-3.....  ___
FDK Priority Order for Application 4-6.....  ___
FDK Priority Order for Application 7-8.....  ___

```

```

=====
F1=CHANGE STATE TYPE  F2=READ    F3=ADD    F4=DELETE  F5=UPDATE
                      F9=NEXT    F10=PRINT F16=EXIT

```

Transaction processing at the ATM is controlled by the configuration of state tables. Refer to the Wincor vendor documentation for descriptions of the fields in the highlighted area on this screen.

Wincor EMV Chip Card State Start/Restart EMV Application Screen

The Wincor EMV Chip Card State Start/Restart EMV Application screen is used to specify the states and screens associated with starting or restarting the EMV application. The Wincor EMV Chip Card State Start/Restart EMV Application screen is shown below.

```
INTERBOLD STATE TABLE
Type e Wincor EMV Chip Card State
State Number 000 Group 0000

Operation 002 (Start/Restart EMV Application)

Application (Re)Started Next State..... 000
Application Not (Re)Started Next State..... 000
EMV Application Error Next State..... 000
EMV Hardware Error Next State..... 000
Starting Application Screen..... 000
Screen Template for EMV Application Label..... 000
Screen Number to Store EMV Application Label.... 000

=====
F1=CHANGE STATE TYPE F2=READ F3=ADD F4=DELETE F5=UPDATE
F9=NEXT F10=PRINT F16=EXIT
```

Transaction processing at the ATM is controlled by the configuration of state tables. Refer to the Wincor vendor documentation for descriptions of the fields in the highlighted area on this screen.

Wincor EMV Chip Card State Initialize EMV Transaction Data Screen

The Wincor EMV Chip Card State Initialize EMV Transaction Data screen is used to configure EMV transaction data initialization. The Wincor EMV Chip Card State Initialize EMV Transaction Data screen is shown below.

```
INTERBOLD STATE TABLE
Type e      Wincor EMV Chip Card State
State Number 000 Group 0000

Operation 003 (Initialize EMV Transaction Data)

Next state..... 000
ISO Transaction Type..... 000
ISO Currency Code..... 000
ISO Currency Exponent..... 0
Amount Source Buffer Conversion Flag..... 0

=====
F1=CHANGE STATE TYPE  F2=READ      F3=ADD      F4=DELETE   F5=UPDATE
                      F9=NEXT      F10=PRINT   F16=EXIT
```

Transaction processing at the ATM is controlled by the configuration of state tables. Refer to the Wincor vendor documentation for descriptions of the fields in the highlighted area on this screen.

Wincor EMV Chip Card State Generate First Application Cryptogram (ARQC) Screen

The Wincor EMV Chip Card State Generate First Application Cryptogram (ARQC) screen is used to specify the states associated with generating the first application cryptogram. The Wincor EMV Chip Card State Generate First Application Cryptogram (ARQC) screen is shown below.

```
INTERBOLD STATE TABLE
Type e Wincor EMV Chip Card State
State Number 000 Group 0000

Operation 004 (Generate First Application Cryptogram (ARQC))

Good Next state..... 000
Declined Next state..... 000
Referral Next state..... 000
No Application Started Next State.... 000
EMV Application Error Next state..... 000
EMV Hardware Error Next state..... 000
EMV Sequence Error Next state..... 000

=====
F1=CHANGE STATE TYPE F2=READ F3=ADD F4=DELETE F5=UPDATE
F9=NEXT F10=PRINT F16=EXIT
```

Transaction processing at the ATM is controlled by the configuration of state tables. Refer to the Wincor vendor documentation for descriptions of the fields in the highlighted area on this screen.

Wincor EMV Chip Card State Generate Second Application Cryptogram (AAC) Screen

The Wincor EMV Chip Card State Generate Second Application Cryptogram (AAC) screen is used to specify the states associated with generating the second application cryptogram. The Wincor EMV Chip Card State Generate Second Application Cryptogram (AAC) screen is shown below.

```
INTERBOLD STATE TABLE
Type e Wincor EMV Chip Card State
State Number 000 Group 0000

Operation 005 (Generate Second Application Cryptogram (AAC))

Good Next State..... 000
No Application Started Next State..... 000
EMV Application Error Next State..... 000
EMV Hardware Error Next State..... 000
EMV Sequence Error Next State..... 000

=====
F1=CHANGE STATE TYPE F2=READ F3=ADD F4=DELETE F5=UPDATE
F9=NEXT F10=PRINT F16=EXIT
```

Transaction processing at the ATM is controlled by the configuration of state tables. Refer to the Wincor vendor documentation for descriptions of the fields in the highlighted area on this screen.

Wincor EMV Chip Card State EMV Language Selection Screen

The Wincor EMV Chip Card State EMV Language Selection screen is used to specify the states and language banks associated with selecting the EMV language. The Wincor EMV Chip Card State EMV Language Selection screen is shown below.

```

INTERBOLD STATE TABLE
Type e      Wincor EMV Chip Card State
State Number 000 Group 0000

Operation 006 (EMV language selection)

No Language Match Next State..... 000
Language 1 Matched Next State..... 000   Language Bank.... 000
Language 2 Matched Next State..... 000   Language Bank.... 000
Language 3 Matched Next State..... 000   Language Bank.... 000
Language 4 Matched Next State..... 000   Language Bank.... 000
Language 5 Matched Next State..... 000   Language Bank.... 000
Language 6 Matched Next State..... 000   Language Bank.... 000
Language 7 Matched Next State..... 000   Language Bank.... 000
Language 8 Matched Next State..... 000   Language Bank.... 000
Language 9 Matched Next State..... 000   Language Bank.... 000
Language 10 Matched Next State..... 000   Language Bank.... 000
Language 11 Matched Next State..... 000   Language Bank.... 000
Language 12 Matched Next State..... 000   Language Bank.... 000

=====
F1=CHANGE STATE TYPE  F2=READ      F3=ADD      F4=DELETE  F5=UPDATE
                      F9=NEXT      F10=PRINT   F16=EXIT

```

Transaction processing at the ATM is controlled by the configuration of state tables. Refer to the Wincor vendor documentation for descriptions of the fields in the highlighted area on this screen.

Wincor AID Table Screen

The Wincor AID Table screen is used to add or update a single application tag information entry in the configuration file. The Wincor AID Table screen is shown below.

WINCOR EMV AID TABLE

Record	Group
ID No.	AID
Lowest version	
Highest version	
TAC denial	
Data Object (TAG) list for Transaction request:	
Data Object (TAG) list for Solicited Status Message:	
AID Options	AID Default Label

=====

F2=READ F3=ADD F4=DELETE F5=UPDATE
F9=NEXT F10=PRINT F16=EXIT

Refer to the EMV vendor documentation for descriptions of the fields in the highlighted area on this screen.

Diebold 10XX Commands Screen

The Diebold 10XX Commands screen is used to enter commands that control the processing messages. The Diebold 10XX Commands screen is shown below.

```

BASE24-DCT   DIEBOLD 10XX COMMANDS   LNET           YY/MM/DD HH:MM   40 OF 46

  ATM TERMINAL ID: _____ FIID: _____

- LOAD GROUP           - LOAD SCREEN           - LOAD MAC KEY
- LOAD GROUP SCREEN    - LOAD STATE           - LOAD PUBLIC KEY
- LOAD GROUP STATE     - LOAD PIN KEY           - LOAD MASTER KEY
- LOAD EMV DATA      - LOAD ALL KEYS

THE FOLLOWING ARE ONLY APPLICABLE FOR 1000 TERMINALS WITH EMG
- COPY FONT           TO LIBRARY
- COPY ICON    0 TO LIBRARY
- COPY CELL DATA FROM LIBRARY    TO LIBRARY
- COPY  0 CELLS FROM LIBRARY  CELL  0 TO LIBRARY CELL  0
- RESTORE STANDARD CONFIGURATION

***** BASE24 *****
NEW PAGE:           FILE DESTINATION:           NEW LOGICAL NETWORK ID:
F1-ENTER  F3-OPEN F4-LOAD  F5-CLOSE  F7-GOTO F 9-RESET  F12-HELP  F14-PROOF LIST

```

When used with a Wincor ATM, the highlighted command field loads the Wincor EMV AID table.

ACI Worldwide, Inc.

D: Shared NDC+ Device Support

The NCR 5XXX Device Handler is an existing BASE24-atm process that provides the interface between NCR 5XXX ATMs and BASE24. The Shared NDC EMV module is an additional licensable module which, when bound with the NCR 5XXX Device Handler process, provides EMV support for ATMs that use either or both of the following Wincor Nixdorf NDC message formats:

- The format configured according to the initial version of the Shared NDC EMV implementation. See appendix B for the configuration screens required to support BASE24-atm EMV transactions with these ATMs.
- The later version as defined in Wincor Nixdorf's *NDC/Diebold D91x Message Format Extension for EMV*. This appendix contains the configuration screens required to support BASE24-atm EMV transactions with these ATMs.

The same Wincor Nixdorf hardware can automatically support both NDC message formats, depending on whether the Wincor EMV Chip Card state or the standard NCR EMV states are downloaded to an ATM. There is no need to load special software or make changes at the ATM site. Likewise, the Shared NDC+ EMV module dynamically determines which format is being used based on the configuration data that is downloaded to the ATM.

There is also an NCR 5XXX EMV module that can be bound with the NCR 5XXX Device Handler process for NCR 5XXX ATMs. An NCR EMV module and a Shared NDC+ EMV module can be bound to the same NCR 5XXX Device Handler process.

NCR ATMs and ATMs adhering to Wincor Nixdorf's shared NDC+ message formats are configured using software and files supplied by their vendors and contained within the ATMs themselves. BASE24 device support emulates the NCR configuration file using the CNFG5070 Configuration File, which can be accessed using the BASE24 Device Control Terminal (DCT). With access to the CNFG5070, you can configure EMV-specific parameters and state table entries and then download these parameters to an NCR ATM or an ATM adhering to either of Wincor Nixdorf's shared NDC+ message formats.

This appendix describes the configuration screens required to support BASE24-atm EMV transactions with the Shared NDC+ EMV module. For general information about DCT functions, see the ***BASE24 Device Control Manual***. See the ***NCR NDC+ Programmer's Reference Manual*** for specific information about the configuration file parameters.

For general information about the CNFG5070 Configuration File, see the ***BASE24-atm NCR 5XXX Device Support Manual***, and see the ***Wincor Nixdorf NDC/Diebold D91x Message Format Extension for EMV*** for specific information about the configuration file parameters.

CNFG5070 Configuration File

To access the configuration file from the DCT, select screen 45 (ATM NCR 5XXX COMMANDS) from the DCT Product Menu. The NCR 5XXX Commands screen appears.

- If you want to download EMV-specific configuration information, select DOWNLINE LOAD DATA and press the **F1** key. The NCR 5XXX Load Commands screen appears.
- If you want to enter configuration parameters or state table data, select ACCESS CONFIGURATION FILE and press the **F1** key. The NCR 5XXX Configuration File Menu appears.

The NCR 5XXX Load Commands screen and the NCR 5XXX Configuration File Menu are described in this appendix, as are EMV-related screens accessed from the NCR 5XXX Configuration File Menu.

NCR 5XXX Load Commands

The NCR 5XXX Load Commands screen allows you to download specified information to NCR 5XXX ATMs. One field on this screen is specific to the BASE24-atm EMV product. The NCR 5XXX Load Commands screen is shown below, followed by a description of the LOAD EMV GROUP field.

```

BASE24-DCT   NCR 5XXX LOAD COMMANDS   LLLL           YY/MM/DD  HH:MM  01 OF 01
ATM TERMINAL ID: _____  LOAD GROUP:      0      FIID: _____

  _ LOAD ALL (INCLUDING KEYS)           _ LOAD PUBLIC KEY
  _ LOAD ALL IN GROUP                   _ LOAD MASTER KEY
  _ LOAD ALL SCREENS                     _ LOAD ALL KEYS
  _ LOAD ALL STATES                      LOAD KEY ENTRY MODE 1
  _ LOAD ALL FITS
  _ LOAD DATE AND TIME
  _ LOAD MAC KEY
  _ LOAD KEY TYPE 1
  _ LOAD CUST SCREEN    0      LANG 0
  _ LOAD C-SCREEN      C  0
  _ LOAD R-SCREEN      R  0
  _ LOAD STATE         0
  _ LOAD FIT           0
  _ LOAD EMV GROUP      0

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
F1-ENTER F2-CNFG F3-CNTRS F4-OPEN F5-CLOSE F6-LOAD F7-RESET F8-CPLF F16-EXIT

```

LOAD EMV GROUP — The number of the EMV group to be downloaded to the ATM from the customized load group specified in the LOAD GROUP field. The range of valid values is 0 through 9999. After entering the group number, place the cursor to the left of the field name and press the **F6** key.

Field Length: 1–4 numeric characters
 Required Field: No
 Default Value: 0

NCR 5XXX Configuration File

Two menu items highlighted on the following NCR 5XXX Configuration File Menu contain fields that are specific to the BASE24-atm EMV product with ATMs that use the version of Wincor Nixdorf's NDC message format that is defined in their *NDC/Diebold D91x Message Format Extension for EMV*. These menu items are described below.

```

BASE24-DCT   NCR 5XXX CONFIG FILE       LLLL       YY/MM/DD  HH:MM  01 OF 01

CONFIGURATION FILE NAME \SYS1.$DATA.PRO1DATA.CNFG5070

GROUP NUMBER      0

  _ ENHANCED CONFIGURATION PARAMETERS
  _ STATE
  _ SCREEN
  _ FINANCIAL INSTITUTION TABLES
  _ SUPERVISOR CARD DATA
  _ PRINTER CONTROL SCREENS (Kxx SCREENS)
  _ CONFIGURATION IDENTITY NUMBER
  _ HOPPER MAPPING TABLE
  _ EMV CONFIGURATION MENU
  _ WINCOR EMV AID TABLE

  _ PRINT

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
F1-SELECT F7-GOTO F10-PRINT F12-HELP F16-EXIT SF16-LOGOFF

```

STATE

Lists the state screens needed to configure NCR terminals.

WINCOR EMV AID TABLE

Configures the Wincor application identifiers.

To select one of these menu items, place the cursor next to the function and press the **F1** key.

State Table Data Menu

The State Table Data Menu lists the state screens needed to configure the NCR terminals. This screen is displayed when you place the cursor to the left of the STATE option on the NCR 5XXX Configuration File Menu and press the **F1** key. The highlighted field shown on the screen below is specific to the BASE24-atm EMV product for ATMs using the version of Wincor Nixdorf's NDC message format that is defined in their *NDC/Diebold D91x Message Format Extension for EMV*, and is described on the following pages. The Global Extension state screen is also described because it is used with this EMV-specific state.

```

BASE24-DCT   STATE TABLE DATA   LLLL   YY/MM/DD   HH:MM   01 OF 42

      GROUP      0      STATE NUMBER      0

2:42 (A-CARD READ)           3:42 (B-PIN ENTRY)           4:42 (C-UNLOCK ENV DISP)
5:42 (D-PRESET OP-CODE)      6:42 (E-4 FDK SELECT)       7:42 (F-AMOUNT ENTRY)
8:42 (G-AMOUNT CHECK)        9:42 (H-4 FDK INFO)        10:42 (I-TRAN REQUEST)
11:42 (J-CLOSE)              12:42 (K-FIT SWITCH)       13:42 (L-CARD WRITE)
14:42 (M-ENH PIN ENTRY)      15:42 (N-CAMERA CONTROL)   16:42 (P-DPM CAV)
17:42 (Q-DPM ACCEPT DOC)     18:42 (R-ENH AMT ENTRY)    19:42 (S-LANG SWITCH)
20:42 (T-CRD READ PIN INIT)  21:42 (V-LANG FROM CARD)   22:42 (W-FDK SWITCH)
23:42 (X-8 FDK INFO)         24:42 (Y-8 FDK SELECT)     25:42 (Z-GLOBAL EXT)
26:42 (ENH FIT SWITCH)       27:42 (i-AUDIO CONTROL)    28:42 (k-SMART FIT CHECK)
29:42 (+-BEGIN EMV INIT)     30:42 (,-COMPL EMV INIT)   31:42 (;-EMV RE-INIT)
32:42 (--AUTO LANG SELCT)    33:42 (?-SET EMV TXN DATA) 34:42 (.-BEGIN EMV SELCT)
35:42 (/ -COMPL EMV SELCT)   36:42 (b-CUST SELCT PIN)   37:42 (>-CASH ACCEPT)
38:42 (w-CHECK ACCEPT)       39:42 (e-WINCOR EMV CARD) 40:42 (m-PIN & LANG SLCT)
41:42 (r-SHARED BNA)         42:42 (q-SHARED CHECK IN)

```

TYPE THE NUMBER OF THE DESIRED FUNCTION IN NEW PAGE FIELD AND KEY F7

```

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP

```

To access a screen, enter the appropriate screen number (the number to the left of the colon) in the NEW PAGE field and press the **F7** key. Each of these EMV-specific screens is described on the following pages.

Wincor EMV Chip Card State Screen Navigation

Seven sets of input fields are available for the Wincor EMV Chip Card state screen. The value you enter in the OPERATION NUMBER field on the Wincor EMV Chip Card state screen controls which set of input fields is displayed for you. The appropriate set of input fields is displayed when you enter a value in the OPERATION NUMBER field and press the **F1** (validate), **F3** (add), or **F5** (update) key. As a result, this appendix contains seven illustrations of the Wincor EMV Chip Card state screen, with the corresponding field descriptions following each illustration. Valid values for the OPERATION NUMBER field are as follows:

- 000 = Contact chip and candidate list creation (default value)
- 001 = Customer EMV application selection
- 002 = Start/restart EMV application
- 003 = Initialize EMV transaction data
- 004 = Generate first application cryptogram
- 005 = Generate second application cryptogram
- 006 = EMV language selection

Wincor EMV Chip Card State with Values for Contact Chip and Candidate List Creation

The values shown on this version of the Wincor EMV Chip Card state are used to contact the chip and create the application candidate list.

Note: This version of the Wincor EMV Chip Card state screen has an extension state associated with it. This extension state is accessed through the Global Extension screen (state Z), which is described later in this section.

The Wincor EMV Chip Card state screen with values for contact chip and candidate list creation is shown below, followed by descriptions of its fields.

```

BASE24-DCT   STATE TABLE DATA           LLLL           YY/MM/DD  HH:MM  39 OF 42

      GROUP      0      STATE NUMBER      0

                                WINCOR EMV CHIP CARD

STATE  OPERATION
TYPE  NUMBER
  e    000 CONTACT CHIP AND CANDIDATE LIST CREATION
        NO ERROR NEXT STATE                0
        NO CHIP/FAILURE NEXT STATE          0
        NO APPL CANDIDATE NEXT STATE        0
        EMV APPL ERROR NEXT STATE           0
        EMV HARDWARE ERROR NEXT STATE       0
        EXTENSION STATE                      0
        (NOT USED)                          0

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP
  
```

STATE TYPE — The value e identifies the Wincor EMV Chip Card state screen.

Field Length: System Protected

OPERATION NUMBER — A code that selects which operation you want to perform. The valid value for this version of the screen is 000.

Field Length: 3 numeric characters

Required Field: Yes

Default Value: 000

NO ERROR NEXT STATE — The number of the state to which the terminal proceeds if EMV processing was successful. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

NO CHIP/FAILURE NEXT STATE — The number of the state to which the terminal proceeds if the card has no chip or if failure conditions such as dirty contacts are detected. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

NO APPL CANDIDATE NEXT STATE — The number of the state to which the terminal proceeds if no application candidate is found to be processed. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

EMV APPL ERROR NEXT STATE — The number of the state to which the terminal proceeds if an EMV application-level error occurs. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

EMV HARDWARE ERROR NEXT STATE — The number of the state to which the terminal proceeds if an EMV hardware-level error occurs. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

EXTENSION STATE — The number of the extension state for this operation. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

Wincor EMV Chip Card State with Values for Customer EMV Application Selection

The values shown on this version of the Wincor EMV Chip Card state screen enable the customer to select an EMV application for further processing.

Note: This version of the Wincor EMV Chip Card state screen has two extension states associated with it. These extension states are accessed through the Global Extension screen (state Z), which is described later in this section.

The Wincor EMV Chip Card state screen with values for customer EMV application selection is shown below, followed by descriptions of its fields.

```

BASE24-DCT  STATE TABLE DATA          LLLL          YY/MM/DD  HH:MM  39 OF 42

      GROUP      0      STATE NUMBER      0

                                WINCOR EMV CHIP CARD

STATE  OPERATION
TYPE  NUMBER
e      001 CUSTOMER EMV APPLICATION SELECTION
      APPL SELECTED NEXT STATE          0
      ONE APPL CANDIDATE NEXT STATE     0
      NO APPL CANDIDATE NEXT STATE      0
      TIMEOUT NEXT STATE                 0
      CANCEL NEXT STATE                  0
      APPL SELECTION SCREEN NUMBER      0
      EXTENSION STATE                    0

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP

```

STATE TYPE — The value e identifies the Wincor EMV Chip Card state screen.

Field Length: System Protected

OPERATION NUMBER — A code that selects which operation you want to perform. The valid value for this version of the screen is 001.

Field Length: 3 numeric characters

Required Field: Yes
Default Value: 000

APPL SELECTED NEXT STATE — The number of the state to which the terminal proceeds if the customer selects an application. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

ONE APPL CANDIDATE NEXT STATE — The number of the state to which the terminal proceeds if there is only one application candidate available. This application is selected automatically without asking the customer. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

NO APPL CANDIDATE NEXT STATE — The number of the state to which the terminal proceeds if there is no application candidate available. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TIMEOUT NEXT STATE — The number of the state to which the terminal proceeds if the customer times out. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

CANCEL NEXT STATE — The number of the state to which the terminal proceeds if the customer presses the cancel key. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

APPL SELECTION SCREEN NUMBER — The number of the screen the terminal displays if a customer has to select an application. This screen is not shown if there are less than two application candidates available.

If this screen number is set to 000, the application with the highest priority is started automatically and the EXTENSION STATE field is not used. Valid values are 0 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EXTENSION STATE — The extension state number for application selection. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Wincor EMV Chip Card State with Values for Starting and Restarting the EMV Application

The values shown on this version of the Wincor EMV Chip Card state screen enable the EMV application currently being selected or the application candidate with the highest priority for processing to be started or restarted.

The Wincor EMV Chip Card state screen for starting and restarting the EMV application is shown below, followed by descriptions of its fields.

```

BASE24-DCT  STATE TABLE DATA          LLLL          YY/MM/DD  HH:MM  39 OF 42

      GROUP      0      STATE NUMBER      0

                                WINCOR EMV CHIP CARD

STATE  OPERATION
TYPE  NUMBER
  e    002 START/RESTART EMV APPLICATION
        APPL (RE)STARTED NEXT STATE      0
        APPL NOT (RE)STARTED NXT STATE   0
        EMV APPL ERROR NEXT STATE        0
        EMV HARDWARE ERROR NEXT STATE    0
        'STARTING APPL' SCREEN NUMBER    0
        EMV APPL TEMPLATE SCREEN NUMBER  0
        EMV APPL LABEL SCREEN NUMBER     0

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                   F12-HELP

```

STATE TYPE — The value e identifies the Wincor EMV Chip Card state screen.

Field Length: System Protected

OPERATION NUMBER — A code that selects which operation you want to perform. The valid value for this version of the screen is 002.

Field Length: 3 numeric characters
 Required Field: Yes
 Default Value: 000

APPL (RE)STARTED NEXT STATE — The number of the state to which the terminal proceeds if an application is started or restarted successfully. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

APPL NOT (RE)STARTED NXT STATE — The number of the state to which the terminal proceeds if the application cannot be started or restarted. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EMV APPL ERROR NEXT STATE — The number of the state to which the terminal proceeds if there is an EMV application-level error such as application blocked or mandatory tags (94, 82, 57, 8C, 8D, 5A, and 5F24) are not available. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EMV HARDWARE ERROR NEXT STATE — The number of the state to which the terminal proceeds if an EMV hardware-level error occurs. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

‘STARTING APPL’ SCREEN NUMBER — The number of the screen to display to the customer while starting the application. If the screen number is set to 000, no screen is displayed. Valid values are 0 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EMV APPL TEMPLATE SCREEN NUMBER — The number of the screen template to be used to format the application label. Valid values are 0 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EMV APPL LABEL SCREEN NUMBER — The number of the screen where the formatted application label should be stored for further use. Valid values are 0 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Wincor EMV Chip Card State with Values for Initializing EMV Transaction Data

The values shown on this version of the Wincor EMV Chip Card state screen provide transaction-specific data for EMV processing.

The Wincor EMV Chip Card state screen with values for initializing EMV transaction data is shown below, followed by descriptions of its fields.

```

BASE24-DCT  STATE TABLE DATA          LLLL      YY/MM/DD  HH:MM  39 OF 42

      GROUP      0      STATE NUMBER      0

                                WINCOR EMV CHIP CARD

STATE  OPERATION
TYPE  NUMBER
e     003 INITIALIZE EMV TRANSACTION DATA
      NEXT STATE                                0
      ISO TRANSACTION TYPE                      0
      ISO CURRENCY CODE                        0
      ISO CURRENCY EXPONENT                    0
      AMOUNT SOURCE BUFFER/CONV FLAG           0
      (NOT USED)                              0
      (NOT USED)                              0

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP

```

STATE TYPE — The value e identifies the Wincor EMV Chip Card state screen.

Field Length: System Protected

OPERATION NUMBER — A code that selects which operation you want to perform. The valid value for this version of the screen is 003.

Field Length: 3 numeric characters
 Required Field: Yes
 Default Value: 000

NEXT STATE — The number of the state to which the terminal proceeds after initialization of all necessary data. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

ISO TRANSACTION TYPE — The decimal value of the ISO Transaction Type tag (9C) to set. Valid values are 0 through 100. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

ISO CURRENCY CODE — The value to set for the Transaction Currency Code tag (5F2A). Valid values are 0 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

ISO CURRENCY EXPONENT — The value to set for the Transaction Currency Exponent tag (5F36). Valid values are 0 through 010. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

AMOUNT SOURCE BUFFER/CONV FLAG — The buffer to be used for amount and if conversion of currency units has to be performed. Valid values are as follows:

000 = Do not set amount
1 or 10 = Set amount to 0
2 or 20 = Set amount from Amount Buffer with no conversion
3 or 30 = Set amount from General Purpose Buffer B with no conversion
4 or 40 = Set amount from General Purpose Buffer C with no conversion

5 or 50 = Set amount from Amount Buffer with conversion
6 or 60 = Set amount from General Purpose Buffer B with conversion
7 or 70 = Set amount from General Purpose Buffer C with conversion

This value specifies the number of trailing zeros to be added to the amount before setting the transaction amount. This field can be zero filled to the left.

The value in the ISO CURRENCY EXPONENT field is used if a currency unit conversion is necessary.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Wincor EMV Chip Card State with Values for Generating the First Application Cryptogram

The values shown on this version of the Wincor EMV Chip Card state screen determine the next action to be performed to generate the first application cryptogram.

The Wincor EMV Chip Card state screen with values for generating the first application cryptogram is shown below, followed by descriptions of its fields.

```
BASE24-DCT  STATE TABLE DATA      LLLL      YY/MM/DD  HH:MM  39 OF 42

      GROUP      0      STATE NUMBER      0

                        WINCOR EMV CHIP CARD

STATE  OPERATION
TYPE  NUMBER
e      004 GENERATE FIRST APPLICATION CRYPTOGRAM
      ARQC RETURNED NEXT STATE      0
      AAC RETURNED NEXT STATE      0
      AAR RETURNED NEXT STATE      0
      NO APPL STARTED NEXT STATE    0
      EMV APPL ERROR NEXT STATE     0
      EMV HARDWARE ERROR NEXT STATE 0
      EMV SEQUENCE ERROR NEXT STATE 0

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP
```

STATE TYPE — The value e identifies the Wincor EMV Chip Card state screen.

Field Length: System Protected

OPERATION NUMBER — A code that selects which operation you want to perform. The valid value for this version of the screen is 004.

Field Length: 3 numeric characters

Required Field: Yes

Default Value: 000

ARQC RETURNED NEXT STATE — Specifies the state number if the chip replied with ARQC. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

AAC RETURNED NEXT STATE — The number of the state to which the terminal proceeds if the chip replies with AAC (decline). Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

AAR RETURNED NEXT STATE — The number of the state to which the terminal proceeds if the chip replies with AAR (referral). Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

NO APPL STARTED NEXT STATE — The number of the state to which the terminal proceeds if no application is started. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EMV APPL ERROR NEXT STATE — The number of the state to which the terminal proceeds if an EMV application-level error occurs. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EMV HARDWARE ERROR NEXT STATE — The number of the state to which the terminal proceeds if an EMV hardware-level error occurs. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

EMV SEQUENCE ERROR NEXT STATE — The number of the state to which the terminal proceeds if the current conditions do not allow processing of this state/operation. This is the case if the EMV transaction data is not initialized. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

Wincor EMV Chip Card State with Values for Generating the Second Application Cryptogram

The values shown on this version of the Wincor EMV Chip Card state screen determine the next action to be performed to cancel an EMV transaction.

The Wincor EMV Chip Card state screen with values for generating the second application cryptogram is shown below, followed by descriptions of its fields.

```

BASE24-DCT  STATE TABLE DATA          LLLL      YY/MM/DD  HH:MM  39 OF 42

      GROUP      0      STATE NUMBER      0

                                WINCOR EMV CHIP CARD

STATE  OPERATION
TYPE  NUMBER
e      005 GENERATE SECOND APPLICATION CRYPTOGRAM
      NO ERROR NEXT STATE                0
      EMV APPL ERROR NEXT STATE          0
      EMV HARDWARE ERROR NEXT STATE      0
      EMV SEQUENCE ERROR NEXT STATE      0
      (NOT USED)                         0
      (NOT USED)                         0
      (NOT USED)                         0

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP

```

STATE TYPE — The value e identifies the Wincor EMV Chip Card state screen.

Field Length: System Protected

OPERATION NUMBER — A code that selects which operation you want to perform. The valid value for this version of the screen is 005.

Field Length: 3 numeric characters
 Required Field: Yes
 Default Value: 000

NO ERROR NEXT STATE — The number of the state to which the terminal proceeds if no error occurs. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EMV APPL ERROR NEXT STATE — The number of the state to which the terminal proceeds if an EMV application-level error occurs. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EMV HARDWARE ERROR NEXT STATE — The number of the state to which the terminal proceeds if an EMV hardware-level error occurs. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

EMV SEQUENCE ERROR NEXT STATE — The number of the state to which the terminal proceeds if the current conditions do not allow processing of this state/operation. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Wincor EMV Chip Card State with Values for EMV Language Selection

The values shown on this version of the Wincor EMV Chip Card state screen are used to set the language offset and group from the EMV preferred language stored in the chip.

Note: This version of the Wincor EMV Chip Card state screen has an extension state associated with it. This extension state is accessed through the Global Extension screen (state Z), which is described later in this section.

The Wincor EMV Chip Card state screen with values for EMV language selection is shown below, followed by descriptions of its fields.

```
BASE24-DCT   STATE TABLE DATA           LLLL           YY/MM/DD  HH:MM  39 OF 42
```

```
GROUP      0      STATE NUMBER      0
```

```
WINCOR EMV CHIP CARD
```

```
STATE      OPERATION
TYPE      NUMBER
```

```
e      006 EMV LANGUAGE SELECTION
      NO LANGUAGE MATCHED NEXT STATE      0
      LANGUAGE 1-2 EXTENSION STATE      0
      LANGUAGE 3-4 EXTENSION STATE      0
      LANGUAGE 5-6 EXTENSION STATE      0
      LANGUAGE 7-8 EXTENSION STATE      0
      LANGUAGE 9-10 EXTENSION STATE      0
      LANGUAGE 11-12 EXTENSION STATE      0
```

```
***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
              F12-HELP
```

STATE TYPE — The value e identifies the Wincor EMV Chip Card state screen.

Field Length: System Protected

OPERATION NUMBER — A code that selects which operation you want to perform. The valid value for this version of the screen is 006.

Field Length: 3 numeric characters
Required Field: Yes
Default Value: 000

NO LANGUAGE MATCHED NEXT STATE — The number of the state to which the terminal proceeds if no language match is found. Valid values are 0 through 254 and 256 through 999. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LANGUAGE 1-2 EXTENSION STATE — The extension state that identifies the next state, language offset, and language group to be set if language 1 or 2 is matched. Valid values are 0 through 254 and 256 through 999. The entry is not used if the parameter is set to 000. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LANGUAGE 3-4 EXTENSION STATE — The extension state that identifies the next state, language offset, and language group to be set if language 3 or 4 is matched. Valid values are 0 through 254 and 256 through 999. The entry is not used if the parameter is set to 000. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LANGUAGE 5-6 EXTENSION STATE — The extension state that identifies the next state, language offset, and language group to be set if language 5 or 6 is matched. Valid values are 0 through 254 and 256 through 999. The entry is not used if the parameter is set to 000. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LANGUAGE 7-8 EXTENSION STATE — The extension state that identifies the next state, language offset, and language group to be set if language 7 or 8 is matched. Valid values are 0 through 254 and 256 through 999. The entry is not used if the parameter is set to 000. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LANGUAGE 9-10 EXTENSION STATE — The extension state that identifies the next state, language offset, and language group to be set if language 9 or 10 is matched. Valid values are 0 through 254 and 256 through 999. The entry is not used if the parameter is set to 000. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

LANGUAGE 11-12 EXTENSION STATE — The extension state that identifies the next state, language offset, and language group to be set if language 11 or 12 is matched. Valid values are 0 through 254 and 256 through 999. The entry is not used if the parameter is set to 000. This field can be zero filled to the left.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Global Extension Screen

The Global Extension screen is a generic screen that is used to access extension state table data for all states that have an extension state. An extension state is used to extend, or expand, the functions of a particular state. The state tables in general only allow eight values to be set. For those states requiring more than eight configuration values, the global extension table is used to hold the excess values.

Fields on the following versions of the Wincor EMV Chip Card state screen use the Global Extension screen:

- The contact chip and candidate list creation version (operation 000) contains the Extension State Number.
- The customer EMV application selection version (operation 001) contains the Extension State Number. In this case, the Global Extension screen contains a field that identifies another Global Extension screen.
- The EMV language selection version (operation 006) contains the Language Extension Number for each of six pairs of languages.

The Global Extension screen is used to input extension state table data for all of the extension states mentioned above. The additional configuration values on the Global Extension screen have generic field names, and the meaning of a specific field changes depends on the original state screen that references this screen.

The table entry fields on the Global Extension screen correspond to the extension state table entries described in the *Wincor Nixdorf NDC/Diebold D91x Message Format Extension for EMV*.

The contents of the table entry fields for each of the extension states are described on the pages following the Global Extension screen.

The Global Extension screen is accessed from the State Table Data Menu by entering a value of 25 in the NEW PAGE field and pressing the **F7** key. The Global Extension screen is shown on the following page, followed by field descriptions for each extension state.

BASE24-DCT STATE TABLE DATA LLLL YY/MM/DD HH:MM 25 OF 42

GROUP 0 STATE NUMBER 0

GLOBAL EXTENSION

STATE TYPE	TABLE ENTRY-2	TABLE ENTRY-3	TABLE ENTRY-4	TABLE ENTRY-5	TABLE ENTRY-6	TABLE ENTRY-7
Z	0	0	0	0	0	0

TABLE ENTRY-8	TABLE ENTRY-9
0	0

***** BASE24 *****

NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:
	F12-HELP	

Extension State on the Contact Chip and Candidate List Creation Version

The following fields are used with the EXTENSION STATE field on the contact chip and candidate list creation version of the Wincor EMV Chip Card State screen. Each field is input as a table entry on the Global Extension screen. For example, the “Please Wait” Screen (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen. The fields TABLE ENTRY-5 through TABLE ENTRY-9 field on the Global Extension state screen are not used with this version of the Wincor EMV Chip Card State screen and must be set to 0.

TABLE ENTRY-2 (“Please Wait” Screen) — The number of the screen to show while the chip is contacted and the candidate list is being created. The value 000 can be used to not show any screen. Valid values are 000 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-3 (Next State if Processing Not Started) — The next state taken when the service code indicates a non-EMV card (code 4). Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-4 (Flag for Service Code Checking) — Indicates whether the service code on the magnetic track 2 should be checked. Valid values are as follows:

000 = Always start EMV processing
001 = Start EMV processing only if the magnetic track 2 data has been read and contains a service code indicating that this card has a chip (code 2 or 6)

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Extension State on the Customer EMV Application Selection Version

The following fields are used with the EXTENSION STATE field on the customer EMV application selection version of the Wincor EMV Chip Card State screen. Each field is input as a table entry on the Global Extension screen. For example, the Extension State Number for FDK Usage (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen.

TABLE ENTRY-2 (Extension State Number for FDK Usage) — The extension state number where the function display keys (FDKs) for the application selection are configured. Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-3 (Next Page FDK) — The FDK to be used to access the next page of selectable applications. Valid values are 001 through 004 and 006 through 009.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-4 (Next Page Active Screen Number) — The screen number that will be displayed when the FDK for the next page is activated. Valid values are 000 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-5 (First Page FDK) — The FDK to be used to access the first page of selectable applications. Valid values are 001 through 004 and 006 through 009.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-6 (First Page Active Screen Number) — The screen number that will be displayed when the FDK for the first page is activated. Valid values are 000 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-7 (FDK Priority Order for Applications 1, 2, and 3) — Specifies the assignment of selectable applications 1, 2, and 3 to FDKs. Valid values are 111 through 999. If TABLE ENTRY-7 through TABLE ENTRY-9 are set to 000, a default value of 123 is used for TABLE ENTRY-7.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-8 (FDK Priority Order for Applications 4, 5, and 6) — Specifies the assignment of selectable applications 4, 5, and 6 to FDKs. Valid values are 111 through 999. If TABLE ENTRY-7 through TABLE ENTRY-9 are set to 000, a default value of 456 is used for TABLE ENTRY-8.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-9 (FDK Priority Order for Applications 7, 8, and 9) — Specifies the assignment of selectable applications 7 and 8 to FDKs. Valid values are 110 through 990. If TABLE ENTRY-7 through TABLE ENTRY-9 are set to 000, a default value of 890 is used for TABLE ENTRY-9.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Extension State Number for FDK Usage on the Global Extension Screen

The following fields are used with the Extension State Number for FDK Usage field on the Global Extension screen that has already been used with the customer EMV application selection version of the Wincor EMV Chip Card State screen. Each field is input as a table entry on the Global Extension screen. For example, the Application Screen Template Number for FDK A (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen.

TABLE ENTRY-2 (Application Screen Template Number for FDK A) — This entry configures the Screen Template number to be used to display the application label (tag 50) for function display key (FDK) A. Valid values are 000 through 999. A value of 000 indicates that FDK A is not activated or used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-3 (Application Screen Template Number for FDK B) — This entry configures the Screen Template number to be used to display the application label (tag 50) for FDK B. Valid values are 000 through 999. A value of 000 indicates that FDK B is not activated or used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-4 (Application Screen Template Number for FDK C) — This entry configures the Screen Template number to be used to display the application label (tag 50) for FDK C. Valid values are 000 through 999. A value of 000 indicates that FDK C is not activated or used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-5 (Application Screen Template Number

for FDK D) — This entry configures the Screen Template number to be used to display the application label (tag 50) for FDK D. Valid values are 000 through 999. A value of 000 indicates that FDK D is not activated or used.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

TABLE ENTRY-6 (Application Screen Template Number

for FDK F) — This entry configures the Screen Template number to be used to display the application label (tag 50) for FDK F. Valid values are 000 through 999. A value of 000 indicates that FDK F is not activated or used.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

TABLE ENTRY-7 (Application Screen Template Number

for FDK G) — This entry configures the Screen Template number to be used to display the application label (tag 50) for FDK G. Valid values are 000 through 999. A value of 000 indicates that FDK G is not activated or used.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

TABLE ENTRY-8 (Application Screen Template Number

for FDK H) — This entry configures the Screen Template number to be used to display the application label (tag 50) for FDK H. Valid values are 000 through 999. A value of 000 indicates that FDK H is not activated or used.

Field Length: 1–3 numeric characters

Required Field: No

Default Value: 0

TABLE ENTRY-9 (Application Screen Template Number

for FDK I) — This entry configures the Screen Template number to be used to display the application label (tag 50) for FDK I. Valid values are 000 through 999. A value of 000 indicates that FDK I is not activated or used.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Language Extension Number on the Wincor EMV Chip Card State Screen for EMV Language Selection

The following fields are used with the LANGUAGE EXTENSION NUMBER fields on the EMV language selection version of the Wincor EMV Chip Card State screen. Each field is input as a table entry on the Global Extension screen. For example, the Next State If Language Matched (shown below in parentheses) is entered in the TABLE ENTRY-2 field on the Global Extension screen. The TABLE ENTRY-8 and TABLE ENTRY-9 fields on the Global Extension state screen for this version of the Wincor EMV Chip Card State screen are not used and must be set to 0.

TABLE ENTRY-2 (Next State If Language Matched) — The state number if the preferred language from the application matched this supported language. Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-3 (Language Offset) — The language offset to activate for this language. Valid values are 000 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-4 (Language Group) — The language group to activate for this language. Valid values are 000 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-5 (Next State If Language Matched) — Valid values are 000 through 254 and 256 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-6 (Language Offset) — The language offset to activate for this language. Valid values are 000 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

TABLE ENTRY-7 (Language Group) — The language group to activate for this language. Valid values are 000 through 999.

Field Length: 1–3 numeric characters
Required Field: No
Default Value: 0

Wincor EMV Application Identifier Table

The Wincor EMV Application Identifier (AID) Table identifies the applications that run in NDC+ emulation and are configured using the Wincor EMV Chip Card State. The Wincor EMV Application Identifier Table screen is shown below, followed by descriptions of its fields. This screen is accessed from the NCR 5XXX Configuration File Menu.

```

BASE24-DCT  WINCOR AID TABLE          LLLL      YY/MM/DD  HH:MM  01 OF 01

EMV GROUP      0          AID NUMBER      0

APPLICATION ID _____

LOWEST  VERSION NUMBER  0000      HIGHEST VERSION NUMBER  0000
TERMINAL ACTION CODE - DENIAL 0000000000
REQUEST DATA OBJECTS
_____
STATUS MESSAGE DATA OBJECTS
_____
AID OPTION 00 PARTIAL SEL - NO TRK2 AGAINST PAN
DEFAULT APPL LABEL _____

```

```

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP

```

AID NUMBER — The number of the application. Valid values are 0 through 255.

Field Length: 1–3 numeric characters

Required Field: Yes

Default Value: 0

APPLICATION ID — The ID of the application. This field is left-justified and cannot contain embedded spaces. Valid values are 0–9 and A–F.

Field Length: 0–32 hexadecimal characters

Required Field: No

Default Value: No default value

LOWEST VERSION NUMBER — The lowest version number of the application. Valid values are 0000 through FFFF.

Field Length: 4 alphanumeric characters
Required Field: No
Default Value: 0

HIGHEST VERSION NUMBER — The highest version number of the application. Valid values are 0000 through FFFF.

Field Length: 4 alphanumeric characters
Required Field: No
Default Value: 0

TERMINAL ACTION CODE - DENIAL — The terminal action code. The valid value range is 0000000000 through FFFFFFFF, blank characters are not allowed.

Field Length: 10 alphanumeric characters
Required Field: Yes
Default Value: 0000000000

REQUEST DATA OBJECTS — The tags of the data objects to be contained in the transaction request. This field is left-justified and cannot contain embedded spaces. Valid values are 0–9 and A–F.

Field Length: 1–40 hexadecimal characters
Required Field: No
Default Value: No default value

STATUS MESSAGE DATA OBJECTS — The tags of the data objects to be contained in the completion and other status messages. This field is left-justified and cannot contain embedded spaces. Valid values are 0–9 and A–F.

Field Length: 1–40 hexadecimal characters
Required Field: No
Default Value: No default value

AID OPTION — A code that identifies whether the application identifier provided on the chip must be an exact match with the value in the APPLICATION ID field and whether to check MasterCard Track2 (Tag 57) against the PAN (Tag 5A). Valid values are as follows:

- 00 = Partial selection and do not check MasterCard Track2 against the PAN
- 01 = Partial selection and check MasterCard Track 2 against the PAN
- 10 = Exact selection and do not check MasterCard Track2 against the PAN
- 11 = Exact selection and check MasterCard Track 2 against the PAN

Field Length: 2 numeric characters
Required Field: Yes
Default Value: 00

DEFAULT APPL LABEL — A label used to identify an application process. This field is left-justified and can contain embedded spaces.

Field Length: 1–16 alphanumeric characters
Required Field: Yes
Default Value: No default value

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ACI Worldwide, Inc.

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